



**GREEN
CLIMATE
FUND**

Meeting of the Board
17 – 20 October 2018
Manama, Bahrain
Provisional agenda item 17

GCF/B.21/10/Add.11

26 September 2018

Consideration of funding proposals - Addendum XI

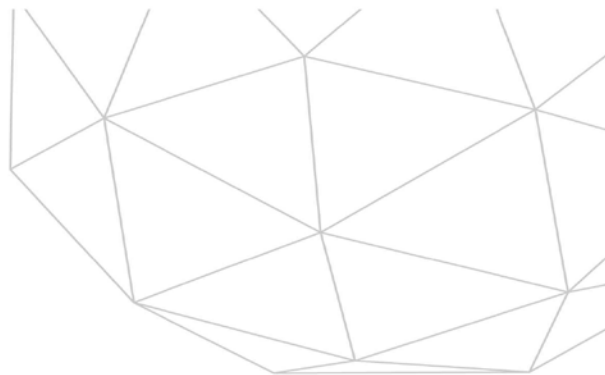
Funding proposal package for FP092

Summary

This addendum contains the following three parts:

- a) A funding proposal summary titled “Programme for integrated development and adaptation to climate change in the Niger Basin (PIDACC/NB)”;
b) No-objection letter issued by the national designated authority(ies) or focal point(s); and
c) Environmental and social report(s) disclosure;

The funding proposal package for FP092 is being submitted for the first time for the Board's consideration at its twenty-first meeting.



Funding Proposal

Version 1.1

The Green Climate Fund (GCF) is seeking high-quality funding proposals.

Accredited entities are expected to develop their funding proposals, in close consultation with the relevant national designated authority, with due consideration of the GCF's Investment Framework and Results Management Framework. The funding proposals should demonstrate how the proposed projects or programmes will perform against the investment criteria and achieve part or all of the strategic impact results.



Project/Programme Title: **Programme for Integrated Development and Adaptation to Climate Change in the Niger Basin (PIDACC/NB)**

Country/Region: **West Africa (Benin, Burkina Faso, Côte d'Ivoire, Guinea, Mali, Niger, and Nigeria), Central Africa (Cameroon and Chad).**

Accredited Entity: **African Development Bank**

Date of Submission: June 25, 2018

Contents

Section A	PROJECT / PROGRAMME SUMMARY
Section B	FINANCING / COST INFORMATION
Section C	DETAILED PROJECT / PROGRAMME DESCRIPTION
Section D	RATIONALE FOR GCF INVOLVEMENT
Section E	EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA
Section F	APPRAISAL SUMMARY
Section G	RISK ASSESSMENT AND MANAGEMENT
Section H	RESULTS MONITORING AND REPORTING
Section I	ANNEXES

Note to accredited entities on the use of the funding proposal template

- Sections **A, B, D, E** and **H** of the funding proposal require detailed inputs from the accredited entity. For all other sections, including the Appraisal Summary in section F, accredited entities have discretion in how they wish to present the information. Accredited entities can either directly incorporate information into this proposal, or provide summary information in the proposal with cross-reference to other project documents such as project appraisal document.
- The total number of pages for the funding proposal (excluding annexes) is expected not to exceed 50.

Please submit the completed form to:
fundingproposal@gcfund.org

Please use the following name convention for the file name:
FP-AFDB-04June2017-AFDB012017
“[FP]-[Agency Short Name]-[Date]-[Serial Number]”

A.1. Brief Project / Programme Information		
A.1.1. Project / programme title		Programme for Integrated Development and Adaptation to Climate Change in the Niger Basin (PIDACC/NB)
A.1.2. Project or programme		programme
A.1.3. Country (countries) / region		West Africa (Benin, Burkina Faso, Côte d'Ivoire, Guinea, Mali, Niger, and Nigeria), Central Africa (Cameroon and Chad)
A.1.4. National designated authority (authorities)		All National Designated Authorities/Focal Persons of the nine member countries of the Niger basin authority (NBA)
A.1.5. Accredited entity		African Development Bank
A.1.5.a. Access modality		☐ Direct ☒ International
A.1.6. Executing entity / beneficiary		Executive Secretary of Niger Basin Authority (NBA) and ministers in charge of water resources of the nine countries Beneficiary: Rural population of Niger river basin
A.1.7. Project size category (Total investment, million USD)		☐ Micro (≤ 10) ☐ Small ($10 < x \leq 50$) ☒ Medium ($50 < x \leq 250$) ☐ Large (> 250)
A.1.8. Mitigation / adaptation focus		☐ Mitigation ☒ Adaptation ☒ Cross-cutting
A.1.9. Date of submission		
A.1.10. Project contact details	Contact person, position	Laouali GARBA, Chief Climate Change Officer Timothy Afful Koomson, Chief Climate Finance Officer
	Organization	African Development Bank
	Email address	l.garba@afdb.org t.afful-koomson@afdb.org
	Telephone number	(225) 20 26 1990
	Mailing address	18ème Etage de l'Immeuble CCIA BPV 316 Abidjan ; Bureau 18-K
A.1.11. Results areas (mark all that apply)		
<u>Reduced emissions from:</u> ☐ Low emission transport (E.g. high-speed rail, rapid bus system, etc.) ☒ Forestry and land use (E.g. forest conservation and management, agro-forestry, agricultural irrigation, water treatment and management, etc.) <u>Increased resilience of:</u> ☐ Most vulnerable people and communities (E.g. mitigation of operational risk associated with climate change – diversification of supply sources and supply chain management, relocation of manufacturing facilities and warehouses, etc.) ☒ Health and well-being, and food and water security (E.g. climate-resilient crops, efficient irrigation systems, etc.) ☐ Infrastructure and built environment (E.g. sea walls, resilient road networks, etc.) ☒ Ecosystem and ecosystem services (E.g. ecosystem conservation and management, ecotourism, etc.)		

A.2. Project / Programme Executive Summary (max 300 words)

Please provide a brief description of the proposed project/programme, including the objectives and primary measurable benefits (see [investment criteria in section E](#)). The detailed description can be elaborated in [section C](#).

High population growth rates, land degradation and use pressures, changes in rainfall patterns, greater frequency and intensity of droughts, and conflicts over natural resources are undermining the resilience of Niger Basin communities—disrupting countless livelihoods and leading to famine and high mortality rates.

To promote climate resilient growth in the Basin, the African Development Bank, European Union, Global Environment Facility and the affected countries have allocated financing for the Programme for Integrated Development and Adaptation to Climate Change in the Niger Basin (PIDACC/NB). Supplemental GCF financing will help forge the required collaboration and provide additional concessional resources for urgent intervention.

Based on preliminary studies undertaken by the Niger Basin Authority (NBA), the PIDACC/NB builds on the Niger Basin Climate Resilience Investment Plan (CRIP). The CRIP aims to enhance climate resilience in Benin, Burkina Faso, Cameroon, Chad, Côte d'Ivoire, Guinea, Mali, Niger and Nigeria, through regional and national water resource investments in reliable information systems, strengthened institutions, and sustainable infrastructure. PIDACC/NB also follows successful implementation of the Multinational Silt Control in the Niger River Basin.

PIDACC/NB aims to contribute to improving the resilience of populations and ecosystems in the Basin through sustainable management of natural resources by: reducing the silting process of the Niger River; enhancing the adaptability of populations to climate change; and improving natural resource management and integrated ecosystem management, the protection of biodiversity and the restoration of soil fertility.

GCF-related outcomes include improvements in:

- management, restoration and protection of natural habitats from climate variability and change;
- generation and use of climate information in decision-making;
- adaptive capacity and reduced exposure to climate risks;
- awareness of climate threats and risk reduction processes; and
- management of land or forest areas contributing to emissions reductions.

The programme is scalable and contributes to transformational change. It is also structured to ensure long-term financial sustainability.

A.3. Project/Programme Milestone

A.3. Project/Programme Milestone

Expected approval from accredited entity's Board (if applicable)	October 2018
Expected financial close (if applicable)	December 2018
Estimated implementation start and end date	Start: <u>January 2019</u> End: December 2024
Project/programme lifespan	6 years

B.1. Description of Financial Elements of the Project / Programme

The detailed costs of the programme, of activities and components are provided in Annex III.

Table 1: Breakdown of costs per component

Components	Components	Million US\$ Amount	In %
C 1	Development of ecosystems and natural resources resilience	86.219	41%
C 2	Development of population resilience	102.603	49%
C 3	Programme coordination and management	21.081	10%
	Total	209.903	100%

Table 2: Breakdown of cost estimates for total project costs and GCF financing by sub-component in local and foreign currency

Component	Sub-component (if applicable)	Amount (Entire project)	Currency	Amount (for entire project)	Local currency	GCF funding amount	Currency of disbursement to recipient
C 1	Sub-component 1.1	70.591	<u>million USD</u>	37,731.9	Million FCFA	25.598	million USD
	Sub-component 1.2	15.627	<u>million USD</u>	8,353.1	Million FCFA	8.705	million USD
C 2	Sub-component 2.1	82.892	<u>million USD</u>	44,306.8	Million FCFA	8.614	million USD
	Sub-component 2.2	19.712	<u>million USD</u>	10,536.2	Million FCFA	14.801	million USD
C 3	Coordination, M & E	21.081	<u>million USD</u>	11,268.2	Million FCFA	10.056	million USD
Total project financing		209.903	<u>million USD</u>	112,196.2	Million FCFA	67.774	

Table 3: Estimated amount of GCF grant and loan activities for each component/sub-component

Component	Sub-component	Amount Entire project [Millions US\$]	GCF Loan Financing [Mill US\$]	GCF Grant Financing [Mill US\$]	GCF Total Financing [Mill US\$]
C 1	Sub-component 1.1	70.591	0.442	25.155	25.598
	Sub-component 1.2	15.627	0.000	8.705	8.705
C 2	Sub-component 2.1	82.892	6.287	2.327	8.614
	Sub-component 2.2	19.712	3.271	11.530	14.801
C 3	Coordination and M & E	21.081	0.000	10.056	10.056
Total		209.903	10.000	57.774	67.774

Table 4: Disbursement table for GCF grant

Year	Grant disbursements (Millions, USD)	(%)
2019	6.777	12%
2020	8.555	15%
2021	8.555	15%
2022	13.555	23%
2023	13.555	23%
2024	6.777	12%
Total	57.774	100%

Table 5: Disbursement table for GCF Loan

Year	Loan disbursements (Millions, USD)	(%)
2019	0.000	0%
2020	5.000	50%
2021	5.000	50%
2022	0.000	0%
2023	0.000	0%
2024	0.000	0%
Total	10.000	100%

A breakdown of cost/budget by expenditure type (project staff and consultants, travel, goods, works, services, etc.) and disbursement schedule in project/programme confirmation (term sheet) as included in section I, Annexes.

B.2. Project Financing Information

	Financial Instrument		Amount	Currency	Tenor		Pricing
(a) Total project financing	(a) = (b) + (c)		209.903	million USD (\$)			
(b) GCF financing to recipient	(i) Senior Loans		10.000	million USD (\$)	40 years	0 %	
	(ii) Subordinated Loans			(\$)			
	(iii) Equity			Options			
	(iv) Guarantees			Options			
	(v) Reimbursable grants *		57.774	Options			
	(vi) Grants *			Options			
				million USD (\$)			
* Please provide economic and financial justification in <u>section F.1</u> for the concessionality that GCF is expected to provide, particularly in the case of grants.							
	Total requested		67.774	Options			
(c) Co-financing to recipient	Financial Instrument	Amount	Currency	Name of Institution	Tenor	Pricing	Seniority
	Loan (46%)	35.924	million USD(\$)	African Development Bank	40 years	0%	Senior
	Grant (54%),	42.171	million USD(\$)	African Development Bank (ADF)			
	Grant	18.081	million USD(\$)	EU			
	Grant	12.980	million USD(\$)	GEF			
	Grant	9.000	million USD(\$)	FIP			
	In-Kind	16.186	million USD(\$)	Countries			
	In -Kind	7.786	million USD(\$)	Beneficiaries			
* Please provide a confirmation letter or a letter of commitment in section I issued by the co-financing institution.							
- Letters requesting financing from the African Development Bank (ADF 14) have been received from all nine Niger Basin Countries. The Bank contribution is estimated at US\$ 78.095 million (comprising US\$. 35.924million as Loan and US\$ 42.171 million as Grant) - The GEF contribution of US\$ 12.980 million was approved early 2017. - The EU funding amounting to US18.081 million ((15 million Euros) was confirmed in December 2017. - Funding from the Forestry Investment Fund (FIP), US\$ 9.000 million was endorsed by the Climate Investment Funds Committee (CIF) in August 2018. - The GCF grant is estimated at US\$ 57.774 million. GCF Grant proceeds will be used to finance technical assistance and to non-revenue generating interventions. Grant proceeds will not be blended with the AfDB loan. - The GCF Loan is estimated at US\$10.000 million and proceeds will be used to mainly finance activities under component 2. These include (i) construction of small hydro-agricultural infrastructure and (ii) development of water points for cattle and development of transhumance pathways infrastructure under Sub component 2-1 and (ii))Community Based Livelihood infrastructure under sub component 2-2. - The country contribution (10%) and beneficiary contribution (up to 5%) were endorsed by Resolution No. 2 of the 11th Summit of Heads of State and Government of the Niger Basin Authority (NBA) held in Cotonou (January 2016)¹.							

B.3. Financial Markets Overview (N/A)

Not Applicable

¹ Actual country and beneficiary contributions are estimated at 8% and 4% respectively of the total project costs

C.1. Strategic Context

Please describe relevant national, sub-national, regional, global, political, and/or economic factors that help to contextualize the proposal, including existing national and sector policies and strategies.

C 1-1 Status of the Niger Basin, including basin strategies

The Niger Basin area is 2,240,000 km² (1,500,000 km² is hydrologically active). The basin cuts across the major climatic zones—which include: the Guinea Savanna vegetation belt in the Fouta Djallon Massif in Guinea, the semi-arid Sahel Savanna zone and the tropical mangrove forests in Nigeria's Niger Delta. An estimated 110 million people live in the Basin. The population is expected to rise to more than 180 million by 2025. Heavy reliance on natural resources, combined with ongoing conflicts and external effects of climate change, make it one of the most fragile river basins in Africa.

Niger Basin Shared Vision. The NBA member countries decided in February 2002 to lead a Shared Vision process for the sustainable development of the Basin. The Shared Vision is an expression of the countries' commitment to promote a framework for enhancing cooperation and sharing benefits deriving from the Niger Basin's resources. The Shared Vision is implemented through a Sustainable Development Action Program (SDAP).

The SDAP with its Implementation Plan (2008- 2027). The SDAP defines and guides the development process in the Niger River Basin. It comprises 350 actions for a total amount of US\$ 7.2 billion. An estimated 89.4% of the investments planned in the Operational Plan are devoted to strengthening resilience to the populations of the basin.

Niger Basin Water Charter (2008). The charter is designed to ensure that NBA Member States share the river's resources equitably. The Charter lays out a series of principles for equitable and reasonable participation and use of the Niger River water.²

Silt Control in the mid-Niger River Basin. The project financed by the African Development Bank (€ 30 million) was located in the loop of the Niger in Mali and Burkina Faso and the right bank of the River in Niger upstream from Niamey. Achievements included: (i) adoption of a Silting Control Master Plan and Investment Programme (**SDLE**); (ii) institutional strengthening of the NBA Executive Secretariat and member countries and (iii) watershed restoration and anti-silting plans ~ 38, 088 ha.

Climate adaptation Initiatives in the Niger basin. The initiatives for the COP21 relating to the Niger River Basin are numerous, particularly in terms of adaptation to climate change. The Intended Nationally Determined Contributions (INDC) of the countries include an adaptation part. The regional institutions (ECOWAS, ECCAS, and CILSS etc.) each have initiatives for adaptation to climate change specific to the water resources sector. These strategic documents define priority areas for intervention at national and regional level to enhance resilience to climate change. The CRIP was developed so that, the measures selected are compatible and complementary to the above initiatives to ensure they are coherent at national and regional level. The CRIP presented at COP 21 is the primary guiding document with respect to climate adaptation within the Niger basin.

The Climate Resilience Investment Plan (CRIP). Under the platform of the NBA, Niger Basin countries also collaboratively developed the Investment Plan for the Strengthening of Resilience to Climate Change in the Niger River Basin (CRIP). The CRIP was developed with the support of the African Development Bank (AfDB), World Bank Water and Energy Global Practices and World Bank Cooperation in International Waters in Africa (CIWA) programme.

The CRIP (2015), is a strategic planning document that comprises 246 climate change adaptation actions for strengthening the resilience to climate change throughout the Niger Basin (see table 4). The CRIP's priority actions were selected from existing planning documents and vetted through an inclusive consultative process with multi-sectoral participation, strategically coupled with exercises to build local capacity. The CRIP, took into account all the actions planned under the PIDACC. The Niger River Basin member States collaborated on the Plan and pledged to advocate for technical and financial support to implement its priority actions.

The plan was presented at the UNFCCC COP21 in Paris to raise the profile of their climate adaptation needs. The confirmed adaptation actions form an integral part of the 2016-2024 NBA Operational Plan (OP), which structures the implementation of the SDAP of the Basin over the 9 nine years in 3-year increments (2016-2018, 2019, 2021, 2022-2024). The CRIP framework is shown in Table 6 below. All PIDACC activities are integrated in NBA's CRIP.

² <http://www.internationalwatersgovernance.com/niger-basin.html> accessed October 30, 2017

Table 6: Niger Crip Focus Areas & Costs

Measures contributing to climate resilience	Number of actions	Costs Millions US\$
Governance and Knowledge Package	54	242
Knowledge. Collection and generation of climatic and hydrological information. Weather-forecasting tools and decision-making tools	26	78
Integration of climate change adaptation into capabilities, bodies and management instruments of national and regional Basin institutions	9	129
Assessment of vulnerability. Communication and awareness-raising	19	35
Sectoral Package:	192	2,869
Vulnerability to water stress	77	1,619
Vulnerability to flooding	7	63
Vulnerability to soil degradation	30	201
Vulnerability to degradation of grazing land	8	127
Vulnerability to ecosystem degradation	47	636
Vulnerability to deterioration of the water quality	5	3
Vulnerability linked to the rising sea-level	1	1
Resilience (alternative livelihoods)	17	219

Details on the 246 CRIP actions can be accessed in the reference: Investment Plan for the Strengthening of Resilience to Climate Change in the Niger Basin Version for presentation at the meeting of the Finance Ministers of NBA Member States Version 5.1 30 September 2015; <http://documents.worldbank.org/curated/en/140841467999087867/Plan-d-investissement-pour-le-renforcement-de-la-resilience-au-changement-climatique-du-bassin-du-fleuve-Niger>

C 1.2 Programme for Integrated Development and Adaptation to Climate Change in the Niger Basin (PIDACC)

The PIDACC. PIDACC is the first operational program of CRIP which will be approved in 2018 and includes all actions of the 1st phase. It is a climate change adaptation program with co-benefits for climate mitigation. It is a program and not an investment plan. The preparation of PIDACC started in 2012 (supported by AfDB and the West African Economic and Monetary Union, UEMOA) on the basis of the actions identified in i) the master plan for silting control of the Niger River in the nine countries (SDLE, 2007) and ii) the Sustainable Development Action Plan (SDAP) of the Niger Basin, 2008.

PIDACC aims at preserving the ecosystems of the Niger River and improving the resilience of the nine Basin countries, of which 7 are ECOWAS countries (Benin, Burkina Faso, Côte d'Ivoire, Guinea, Mali, Niger and Nigeria) and two belong to the ECCAS (Cameroon and Chad). The nine countries endorsed the PIDACC/NB and the two economic communities included it in the list of priority regional investment programs. The sub-basins covered by PIDACC interventions are presented in **Annex I** and include: (i) Niger Upper Basin (Guinea–Kankan); (ii) Bani Basin (Côte d'Ivoire- Folon); (iii) Niger Inland Delta (Mali – Koulikoro – Mopti – Ségou – Sikasso and Bamaba); (iv) Middle Niger (Burkina Faso; Mali – Gao, Timbuktu, Ansongo and Bourem; Niger - Tillabéri; Niger - Tahoua; Niger – Dosso; Benin – Karimama / Malanville); and (v) Benue Upper Basin (Nigeria - Upper Benue (RBDA); Cameroon - Guider ; Mokolo; Hina and Mogodé and Chad– Department of Kabbia).

In prioritisation of measures for the PIDACC the following criteria (standard for all the countries), were considered:

Measures must contribute to strengthening resilience to climate change and fulfil at least one of the following 2 criteria.

- Institutional measures, knowledge enhancement measures or capacity-building measures contributing to enhancing resilience to climate change (criterion 1).
- Sector-specific measures contributing to strengthening resilience of the populations and ecosystems of the basin to the climate changes identified in the Niger Basin (criterion 2). Sub criterion include: (measures aiming to reduce vulnerability linked to water stress conditions, flooding, deterioration of the water quality, soil degradation, degradation of the grazing land, degradation of the ecosystems and measures aiming to strengthen resilience (e.g. generation of jobs, revenues, etc.)

Moreover, it must also satisfy all the other 8 criteria (3 to 10):

- Contribute to the sustainable development and management of the Niger Basin (criterion No.3)
- Avoid maladaptation (criterion No.4)
- Contribute to the protection, management and development of the water resources in the Niger Basin, as well as of the related ecosystems (criterion No.5)

- Be compatible with the planning documents and agreements concerning the Niger Basin, CRIP, Masterplan for silt control (SDLE) and the SDAP (criterion No.6)
- Be-adopted by the Member States (taking into account national circumstances —criterion No.7)
- Demonstrate the level of preparation required (criterion No.8)
- Pre-identified funding (criterion No.9)
- Economic feasibility (criterion No.10)

C-1-3: African Development Bank experience, lessons reflected in PIDACC programme design

The African Development Bank's experience from multinational programmes in shared basins have informed PIDACC programme design; in particular the programmes financed in Lake Chad (PRODEBALT and PRESIBALT), Lake Victoria, Lake Tanganyika and the Songwe River Basin. Lessons include the following:

- (i) Flexibility and adaptability of development support instruments like in the Niger Silting Programme determine the achievement of programme targets. In activity design, emphasis should be placed on developing decentralized services to support grassroots organizations and investments.
- (ii) Long-term commitment is needed for achieving sustainable results and is important for institutional development programs like the NBA and the National Focal Institutions as well.
- (iii) Programmes must be firmly anchored to the local economy. Direct relations must be facilitated among farmers' organizations and local CSOs, enterprises, giving the former control over implementing most of the investments financed by the loan. Communities should have greater freedom to choose their service providers, suppliers and construction enterprises.
- (iv) To support the implementation of a multi-country project, it is important to strengthen the capacity of the regional organization whose role is to support the countries to manage the shared resource. The African Development Bank mobilized GEF financing to support the NBA in playing its role of monitoring/advising the nine countries for the implementation of PIDACC. Similar efforts have been applied to the Lake Chad and the Songwe River basins.

C.2. Project / Programme Objective against Baseline

Describe the baseline scenario (i.e. emissions baseline, climate vulnerability baseline, key barriers, challenges and/or policies) and outcomes and the impact that the project/programme will aim to achieve in improving the baseline scenario.

C-2-1: Climate vulnerability baseline

C-2-1-1: Climate drivers for vulnerability (weather and climate variability) in the Niger Basin

Climate drivers are involved in the shocks that push households in the Niger Basin into poverty. A map showing climate hotspots in the Sahel is shown in Figure 1.

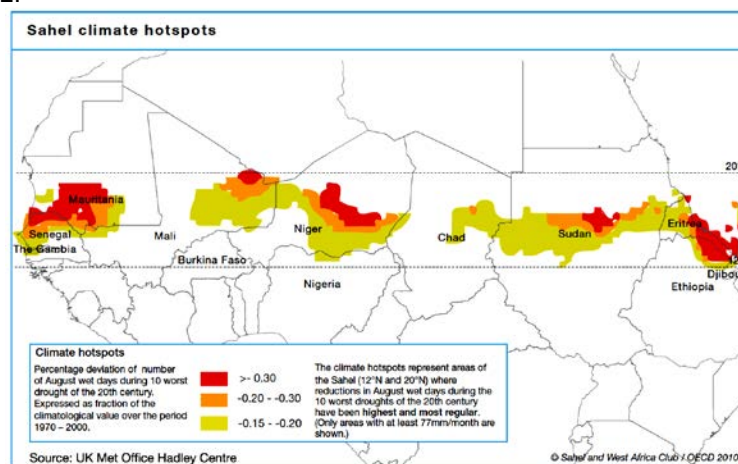


Figure 1: Map of Sahel showing climate hotspots (Source UK Met Office Hadley Centre)

The drivers include: extreme weather events (major floods and droughts resulting in loss of assets and disability for communities); weather related risks; crop losses (disruption of production due to adverse weather); and food price shocks. The PIDACC program will contribute towards efforts to address climate variability in the Niger Basin.

Temperature trends in the Niger Sub-Basins

The temperature trends in the Basin are all positive (Figure 2) and significantly higher than the global average rates of temperature increases reported by the IPCC, Kundzewicz et al (2007).

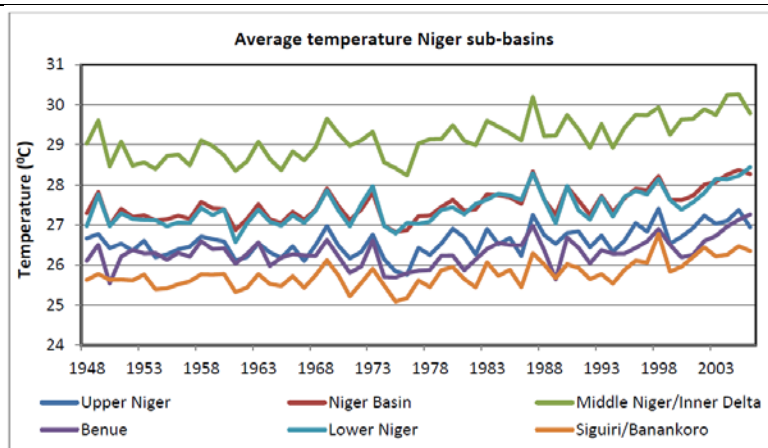


Figure 2: Long term average annual temperature (top panel) and rainfall (bottom panel) trends in sub catchments of the Niger Basin (data source: Hirabayashi, 2008)³

Temperatures across the Sahel have increased by nearly 1°C since 1970, at a rate nearly twice the global average.⁴ As may be expected, the largest trend (0.023°C/year) occurs in the Inner Delta-Middle Niger Sub-Basin. This rate translates to a change in the average temperature of 0.85°C between 1970 and 2006 or, using the trend since 1901, an increase of 1.91°C.

Rainfall trends

The Sahel represents a hotspot as far as annual rainfall variability. It is characterized by a great inter-annual rainfall, which has been expressed by multi-year droughts since 1968, as shown in Figure 3.

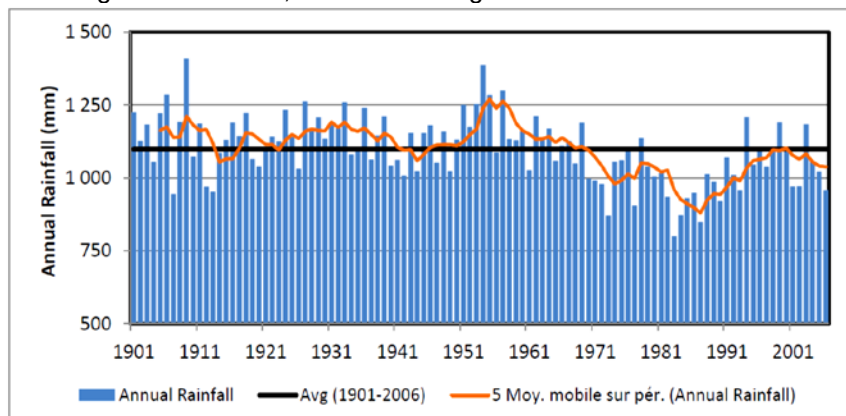


Figure 3: Annual rainfall for the Niger Basin (CRU data)

Instrumental rainfall records in the Sahel show the strong variability of rainfall, in addition to the well-documented decline in rainfall since the late 1960s (Hulme, 2001). The rainfall record reveals a shift in climate around 1950, changing from predominantly year-to-year variability to variability visible over longer periods of decades. The last 40 years since 1969 have witnessed reductions in mean annual rainfall, with a decrease of 20–40% noted between the periods 1931–1960 and 1968–1990 (IPCC 2007; Chappell and Agnew 2004).⁵ From the mid-1990s, rainfall recovered somewhat in the Niger Basin, with rainfall in a number of years being above the long-term mean. This has been followed by an increase in vegetation in some areas, in part due to rainfall increases (Olsson et al., 2005). Measures are required for strengthened resilience against rainfall variability.

C-2-1-2: Impacts of extreme climate events (droughts, floods) in the Niger Basin

Trends observed over the last 40 years in the Basin show that overall temperatures have risen, droughts have been recurrent and severe, rainfall has increased, and floods have occurred more frequently and with more intensity. (Figure 4). The most

³ May 2014; Niger River Basin Sustainable Development Action Plan' Niger River Basin Climate Risk Assessment; Final Report; Volume 2; Water Resources Profile and Historical Climate Variability; A Joint Initiative of the Niger Basin Authority (NBA) and the World Bank

⁴ June 2015; Alec Crawford; Climate change and state fragility in the Sahel;

⁵ November 2010; Keffing Sissoko Herman van Keulen et al. Agriculture, livelihoods and climate change in the West African Sahel

severe droughts occurred in 1912-1915, 1970-1974, and 1979-1993 and recently in 2014.⁶ The dry episode that started in the 1960s culminated in severe droughts in 1973, 1984 and 1990, and resulted in a decline in rainfall in the range of 20–30% in the Sahel region.

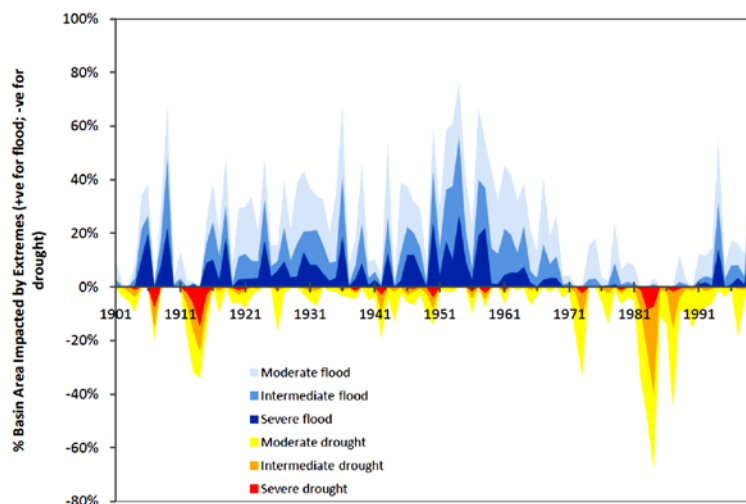


Figure 4: Niger Basin Weighted Anomaly of Statistical Precipitation (WASP) Index⁷

Droughts are particularly harmful to rain-fed agriculture and pastoralism. In the Niger basin, droughts killed up to 250,000 people and decimated over two million livestock in the 1970–1973 period (Kandji, Verchot, and Mackensen 2006). Benson and Clay (1998) assessed the impact of drought on the GDP of several African countries including Nigeria, Burkina Faso, and Niger. They did this by comparing the performance of GDP in the pre-drought and drought years. Their results showed that in 1984, one of the worst drought years on record, Niger's economy suffered a 6% decline. The corresponding values for Nigeria and Burkina Faso were 4.5% and 1.5% decline in GDP.

In the Niger Basin, floods in 2012 destroyed hundreds of thousands of farm acres of land in Nigeria, displacing 1.3 million people and causing 431 deaths (Kandji, Verchot, and Mackensen 2006). Paeth et al. (2009) analysed the conditions associated with widespread flooding in the Niger Basin in 2007 and associates flooding with a number of factors, including anomalous heating in the tropical Atlantic and to La Niña in the tropical Pacific. Other climate-related hazards of significance in the Niger basin include extreme temperatures, sand and dust storms, strong winds, brush fires and locust swarms.⁸

Extreme climate events not only impact livelihoods through disaster shock, they also—push poor households to adopt low-risk strategies that have low returns, and leave no prospect of rising above the poverty line (Hallegatte et al. 2016).

C-2-1-3 Uncertainty with respect to climate risk⁹

In the Niger Basin, all GCMs project significant increases in temperature, mostly between 1°C and 3°C for 2050 and 4.0°C by the end of the 21st century for both CMIP3 and CMIP5 Global Circulation Models (GCMs) under the A2 and A1B scenarios (CMIP3) and RCP4.5 and RCP8.5 (CMIP5).¹⁰ Two out of three climate models project small to moderate increases in precipitation and 90% of the analysed climate projections for the 21st century indicate changes between -6% and +7% for 2050 (Figure 5).

GCMs do not agree on whether the Niger Basin is likely to become wetter or drier in the 21st Century. 50% of the models used by the IPCC (Christensen et al., 2007) predict increased rainfall, while the other half predict decreased rainfall. However a high degree of climatic variability is projected to continue, and may become more pronounced on seasonal, annual and decadal timescales, to the scale that resulted in the severe droughts of the 1970s and 1980s.¹¹ This underscores the importance of strengthening the capacity of institutions in the basin to plan investments that can deliver the intended benefits under a wide range of future climates at various timescales. The PIDACC will factor this in the project design

⁶ February 2010; Hadley Centre Climate Change Consultancy; Sahelian climate: past, current, projections

⁷ IRI (2009) for World Bank Africa Region Water Resources & Climate AAA.

⁸ December 2011; Marisa Goulden and Roger Few; Climate Change, Water and Conflict in the Niger River Basin; USAID

⁹ February 2010; Hadley Centre Climate Change Consultancy; Sahelian climate: past, current, projections

¹⁰ June 2015; Alec Crawford; Climate change and state fragility in the Sahel

¹¹ Juddy N. Okpara, Aondover A.Tarhule and Muthiah Perumal; Study of Climate Change in Niger River Basin, West Africa: Reality Not a Myth

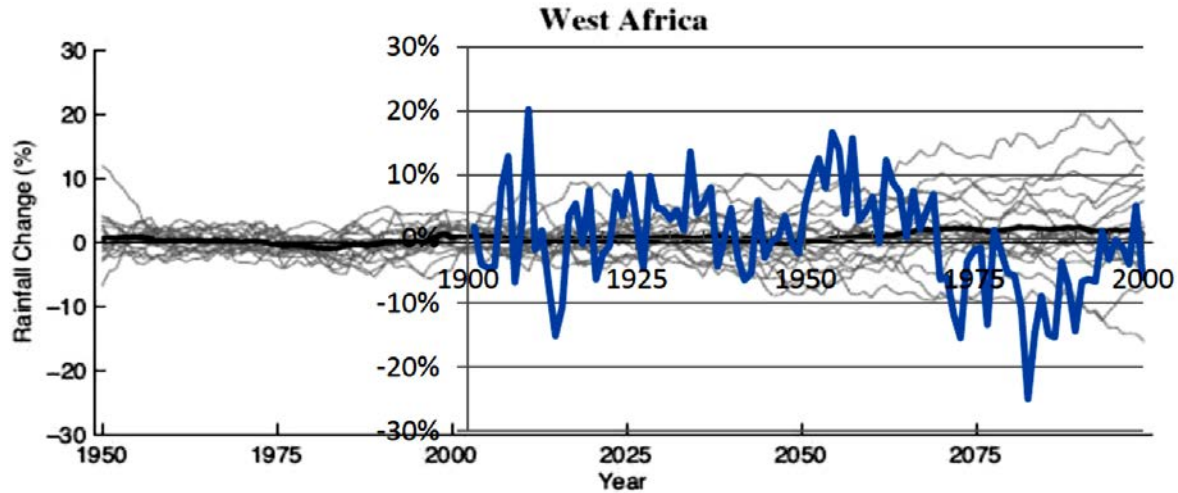


Figure 5: Ensemble of regional climate projections for West Africa (Source: Giannini et al., 2008; bold black line is the ensemble mean for the A1B emission scenario; bold blue line reflects the historical 20th century rainfall variability for the Niger Basin)¹²

C-2-2: Climate-related sectoral challenges in the Niger Basin

C-2-2-1: Biodiversity loss in the Niger Basin

Climate change increases the threats to biodiversity. Biodiversity and ecosystem services help the Niger basin countries to adapt to and mitigate climate change. Ecosystems in the Niger basin are already degraded due to prolonged drought, agricultural expansion, deforestation, erosion, the proliferation of invasive species, and biodiversity loss from poaching. As a result of long-term declines in rainfall from the 1970s to 1990s, the Sahel ecological zone has shifted 25–35 km southward, resulting in biodiversity loss and conversion of arable land to sand dunes. Climate change poses additional risks, such as habitat and biodiversity loss, due to increased temperatures and extreme weather events all of which alter food chains and the habitats that support them. Endangered mammals such as the cheetah, giraffe, and lion, once common in the region, are now largely absent outside of protected areas (e.g., Air and Ténéré National Nature Reserve in Niger and Sahel Partial Faunal Reserve in Burkina Faso). Biodiversity loss will lead to the erosion of ecosystem services and will exacerbate vulnerability to the impacts of climate change. Among the main threats to biodiversity are climate change, loss of natural habitats and environmental degradation. Invasive species, with their high evapotranspiration rates, are an immense burden to already water-scarce regions like the Niger basin. Climate change threatens to further degrade land, vegetation and water resources through increased incidence of drought, desertification and floods and projected shortening of the rainy season. A drier climate could result in 3,000 km² reduction of the Niger Inner Delta, which will impact aquatic biodiversity, fisheries, and rice production (Zwarts et al. 2005). Bush fires are predicted with increasing frequency as temperatures rise in the Sahelian part of the Basin leading to the release of more carbon to the atmosphere. Enhanced management of biodiversity could mitigate these impacts and contribute to climate adaptation. By conserving nature and restoring ecosystems PIDACC interventions will reduce vulnerability and increase resilience. With strengthened protection and improved fire management, millions of tons of carbon could be stored in the woodlands.¹³

Social and economic vulnerability is a concern in communities that are dependent on fisheries and aquaculture. Many aquatic species depend on the timing of rainfall and flood events for important migrations (e.g. spawning and feeding). Inland fisheries are at risk because of water shortages, drought and impacts from other sectors. Freshwater fisheries throughout the Niger basin are based largely on species that migrate to wetland systems within the basin for breeding and spawning. The combination of changes in rainfall and runoff resulting in falling water levels during droughts, is likely to affect habitats and fisheries in the basin, impacting the movement of fish and affecting nursery ground function. Changes may occur in timing and success of migrations, spawning and peak abundance, as well as in sex ratios. In addition there may be potential loss of species or a shift in composition in capture fisheries and impacts on seed availability for aquaculture, thereby threatening livelihoods. Stability of supply will be impacted by changes in seasonality, increased variance in ecosystem productivity and increased supply variability and risks.¹⁴ Increased storage could promote integration of fish farming through cage culture and enhance fisheries production.

¹² World bank 2014; Niger River Basin Sustainable Development Action Plan Niger River Basin Climate Risk Assessment; Final Report; Volume 1; Main Report; A Joint Initiative of the Niger Basin Authority (NBA) and the World Bank

¹³ Protected areas: providing natural solutions to 21st Century challenges N. Lopoukhine, N. Crawhall, N. Dudley, P. Figgis, C. Karibuhoye, D. Laffoley, J. Miranda Londoño, K. MacKinnon and T. Sandwith

¹⁴ FAO, 2016; Fisheries, Aquaculture And Climate Change; The role of fisheries and aquaculture in the implementation of the Paris agreement

C-2-2-2 Climate change and land degradation:

Unsustainable land use practices and watershed degradation. The Sahelian part of the Niger basin has witnessed land degradation over the past several decades, triggered by incessant droughts and increasing desertification. Slash and burn farming, forest fires, deforestation, and mining concessions within selected watersheds contribute to severe degradation adversely impacting their ecological functions, and compromising their integrity. Higher temperatures and variable rainfall are additional stressors that reduce water availability and contribute to land degradation (USAID 2017). These degraded environmental conditions have fostered rural-urban migration in the region, especially in Niger, Burkina Faso, and Mali. Land degradation in the Sahel is characterised by soil degradation, mainly due to wind erosion. This is caused by climatic factors such as drought and diminishing rainfall, compounded by anthropogenic factors, including population growth, agricultural intensification and overgrazing. These factors result in the reduction of vegetation cover, decrease in fallow periods and a reduction in the balance between fallow areas and cultivated fields, which are vital to maintaining soil fertility and reducing losses from erosion. Should this trend continue, it would bring about a sharp decline in soil nutrients, accelerating the loss of vegetation cover. Agroforestry, integrated farming and practices that promote vegetation cover are proposed as sustainable land practices in the project. These will provide soil cover to protect soils against agents of erosion, increase agricultural productivity per unit land area and diversify farmers' sources of income, resulting in benefits for agricultural production and addressing land degradation¹⁵.

C-2-2-3: Water resources scarcity and increasing environmental degradation in the Niger Basin

Severe water stress across the agro-pastoralist region of West Africa.¹⁶ Water is a limiting factor for development in the Sahel. The region's water supply is unevenly distributed, poorly accessible due to undeveloped hydraulic supply systems, and crosses national boundaries, creating significant challenges (USAID 2014). In the Niger basin, where communities are faced with highly seasonal and often unpredictable rainfall, a lack of adequate water storage already causes large and avoidable economic losses from floods and droughts, and constrains long-term growth. Recent studies suggest that increased temperatures, irregular rainfall patterns and drier conditions will amplify future drought severity and temperature extremes, increasing resource scarcity and exacerbating conflict levels and increasing migration and conflict. Countries like Burkina Faso, and Niger that have a low water resources base per capita are expected to experience physical water scarcity by 2025.

Water quality degradation. Precipitation and associated river discharge are major drivers of water pollution in the form of excess nutrients and sediment, transport into the Niger river. Changing climate conditions will affect water and wastewater treatment and disposal in the Niger basin in several ways. Elevated stream temperatures, combined with lower flows, may require wastewater facilities to increase treatment to meet stream water quality standards. When drought frequencies increase, water quality in rivers may suffer because of reduced dilution of pollutant concentration, with potential health impacts. More intense precipitation and floods, combined with urbanization and associated increasing impermeable surfaces on the other hand, may amplify the likelihood of contaminated overland flow or combined sewer over flows contributing to declines in water quality in those areas. If flood frequencies increase, human health may be impacted by sewage contamination during flood events.

Navigation constraints. Navigation on the Niger River will become increasingly vulnerable to more varied precipitation patterns. The impacts from extreme weather events, higher temperatures and heat waves, precipitation changes, and low flows and other climatic conditions are affecting the reliability and capacity of navigation in the Middle and Lower Niger. Drought often lowers vessel drafts on navigable sections of the river in rivers Cameroun, Mali, Niger and Chad. Continued low water flows have limited the duration of the navigation period in Cameroun. In Chad, the Niger River carries a heavy sediment load, making navigation difficult, during low flows. In the Inland Delta (Mali) and other river areas experiencing progressive narrowing of secondary branches, the drought that dried these channels has not allowed the annual "rinsing" of the seasonal barriers created by deposits of windblown sand and dunes.

C-2-3: Barriers to climate adaptation in the Niger basin

#1—Policy and institutional barriers. Implementing a climate resilient program in the Niger basin requires supportive policies, as well as coordination across processes and institutions responsible for climate change to avoid inconsistencies. Issues of climate adaptation are communicated within environmental policy, isolating the issue from the development and disaster risk reduction agendas. Institutional barriers contribute to weak coordination between those working on climate change, development and disaster risk reduction. There is limited awareness among policy-makers and development practitioners about the risks posed by climate change, and how these relate to development priorities. Government departments responsible for poverty reduction and disaster risk reduction have limited means of coordination, which undermines adaptation efforts.

¹⁵ <https://europa.eu/capacity4dev/public-environment-climate/document/land-degradation-and-agriculture-sahel-africa-causes-impacts-and-recommendations>

¹⁶ IPCC (2007): Summary for policy-makers. In: Climate Change 2007: The physical Science Basis. Contributions of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change

#2—Weak adaptive capacity.¹⁷ The capacities for implementing Early Warning systems (EWS) in each of the countries remains varied, with weaknesses in their ability to issue critical warnings. Left unaddressed, climate change will have an increasingly negative impact on livelihoods across the region. The development of EWS, effective communication of climate information and an understanding of the local context within which adaptations take place, are pre-requisites to enhancing climate adaptations and rural livelihoods. EWS could help defuse livelihood insecurities by providing the information required to mitigate disaster risk, food insecurity, and related conflict and migration outcomes.

#3—Inadequate Climate Risk Information. Reliable hydro meteorological systems and services are needed to ensure timely early warning and preparedness, yet such systems and services are lacking in many of the Niger basin countries. The network of hydromet stations is sparse and deteriorating, and hydromet data are often inaccurate. Further, hydromet data are not freely shared (inter-country and inter-agency), despite WMO mandates (Peterson and Manton 2008). In part, this is because underfunded hydromet agencies sell data to support their operations. At the regional level, AGRHYMET, ACMAD and the NBA, continue to face several challenges, some of which are related to data acquisition in member countries, their timely transmission to the processing centres and the small number of observation points. Strengthening of the basin hydromet system is required.

#4—Inadequate Climate Adaptation Finance. Long-term financial shortages give rise to and magnify other barriers, such as those stemming from lack of information, inadequate interagency coordination and natural resources investments. Short funding horizons also make full integration of adaptation measures difficult. Investment to increase the robustness of infrastructure to more frequent extreme weather events in the Niger basin is also delayed due to a lack of financial resources. Institutional effectiveness falters when funding is cut back, as occurred with the Niger Basin Hycos program. Financing of adaptation is also difficult because the economic benefits of many adaptation investments are hard to quantify.

#5—Uncertainty. Effective climate change adaptation in the Niger basin is also complicated by a number of cascading uncertainties that limit the ability to predict the timing and magnitude of future climate change impacts. Uncertainty undermines the motivation of actors to become more aware, while lack of awareness inhibits efforts directed towards reducing uncertainty.

C-2-4: Overcoming climate change-induced extremes and barriers in the Niger Basin

The Niger basin countries face challenges in basin natural resources management due to overexploitation of limited and unevenly distributed water resources. These challenges will be exacerbated by rapid population growth and increasing urbanisation on one hand and by the impacts of climate change on the other. The changing climate will bring greater extremes – ranging from longer periods with droughts to more extreme rainfalls and thereby, more frequent and severe flooding. This means that adaptation measures in the water resources and agricultural domain must be robust and flexible to adjust to these uncertainties. Early warnings, climate outlooks for the rainy season, and extreme weather event forecasts are important elements for reducing exposure and sensitivity to extreme weather and climate events in drylands of the Niger basin. A holistic approach for adaptation in the basin will improve the ability to cope with present variability, as well as enhance the adaptive capacity to respond to future climate change. This will involve both disaster risk reduction and climate change adaptation. Such adaptation measures within a framework of Integrated Water Resources Management (IWRM) include improving disaster risk management, building adaptive capacity, incorporating water resources planning and ensuring that infrastructure is resilient to climate change.

C-2-5: Existing Tools for basin monitoring, hydrological forecasting and decision making

A number of tools have been developed by the Niger basin, however, they have limitations in translation of science to policy and decision making. A description of existing tools is given in sections C 2-4-1 and C 2-4-2.

C-2-5-1: Early warning systems and Regional Climate Outlook Forums in the Niger Basin

Despite the uncertain climate, early warning systems (EWS) in the Niger Basin are expected to comprise part and parcel of disaster risk reduction and management of responses to climate extremes in the Basin. Interface of the PIDACC with Regional Climate Outlook Forums (RCOFs) like the African Centre of Meteorological Application for Development (ACMAD) and coordination with National Climate Outlook Forums (NCOFs) is expected to play a key role in reducing the uncertainty in climate projections and mainstreaming outcomes into PIDACC. RCOFs like ACMAD work to improve the provision of weather and climate information in the fields of agriculture and food security, water resources and disaster risk management. The ACMAD-AGRHYMET-NBA consortium which is the designated WMO West Africa Regional Climate Centre Network plans to generate regional tailored climate products, providing online access, and coordinating training for users, among other tasks.

¹⁷ June 2015; Alec Crawford; Climate change and state fragility in the Sahel

C -2-5-2 System for monitoring water resources and flows forecasts using satellite imagery in the NIGER Basin.

With financing contribution from the African Development Bank –Climdev fund (Euro 1 million¹⁸), the NBA has effective 2017, set up a system for monitoring and disaster risk prevention related to floods and droughts based on satellite observations. The 3 year project supports the collection of hydro climatic data, capacity building and the establishment of a Water resource tracking system and satellite flow forecasts. Results include: (i) Users and decision makers trained and sensitized on disaster risks (ii) Flood and drought forecasts developed (iii) operational equipment for data collection, processing and archiving acquired and (iv) operational hydro climatic databases at national and regional level. This project has enabled the acquisition of hydro climatic data in real time with a grid of 3km X 3km and the diffusion of hydrological information products for enhanced hydrological forecasting.

Schemes of the basin forecasting tools are shown in figures 6 and 7. The tool in figure 6 operates in two modes: (i) a rain-flow simulation mode to generate daily flow to stations and (ii) flow forecast mode to calculate daily flow up to 6 days in advance.

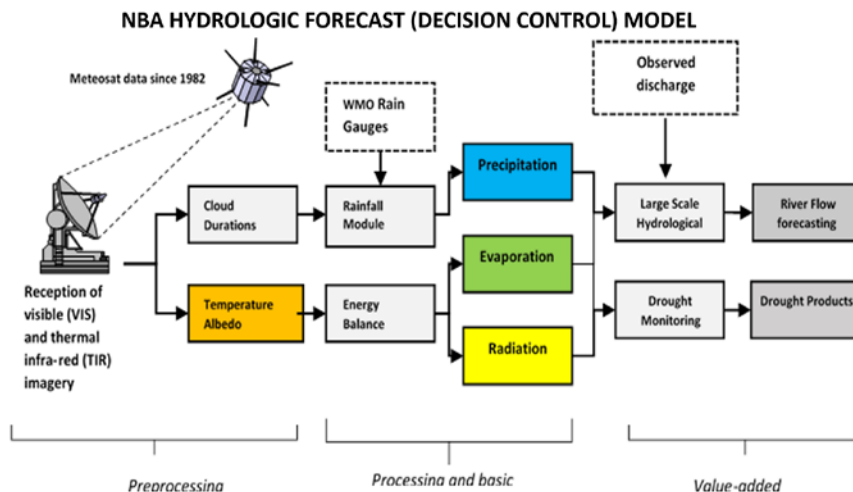


Figure 6: Scheme of the NBA hydrological forecasting model(source NBA 2018)

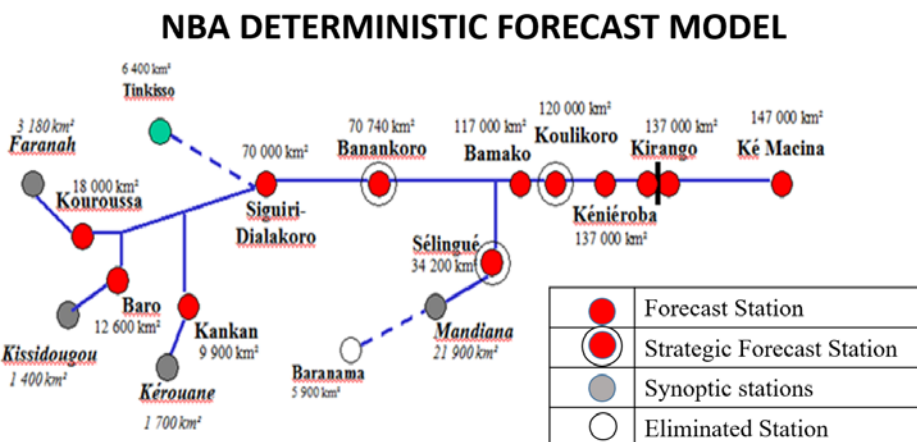


Figure 7: Scheme of the computerised System for stream flow forecasting (Source NBA, 2018)

The deterministic model (figure 7) is built around data from 38 Hydrometric stations in the basin. Forecast Horizons range from 3 days to 150 days for trend forecasts. Both models are used for the development of alerts (floods and severe floods). These forecasting tools are managed by the NBA Modeling Expert and the Hydrological Expert in charge of the hydrological database.

C -2-5-3 Existing Decision Support Tools for Allocation of Water resources in the NIGER River basin

In 2007, the NBA developed a model for the allocation of water resources in the Niger basin using the Mike Basin software. The model was used for the preparation of the Niger Basin Sustainable Development Action Plan (SDAP). A macro-economic model was also developed. The model was later updated to MIKE HYDRO (see figure 8).

¹⁸ The estimated project cost is €3 016 227, with the contribution from the bank estimated at €1,000,000

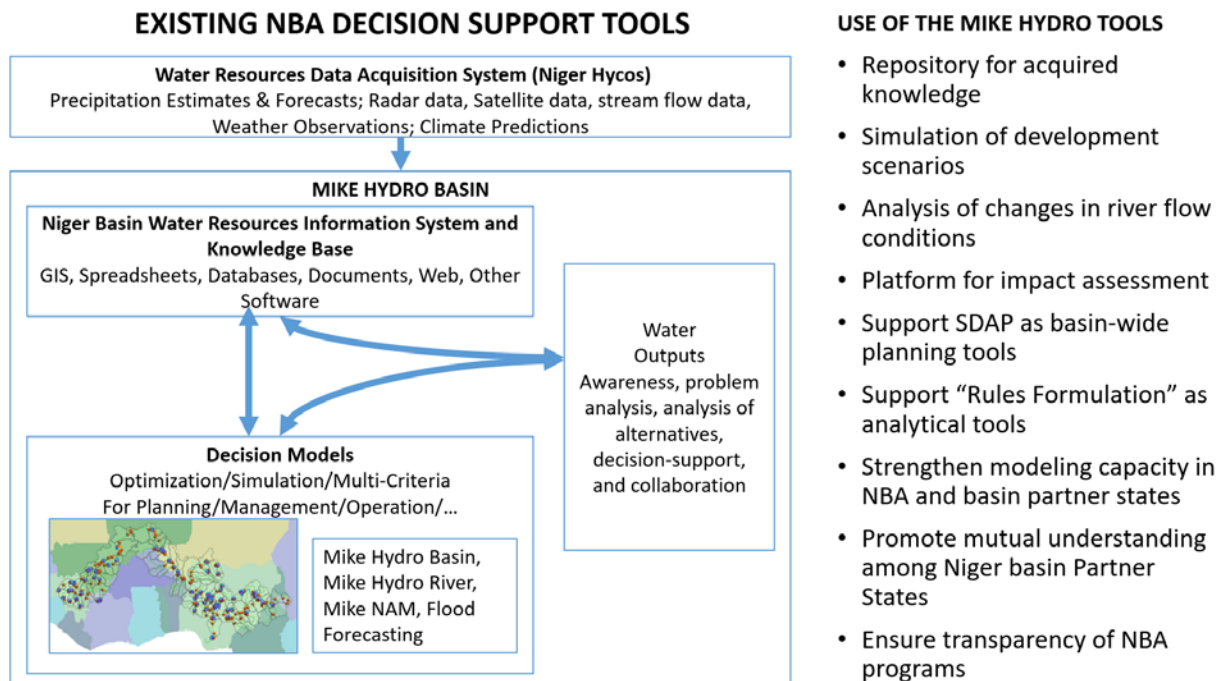


Figure 8 Niger basin decision support tools (Mike Hydro Basin), Source (African Development Bank, 2018)

The modules include (i) a Basin module (MIKE HYDRO Basin) and (ii) a rainfall runoff model MIKE NAM. MIKE HYDRO Basin used for WRM and planning in the basin. It is a semi-distributed model that includes: (i) 66 Hydrometric stations; (ii) 161 sub-basins; (iii) 481 hydrological nodes (hydrological stations, dams, sampling points); (iv) 304 sampling points and (v) 37 existing dams and 36 dams in project. The model was calibrated from two databases: (i) the Regional hydrological database of 105 stations and (ii) the sampling database. The model does not take into account the climate projections.

NBA experts have been using MIKE Hydro since 2012 to assess the impacts of projects on water resources, including the following projects: (i) IWRM studies like the development of the Niger Basin SDAP (ii) Provision of multi-sector solution alternatives to water allocation and shortage problems (iii) Reservoir operation optimization (Fomi, Taoussa and Kandadji) and other development projects (levies, works on tributaries) (iv) technical support for the implementation of the notification procedure for planned measures (annex No. 3 of the Water Charter); (v) Exploration of conjunctive use of groundwater and surface water and (v) Irrigation scheme performance improvements. The Mike Hydro Basin model is used independently by NBA experts. However, major updates are done by international consultancy as was the case in 2012. The same is true for forecasting models.

The NBA therefore has requisite capacity in running the Mike Hydro Models, however the capacity of the National Meteorological and Hydrological services is inadequate. Thus considerable capacity building is needed for all NBA countries through an intensive capacity development program in all partner States.

C -2-5-4 Strengthening Decision Support Tools for Climate Adaptation in the NIGER River basin

The existing tools and capacity are inadequate to assess climate adaptation at the basin scale. This is critical given that the water resources are shared between countries and adaptation measures, depending on the type of interventions considered may have important implications downstream. The PIDACC project will strengthen the application decision support tools through capacity development and integration of climate risk management, strengthening of hydrological forecasting based on rainfall-runoff models driven by projections of future precipitation and application of freeware such as the Soil and Water Assessment tool (SWAT) to assess impacts of landscape measures. The enhanced tools will enable decision-makers to evaluate and implement climate adaptation measures

Examples of capacity needs include: (i) major model updates; (ii) possible changes to the Mike Hydro Basin tool produced by DHI and (iii) to train NMHS on these tools to ensure their sustainable operation by the NBA at regional national levels. Trainings include: IWRM and Climate Change Adaptation; Use of satellite data for drought and flood management; Real-time operational flood forecasting and early warning; Real-time data and information management; Agro-Climatic Drought Monitoring & Early

Warning; Configuration of systems to support specific needs, including creating data interfaces to external data sources; Configuration of early warning system; Running flood mitigation scenarios during emergency situations.

C-2-4: Outcomes and impact of planned program

C 4-2-1: An IWRM Approach towards Climate Resilient Growth in the Niger basin

Adaptation to climate change-induced stress on water resources requires a holistic approach. IWRM approaches — combined with tools forecasting the changed availability and use of water — will therefore play an important role in managing water resources. The planned PIDACC project therefore follows an IWRM approach, to sustainable development, allocation and monitoring of water resource use in the context of social, economic and environmental objectives (see figure 9).

The IWRM approach to the PIDACC is the encompassing paradigm for adaptation to contemporary climate variability, and it is the prerequisite for coping with the still uncertain consequences of global warming, climate changes associated with it and their repercussions on the water cycle. IWRM in the Niger basin, combined with the physical models will offer a framework, which enables a better understanding of the impacts of climate change on water resources in the Niger basin as well as the interaction between the atmosphere and the hydrological cycle. This will help in planning for climate change adaptation in the basin.

The focus on the river basin as a spatial unit, addressing multiple goals and objectives through cross-sectoral and regionally driven planning and actions, stakeholder participation, vertical integration from the lowest level of user to the top policymakers, co-ordination and collaboration among all the institutions responsible for resources management at a watershed (catchment) scale, and incorporation of Climate Risk management will address the basin challenges and strengthen the project outcomes

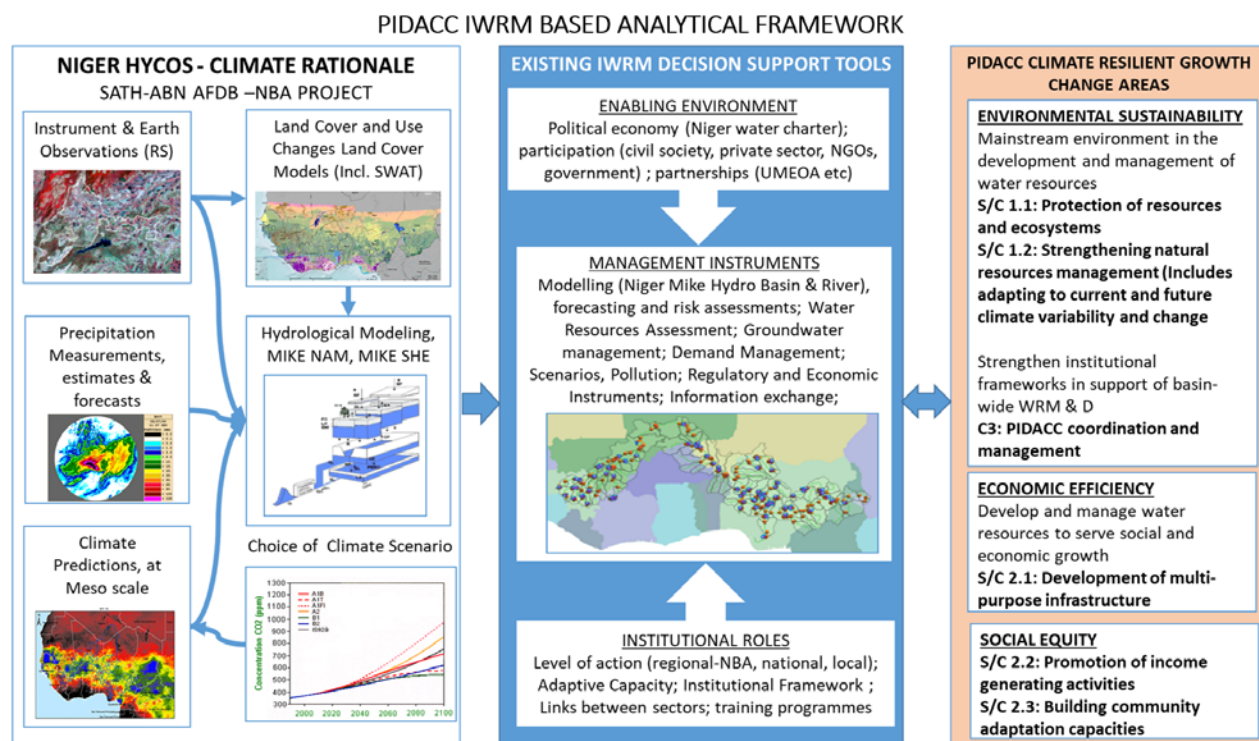


Figure 9 integration of PIDACC core elements using IWRM conceptualization approach – optimizing social, economic and environmental co-benefits (African Development Bank, 2018)

C 4-2-2: Outcomes and impact of the planned PIDACC program

The programme's objective is to contribute to improving the resilience of populations and ecosystems in the Niger Basin through sustainable management of natural resources. Specifically, the program aims to: i) reduce the silting process of the Niger River; ii) improve the adaptive capacity of the population and iii) improve natural resource management and integrated ecosystem management. The programme's objectives and impacts at all levels are presented in the consistency framework in Table 7.

Table 7: Consistency framework

Objectives	Components	Effects	Impacts at the basin level	African / International Impact
Reduce the silting process of the Niger River	S/C 1.1: Resource and Ecosystems Protection	Silting of the Niger River has decreased	<u>Inter-Regional</u> The Niger Basin's ecosystems are preserved and living environment of people is improved	Climate change resilience is improved
Improve natural resources management	S/C 1.2: Strengthening natural resources management	The sustainable natural resources management is improved		Rural exodus and illegal migration has reduced
Improve population adaptation capacity to climate change	S/C 2.1: Implementation of Climate Resilient Growth Infrastructure	Agro-pastoral and fishery productions have increased	<u>National</u> Income from agricultural and resilience of local population to climate change are improved	The fight against global warming is strengthened
	S/C 2.2: Promotion of income generating activities Building community adaptation capacities	Adaptation to CC capacities are built		
	C3: PIDACC coordination and management			

C-2-5: Alignment of PIDACC outcomes to GCF outcomes

Proposed GCF financed PIDACC outcomes include: (i) strengthened management, restoration and protection of natural habitats from climate variability and change; (ii) increased generation and use of climate information in decision-making; (iii) strengthened adaptive capacity and reduced exposure to climate risks; (iv) strengthened awareness of climate threats and risk reduction processes; and (iv) improved management of land or forest areas contributing to emissions reductions. The outputs are elaborated below. The implementation plan and log frame are elaborated in Section C-8-1 and H1 respectively.

Table 8: Alignments of PIDACC Outcomes with GCF outcomes, and implementation role.

Component/Sub component	Outputs	Nationally Implemented ¹⁹	Regionally Implemented ²⁰
C1: Development of ecosystems and natural resources resilience			
S/C 1.1, Resource and ecosystems protection	Output 1: Strengthened sustainable land management	√	
	Output 2: Strengthened sustainable forest management	√	
	Output 3: Strengthened sustainable agroforestry management	√	
	Output 4: Strengthened sustainable protection of natural habitats	√	
S/C 1.2. Strengthening the shared management of natural resources	Output 5 Strengthened climate and weather information systems	√	√
	Output 6: Regional pollution prevention plan developed		√
	Output 7: Payment mechanism for environmental services mechanism and adaptation fund set up	√	√
	Output 8: Strengthened capacity for IWRM implementation at national and regional (NBA) levels	√	√
C 2: Development of population resilience			
S/C 2.1, Implementation of Climate Resilient Growth Infrastructure	Output 10. Strengthened climate resilient hydro-agricultural infrastructure	√	
	Output 11. Water points for cattle and development of transhumance pathways	√	
S/C 2.2. Promotion of income generating activities Building community adaptation capacities	Output 14: Community-based Climate Adaptation Action Plans prepared and implemented	√	
	Output 15. Increase in local economic development arising from the upgraded Community based Livelihood infrastructure	√	
C3 PIDACC coordination and management	Output 16: Dissemination of good adaptation practices and climate information	√	√
	Output 17: Strengthened Monitoring and Evaluation (M&E) by the NBA Environment Observatory	√	√

Table 9: Financial Contributions by Co-financier and Outputs

Component/Sub component	Output	Total Millions (USD)	Amount GCF, Millions (USD)	Amount AE (Millions USD)	Amount EU (millions USD)	Amount GEF (millions USD)	Amount FIP (millions USD)	Amount Country (million USD)	Amount Beneficiary (millions USD)
	Project Cost	209.903	67.774	78.095	18.081	12.980	9.000	16.186	7.786
C 1: Development of ecosystems and natural resources resilience		86.219	34.302	1.270	17.486	8.771	8.181	9.352	6.856
S/C 1.1 Resource and Ecosystems Protection		70.591	25.598	0.168	16.087	4.516	8.013	9.352	6.856
	Output 1: Strengthened sustainable land management (SLM)	39.622	15.208	-	13.881	1.548	-	5.609	3.376
	Output 2: Strengthened Sustainable forest management	13.553	4.529	-	0.612	2.325	3.071	1.760	1.256
	Output 3: Strengthened sustainable agroforestry management	8.713	1.580	-	0.382	0.644	4.942	0.430	0.735
	Output 4: Strengthened sustainable protection of natural habitats	8.704	4.281	0.168	1.213	-	-	1.553	1.489
S/C 1.2: Strengthening shared management of natural resources		15.627	8.705	1.102	1.399	4.254	0.167	-	-
	Output 5 Strengthened climate and weather information services	8.881	6.306	-	0.752	1.822	-	-	-
	Output 6: Regional pollution prevention plan developed	1.184	0.452	-	0.112	0.620	-	-	-
	Output 7: Payment Mechanism for environmental services (PES) and RAFFC Fund set up	1.145	0.597	-	0.254	0.125	0.167	-	-
	Output 8: Strengthened Capacity for IWRM implementation at national/ regional (NBA) levels	3.648	1.349	0.611	-	1.687	-	-	-
	Output 9: Master Plan for the river navigability developed	0.771	-	0.490	0.281	-	-	-	-
C 2: Development of population resilience		102.603	23.415	68.161	0.595	3.416	-	6.086	0.930
S/C 2.1 Implementation of Climate Resilient Growth Infrastructure		82.892	8.614	67.864	0.595	1.515	-	4.276	0.028
	Output 10: Strengthened Climate Resilient Hydro-Agricultural Infrastructure	59.831	2.424	53.799	-	-	-	3.608	-
	Output 11: Water points for cattle and development of transhumance pathways developed	15.663	6.190	6.874	0.595	1.515	-	0.461	0.028
	Output 12: River Sections trained for improved navigation	1.521	-	1.521	-	-	-	-	-
	Output 13: Fisheries Infrastructure constructed for improved productivity	5.877	-	5.670	-	-	-	0.207	-
S/C 2.2: Promotion of income generating activities Building community adaptation capacities		19.712	14.801	0.297	-	1.902	-	1.810	0.902
	Output 14: Community-based Climate Adaptation Plans prepared and implemented	5.969	4.129	-	-	0.863	-	0.543	0.434
	Output 15: Increase in local economic development arising from the upgraded Community based Livelihood infrastructure	13.743	10.672	0.297	-	1.038	-	1.267	0.468
C 3 PIDACC Coordination, monitoring-evaluation and communication		21.081	10.056	8.664	-	0.793	0.819	0.748	-
	Output 16: Dissemination of good adaptation practices and climate information	1.742	1.154	0.423	-	0.165	-	-	-
	Output 17: Strengthened Monitoring and Evaluation (M&E) by the NBA Environment Observatory	7.569	6.044	0.976	-	0.082	0.325	0.143	-
	Project Management and coordination	11.771	2.859	7.266	-	0.546	0.495	0.606	-

Annex III - PIDACC Budget -FD+FIP+AfDB v10-09-18 provides details of the GCF Funded activities at national and regional level.

¹⁹ Ministries responsible for water Resources and Environment

²⁰ Niger Basin Authority

C.3. Project / Programme Description (PIDACC-GCF Project)

Describe the main activities and the planned measures of the project/programme according to each of its components. The PIDACC framework is summarized in Figure 10. The PIDACC programme focuses on three components: (i) Development of resources and ecosystems resilience; (ii) Development of population resilience, and (iii) Programme coordination and management. The component description for the PIDACC- project follows thereafter.



Figure 10: Intervention Framework of the PIDACC²¹

Notes to Component, Output and Activity Description :

Section C3, describes the project activities by output across the Niger Shared River basin. Table 10: Shows the Regions where PIDACC Measures will be implemented by country as well as the executing agencies in each of the countries.

Table 10: Showing the Regions of where PIDACC Measures will be implemented by country as well as the Country EES

Country	Region	Executing Entity
Benin	Departments of Alibori, Atacora and Borgou)	Ministry of Mines, Energy and Water
Burkina Faso	East, Center-East, Central Plateau, North-Central, Sahel and Upper-Basins	Ministry of Water and Sanitation(MEA)
Cameroon	North and Far North	Ministry of Economy, Planning and Regional Development
Côte d'Ivoire	Bagoué Kabadougou, Folon	Ministry of Environment, Water and Forestry
Guinea	Faranah, Kankan and Nzérékoré	Ministry of Environment, Water and Forestry
Mali	Koulikoro, Sikasso, Ségou, Mopti, Timbuktu and Gao	Ministry of Water and Energy (MEH)
Niger	Dosso, Niamey, Tahoua and Tillabery	Ministry of Water Resources, Environment & Desertification Control
Nigeria	Sub-basins of Upper Niger basin, Upper Benue River basin, Anambra-Imo River basin)	Federal Ministry of Water Resources
Chad	Regions of East Mayo Kébbi and West Mayo Kébba	Ministry of Environment and Water Resources
Niger Basin Authority	Basin wide	Niger Basin Authority (PIDACC Coordination Unit)

Appendix VI Climate Rationale for the GCF PIDACC Project, a detailed description of the climate stressors, impacts of the climate change stressors and proposed interventions, by output, activity and country. Not all outputs are implemented in each country. The Executing Entity (EE) by country and at regional level is included as part of the Intervention Matrix. Broader institutional arrangements at regional and country level are included in Annex II-Institutional-implementation-arrangements

²¹ Modified from preliminary studies of multi-purpose infrastructure of the Programme for the Integrated Development and Adaptation to Climate Change in the Niger Basin (PIDACC/NB); Regional Synthesis Report

Component 1. Development of the Resilience of Ecosystems and Natural Resources

Component 1 is aimed at enhancing the resilience of agro-ecosystems by increasing ecosystem services through the use of agro-ecology principles and landscape approaches. The location of activities for development of ecosystems and natural resources resilience under component 1 are shown in Figure 11. Under this component, the intervention strategy will focus on IWRM as a set of tools and instruments that can assist communities to adapt to changing climatic conditions that limit water availability or may lead to excessive floods or droughts in the Niger Basin.

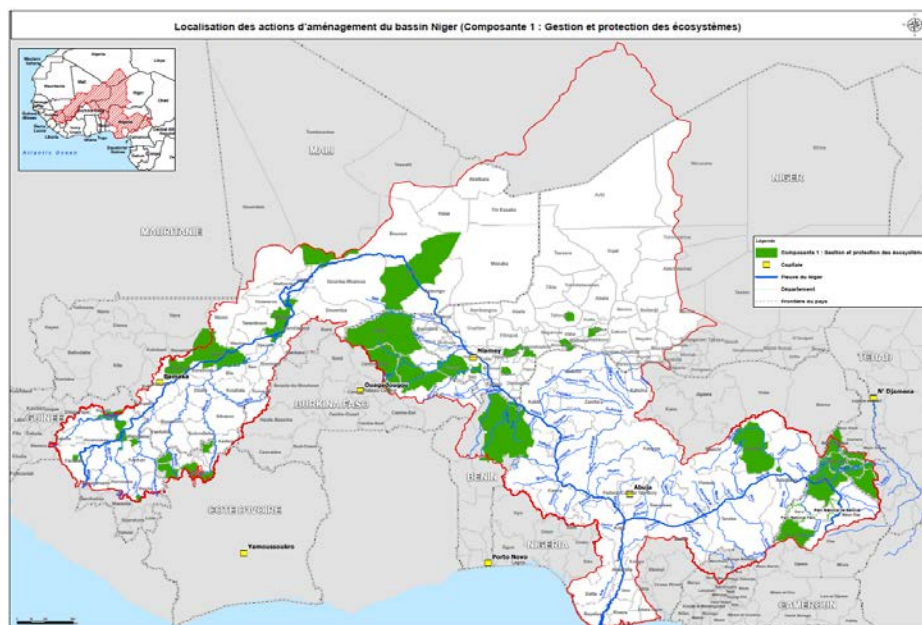


Figure 11: Location of the activities of Component 1 of PIDACC^{Ibid}

Outputs related to this component are elaborated below.

Sub-component 1.1. Resource and Ecosystems Protection²²

Output 1: Strengthened sustainable land management

This output includes reducing risks related to water shortages and land degradation. It addresses land degradation and climate change risks in the Niger basin at the regional level by linking national-level efforts for restoration across the basin countries. The scope covers erosion and silting control (stabilization of 10000 ha of dunes; restoration of 110,000 ha of degraded land and mechanical and biological management of 45,000 ha of ravines) and includes strengthened agro-sylvo-pastoral management in selected watersheds. Measures (technologies) to address land degradation under this output include: Mechanical and biological Soil Defence Restoration works on communal land; Mechanical and biological treatment of ravines; Water and soil conservation works in cultivated plots and shoreline protection. These practices will contribute to climate change adaptation and mitigation, and food security in the Basin.

Activities

Activities will include: (i) Consultation with partner states (agronomists, land officers on zoning of land for agriculture etc); (ii) Development and implementing of 9 sub basin masterplans/watershed plans (iii) Stabilizing dunes; restoring degraded land, and mechanical/biological management of ravines (iv) awareness campaigns on erosion control and improved soil management practices; (v) expansion of sustainable land and water management interventions and technologies —includes gender-responsive CSA technologies (vi) supporting the fight against bushfires; (vii) training men and women farmers in gender-responsive CSA technologies on-farm or at functional farmer training centres (with demonstration plots, trained extension officers in gender and CSA, appropriate equipment and inputs, business development and marketing and selected value chains in agriculture and watershed management and (viii) Building of a one lane 60 m span bridge in Cameroun. For the sub basin masterplans, vulnerability assessments will be undertaken and will allow adaptation interventions to be targeted to areas that are the most at-risk of catastrophic climate change impacts in each of the sub basins. Climate data (trends) and socioeconomic data will be used to assess human and ecological vulnerability to climate change, which are both relevant at the watershed level

²² (i) Preliminary studies of multi-purpose infrastructure of the Programme For The Integrated Development And Adaptation To Climate Change In The Niger Basin (PIDACC/NB) and (ii) Investment Plan For The Strengthening of Resilience to Climate Change in the Niger Basin; Final Version; November 2015

Output 2: Strengthened sustainable forest management

This output involves support to regeneration of forests and increasing density of trees on farms through farmer-managed natural regeneration. It focuses on increment of forest productivity and SFM (40,000 ha of forests). Climate change is adding to the existing stresses on forests, contributing to decreased productivity of trees from drought and temperature stress, increased wind and water erosion, increased storm damage, increased frequency of forest fires, pest and disease outbreaks, changes in ranges of forest plants and animals and inundation and flood damage²³ Key strategies for increasing resilience of forests to climate change include: maintaining healthy forest ecosystems for resilience; restoring degraded forests and conserving, enhancing and using biodiversity. Improving the sustainability of forest management will not only increase the forest's resilience, but contribute to improving water management, reducing soil erosion, conserving biodiversity and improving the resilience of farming systems. GCF interventions are planned in: Cameroon; Côte d'Ivoire; Guinea; Mali; Niger; Nigeria and Chad.

Activities

Activities will include: (i) Support to development of private nurseries and service providers; (ii) Support provision of nationally available sources of quality seed and plant material (iii) Demonstration/trial forestry plots and re-establishment of forest tree species (includes increasing the availability of wood for other uses for communities and improvement of wildlife habitats) and (iv) Building capacity in SFM approaches, includes training of 200 community agents and associations for REDD+

Output 3: Strengthened sustainable agroforestry management

The output includes integration of agro-forestry into farming systems on 26,000ha of selected watersheds. Increasing frequency, intensity and duration of severe weather events are posing major challenges to food security and livelihoods of rural people in the Niger basin. Without action, the basin's agriculture sector will continue to be extremely vulnerable to climatic shocks, especially droughts. Agroforestry is important both for climate change mitigation (carbon sequestration, improved feed) and for adaptation in that it improves the resilience of agricultural production to climate variability by using trees to intensify and diversify production in the farming systems, amelioration of soil fertility, control of erosion, diversify income for communities and diminishing the impacts of extreme weather and buffering farming systems against hazards. The output will include support to increase the use of agro-forestry systems to reduce soil erosion and increase organic matter, soil fertility and soil water holding capacity, thereby enhancing climate resilience and productivity. Agroforestry practices will include: (i) contour hedgerows; (ii) boundary planting; (iii) trees scattered in crops; (iv) home gardens; and (v) fodder banks. Agro forestry would also offer diversified sources of income, through selling wood for various uses, which will provide a buffer against economic shocks. GCF funded interventions are planned in: Burkina Faso, Niger and Nigeria.

Activities

Activities will include: (i) Support to development of private nurseries and service providers; (ii) Establishing fodder banks with Napier grasses (iii) Planting trees/shrubs in home gardens (fruit trees); on existing progressive terraces; boundary planting and planting trees within crops (non-fruit trees); (iv); Training for men and women farmers on gender-responsive CSA technologies on-farm or at functional farmer training centres (agro-meteorological data use supporting 10,000 women in their agricultural activities, sensitizing 500,000 women on climate change adaptation techniques and other resources); and (v) Land certification/titling policy enacted and approved and implemented— provision of access to restored land for 9,000 women.

Output 4: Strengthened sustainable protection of natural habitats

The output aims at reducing climate change-related risks and the vulnerability of natural ecosystems and biodiversity. This output includes participatory management and development of 10,000 ha of natural habitats and rehabilitation of 5,500 ha of spawning grounds. Mechanisms will be developed to promote sustainable land-use patterns that maintain the functional integrity of wetland ecosystems as spawning grounds. Linking wetland restoration with the sustainable use of natural resources in the region will help increase the well-being of the local population by enabling them to increase their economic opportunities for fishery, agriculture, and eco-tourism as well as by allowing downstream communities to enjoy cleaner water supplies. PIDACC interventions will reduce vulnerability and increase resilience. Interventions under this output include: Control of Invasive species in Cote D'Ivoire; Restoration of habitats in wetlands in the Inner Delta in Mali; Development of spawning grounds and flood plains in Guinea; Protection of biodiversity in Burkina Faso and Nigeria and Improved productivity of aquatic environments in Benin, Burkina Faso and Niger (refer to annex VI, for detailed measures by country).

Activities

Activities include: (i) Identifying areas for development of the nature reserves that target local communities; (ii) Establishing corridors for wildlife to access water and biomass during droughts; (iii) Improving management within protected areas: to reduce degradation of habitats (4,200 ha) and (iv) Rehabilitation of bare land and spawning grounds in conservation areas (4,700 ha).

²³ Building resilience for adaptation to climate change through SFM Susan Braatz Forestry Department, FAO, Rome

Sub-component 1.2. Strengthening the shared management of natural resources

Output 5: Strengthened climate and weather information services including last mile EWS

This output involves the strengthening of the decision support tools through major model updates; integration of climate risk management, strengthening of hydrological forecasting based on rainfall-runoff models driven by projections of future precipitation and application of freeware such as the Soil and Water Assessment tool (SWAT) and capacity development of NMHS agencies. It will enhance ongoing programs by the NBA Environment observatory, ACMAD and AGRHYMET.

Last Mile EWS

The program will reduce vulnerability by focusing on the transformation of “last-mile” community EWS as illustrated in Figure 12. Climate and weather services information will be mainstreamed across decision timescales as shown in Figure 13.

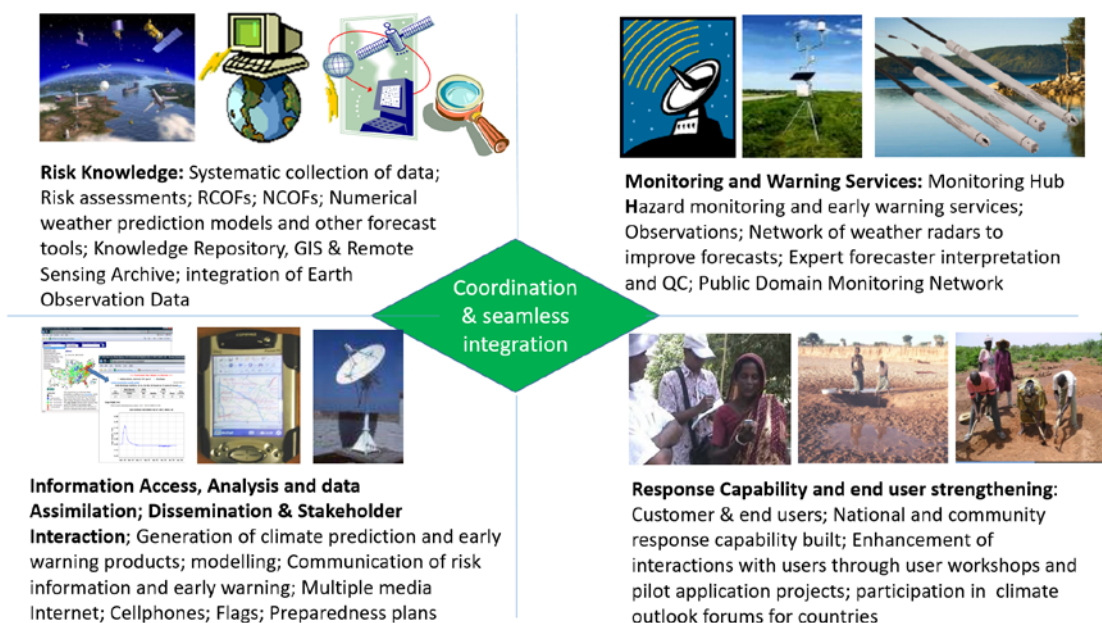


Figure 12: Integration of climate information and early warning into the various components including end to end multi hazard EWS



Figure 13: Range of services for planned climate and weather information services (modified from WMO). The idea is to provide “seamless” forecasts, whereby a range of products are offered through the climate service, meeting the users’ needs.²⁴

²⁴ Informed by World Meteorological Organization and World Health Organization, 2016; Use of Climate Predictions to Manage Risks

The NBA through the observatory will bear primary responsibility for the regional aspects of the Climate and Weather Information system, including developing and implementing tools that permit national data to be transmitted to a regional data management system. The NBA will also be responsible for the regional data management system that will permit storage, retrieval, and analysis of the data and encourage use of the data through appropriate web interfaces to the database. Embedded within the national and regional systems will be a national-regional data communications component as well as information products.

The role of the National Climate and weather systems will be to: (i) Manage the hydro-met network data and meet national hydromet services needs in terms of data analysis, reporting, execution of models, data ingest and information dissemination for the sustainable management and development of national water resources, and multi hazard warning systems; and (ii) Permit and facilitate the exchange of approved and quality assured data with the regional system. It will also be responsible for the last mile community multi hazard warning systems.

Strengthened end to end forecasting

PIDACC will contribute to the development of management strategies based on downscaled climate products, existing hydrologic observations and data taking into account the spatial analysis of exposure and vulnerability to floods and droughts. The forecast process will build onto the system for monitoring water resources and forecasts using satellite imagery in the basin (SATH-ABN) and the NBA Mike Hydro Basin Model (see figure 14). Benefits will include (i) Near-real time monitoring of basin water resources (ii) Historical and regional perspective on local hydrological variability (iii) Rapid assessment of the severity and extent of droughts and floods (iv) strengthened tools for agricultural planning, including irrigation and (v) strengthened potential to assess hydrological impacts of climate change

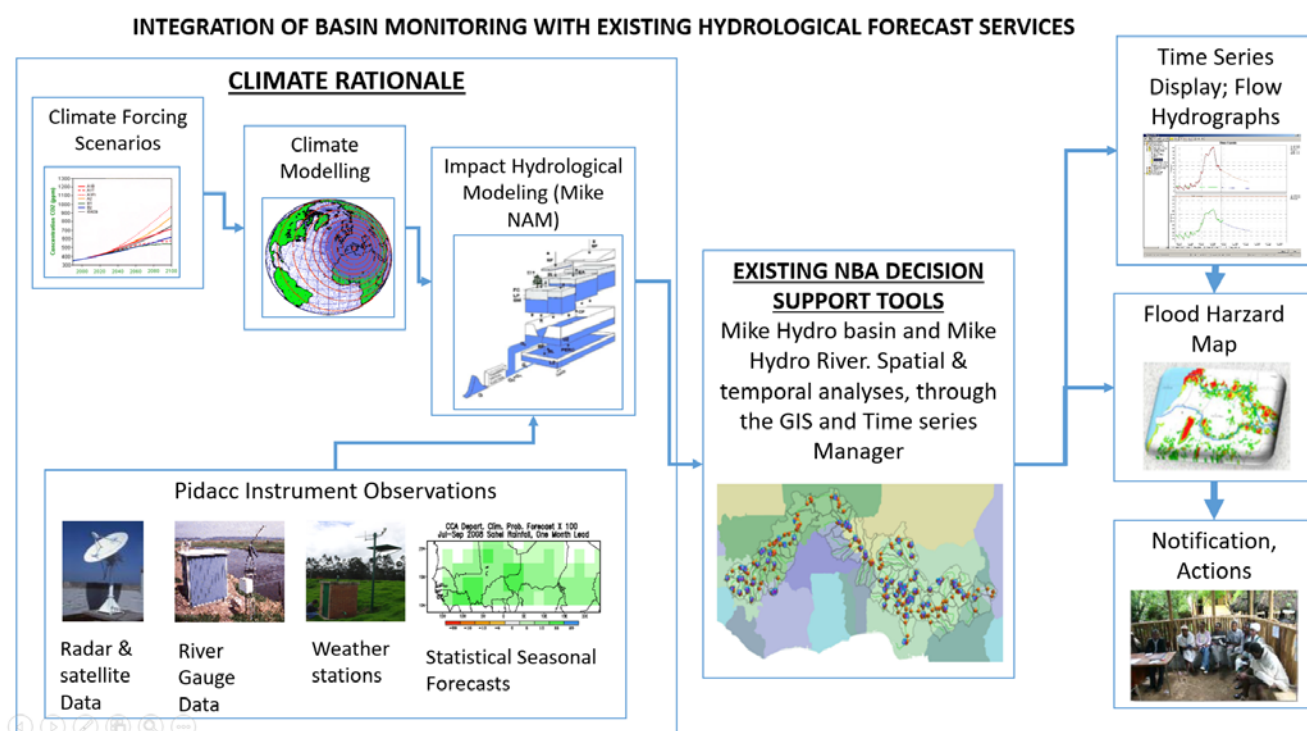


Figure 14 shows Use of the basin monitoring system, forecasting products and existing Mike basin decision support tools (African Development Bank, 2018)

Activities:

- (i) **Strengthening the decision support tools (Mike Hydro basin, Mike Hydro flood etc) and capacity development.** This activity will involve: Strengthening the tools for flood and drought forecasting building onto the existing Mike Hydro tools at the NBA Secretariat. For floods, the strengthened forecasting systems will estimate expected water level rise using data inputs from simulation tools and models that predict precipitation levels and stream flow, as well as from hydrometric stations measuring water levels at selected points along the Niger River. For droughts, strengthening drought forecasting systems will make use of models fed by climatic and atmospheric data (historical/seasonal weather patterns, real-time meteorological monitoring, and weather forecasts) to predict the probability of a drought occurring in particular parts of the Niger basin in the future (up to approximately three months). This activity covers the countries of: Cameroon, Guinea, Mali, Niger and Chad. The rest of the countries are supported through the ongoing Bank financed hydromet program. For sustainability, the

strengthened tools and capacity will be based on a real 'need' for the services to be provided (basin planning and flood and drought monitoring and warning). Flexibility to adjust to emerging requirements support like the use of new or emerging technologies for data and information capture and accommodating different levels of data availability. The tools will also be anchored in an appropriate water agency (NMHS) with qualified staff.

- (ii) **Long-term and on-demand capacity development and training program for NMHS staff on O&M, data modeling, and forecasting.** GCF financing will be used to procure and install desk tops and modelling and data base software for equipping the NMHS in Côte d'Ivoire; Guinea and Nigeria. Other countries are covered with financing from the ongoing Afdb financed project on strengthening hydrological monitoring in the Niger basin. Capacity will be built using a combination of both regional and international experts. Trainings will involve short-term experts and the NBA-AGRHYMET staff in order to further strengthen the modeling teams at regional and national levels. The activity includes TA for support to certification of meteorological agencies in line with WMO requirements.
- (iii) **Expansion of coverage of Hydrological and meteorological monitoring networks.** This activity includes modernizing the NMHSs' observation networks, improving the meteorological and hydrological forecasting systems, and refurbishing offices and facilities. GCF financing will be used to finance activities in Benin (Procurement and Installation of 4 Hydromet stations); Burkina Faso (Procurement and Installation of 13 Hydromet stations); Guinea (Procurement and Installation of 14 Hydromet stations); Niger (Procurement and Installation of 4 Hydromet stations) and the Niger Basin Authority (TA for Design and supervision of Hydromet stations). Other countries will be covered with financing from the European Union and the ongoing Afdb financed project on strengthening hydrological monitoring in the Niger basin
- (iv) **Community based Early Warning Systems Established.** This activity involves procurement and installation of equipment for community based early warning systems. It will cover: Benin; Cameroon; Côte d'Ivoire; Guinea, Mali, Niger, Chad and the NBA. The community members will use simple equipment to forecast potential natural disasters such as floods and drought, and operate a communication/dissemination system to inform other local residents of impending threats. Effective EWS will increase the lead-time before a potential disaster and allow for community and disaster response teams to prepare for the disaster and mitigate its impacts. GCF financing will be used to Procure and install equipment for community based early warning systems in Benin, Cameroun, Guinea, Mali, Niger, Chad and the NBA.
- (v) **Development of Knowledge products on climate and weather information services.** GCF funds will support the development of new information and value-added products for vulnerable communities and the main meteorology- and hydrology-dependent sectors in Burkina Faso. This will involve development of new information products (dedicated bulletins, model outputs, distribution and dissemination of information, agro-meteorological data use support for 10,000 women in farming activities and warnings for vulnerable communities and the meteorology and hydrology-dependent sectors. Beneficiaries will be targeted following a gender-disaggregated approach. The project will ensure that women are adequately involved in the implementation of the EWS and that they benefit from climate information which is relevant to them, presented and transmitted in a way that is accessible and easy to understand
- (vi) **Undertake capacity-building, in the generation, & dissemination of climate products;** includes: (i) training of 30,000 producers in the timely dissemination of early warning products (including agro-climatic information); and (ii) raising awareness among 1,000,000 producers on climate adaptation techniques. Gender impact analyses will be conducted to identify gender-specific needs and protection measures related to floods, droughts, and other climate change related disasters in the basin. A toolkit for the dissemination of early warnings will be developed and it will include a gender-based analysis of the national and local media that are used to disseminate weather and climate alerts. Based on gender-sensitive consultations with end-users, the project will supply climate information that is tailored to the specific needs of decision makers and local communities.

Output 6 Regional pollution prevention plan developed ²⁵

This output will comprise updating the existing pollution study. Precipitation and associated river discharge are major drivers of water pollution in the form of excess nutrients and sediment, transport into the Niger river. Increasing temperatures, intense precipitation and runoff, and intensifying droughts will lower river water quality in many ways, including increases in sediment, nitrogen, and other pollutant loads. The regional pollution control plan, will therefore enhance making water supply decisions in the context of changes in precipitation, increased temperatures, and changes in water quality, quantity, and water use.

Activities:

Activities will include: Undertaking a Diagnostic study leading to preparation of plan for strengthening the conventional network for water quality monitoring in the Niger River. This includes: updating of sources of diffused pollution at country level and characterization of water pollutants; and preparation of a regional plan for strengthening basin water quality management and strengthening a water quality monitoring network; The activity will make use of: (i) historic climatological data for estimation of evapotranspiration; (ii) regular updating of climatological data; and (iii) seasonal and long term weather forecasts

²⁵ Preliminary studies of multi-purpose infrastructure of the Programme for the Integrated Development and Adaptation to Climate Change in the Niger Basin (PIDACC/NB); Regional Synthesis Report

Output 7: Payment mechanism for environmental services (PES) and FRACC Fund set up

This output involves the establishing of payments for biodiversity conservation, wildlife corridors and watershed protection in the Niger Basin. NBA member countries have adopted the Water Charter which establishes in principle the payment of fees for water use. In line with the need to restore the integrity of the degraded lands in the Niger basin, the PIDACC will create an enabling environment by establishing a financing mechanism (environment funds) for promoting PES. The PES will establish a system for the funding of activities that contribute to the protection of the natural resources in the watersheds for the benefit of the users of these resources. PIDACC plans to set up a Regional CC adaptation fund (FRACC), which will be fed by the royalties that will be paid by the heavy users of the water. The FRACC, will be used to finance PES and to foster adaptation, with proceeds from the PES and other donors. Only measures that enhance climate resilience and climate change adaptation actions will be eligible for FRACC financing. This modality for establishing the FRACC and PES will be approved by the Council of Ministers and the Heads of state summit of the NBA countries by year 3 of PIDACC implementation. FRACC and PES will be operationalized at the in year 4 of PIDACC. The funding for the operationalization of the PES mechanism and the FRACC covers the 9 basin countries. It is shared between i) European union which finances this activity in 5 countries (Cameroon, Côte d'Ivoire, Mali, Nigeria and Chad) and ii) GCF which finances the activity in 4 countries (Benin, Burkina Faso, Guinea and Niger) through the NBA.

Activities:

Activity: A97:Activities will include: (i) a diagnosis study (legislation, technical and institutional) and FRACC and PES feasibility studies; (ii) Establishing rewards or payments for biodiversity conservation, wildlife corridors etc includes—establishing payments for biodiversity conservation, wildlife corridors and, in steep areas, watershed protection; and (iii) elaboration, validation and endorsement of the PES system and its diffusion and extension. Development of the PES will consider participatory vulnerability assessment to identify adaptive capacity, adaptation strategies, dependence on ecosystem services of PES actors and provision of incentives for adaptation strategies through PES. The established FRACC, will be used to finance PES and to foster adaptation, with proceeds from the PES and other donors. The PIDACC project will contribute towards steps (i) and (ii).

Output 8: Strengthened capacity for IWRM implementation at national and regional (NBA) levels

This output involves institutional capacity building activities under the PIDACC to strengthen the NBA and national level coordinating structures for implementing IWRM. The activities will provide participants with tools for using IWRM as a tool for climate change adaptation. Capacity-building activities will build knowledge and enhance communication and collaboration within and across sectors in various parts of the Niger basin.

Activities:

Activities will include: (i) Preparation of Regional IWRM action Plan (ii) Operationalization of IWRM plans in the 9 countries and improved regional coordination in the implementation of the OP (2016-2025); (iii) Preparation of a capacity development program and (iv) provision of capacity building to NBA executives, partner institutions to plan and implement climate resilient land and water management regimes and scale up effective adaptation strategies at the national and local levels. Content will include but not limited to the following topics: Introduction to IWRM and climate change; Drivers and impacts of climate change; Strategy development and planning for adaptation; Impacts of climate change on water use sectors; Dealing with uncertainties; Instruments and measures for adaptation; and Adaptation to climate change in water management.

Output 9: Master Plan for the river navigability developed

The aim of this activity is to improve the navigation conditions on cross-border sections and to strengthen trade and socio-economic links on the Benoué river (Nigeria and Cameroun) including the port of Garoua). The impacts from extreme weather events, higher temperatures and heat waves, precipitation changes, and low flows and other climatic conditions are affecting the reliability and capacity of navigation in the Middle and Lower Niger. Drought often lowers vessel drafts on navigable sections of the river in rivers Cameroun, Mali, Niger and Chad. Continued low water flows limit the duration of the navigation period in Cameroun. The increasing low flows due to drought and the high degree of climate variability across the basin necessitate a comprehensive system analysis²⁶. A feasibility study will be undertaken focusing on definition of measures for improving navigation in the navigable river sections in Cameroun, Mali, Niger and Chad, with financing from the EU & AfDB.

²⁶ Niger River Basin Sustainable Development Action Plan; Niger River Basin Climate Risk Assessment; Final Report; Volume 5; Climate and Runoff Projections and Climate Risk Assessment; May 2014
A Joint Initiative of the Niger Basin Authority (NBA) and the World Bank

Activities:

Activities will include a diagnostic analysis including: Sediment yield analysis; Hydrologic and hydraulic analysis of the Niger and Benoué course, Definition of a regional soil conservation policy and a navigation masterplan.

Component 2. Development of population resilience

This component will be implemented at national level and will contribute to the development of socio-economic infrastructure and the protection of the Basin's resources and ecosystems in each country. It comprises: (i) S/C 2.1: Implementation of Climate Resilient Growth Infrastructure and (ii) S/C 2.2: Promotion of income generating activities Building community adaptation capacities. The intervention (activity) locations are shown in Figure 15. The GCF project will contribute towards sub component 2-2. Investments will be drawn from the sub-basin development plans and community climate adaptation plans. Interventions (Activities) will be financed through a call for proposals for grassroots community organizations, for agro-pastoral and fish production activities or income generating activities for vulnerable groups.

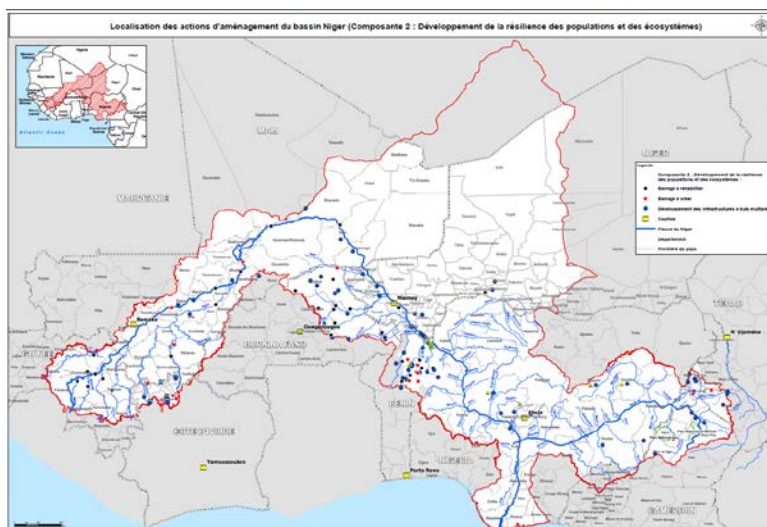


Figure 15: Location of the activities of Component 2 of the PIDACC

Sub-component 2.1, Implementation of Climate Resilient Growth Infrastructure

Output 10. Strengthened climate resilient hydro-agricultural infrastructure

This output includes rehabilitation / construction of small hydraulic infrastructure comprising 134 small multi-purpose dams, the development of 19,000 ha of irrigated land). In the Niger basin, where communities are faced with highly seasonal and often unpredictable rainfall, a lack of adequate water storage causes large and avoidable economic losses from floods and droughts, and constrains long-term growth. Recent studies suggest that increased temperatures, irregular rainfall patterns and drier conditions will amplify future drought severity and temperature extremes, increasing resource scarcity and exacerbating conflict levels and increasing migration and conflict. Implementation of hydro agricultural infrastructure will enhance the resilience to droughts and temperature increase. This output therefore involves external water harvesting systems to mitigate dry spells, recharge groundwater, and permit multiple uses of water. The storage of excess rainfall and the efficient use of small holder irrigation are critical in view of growing water scarcity and will strengthen resilience against climate variability. Measures will be undertaken in all the Niger basin Countries, financed by the Afdb. Measures financed using GCF resources are mainly in Chad and involve Construction of Community based Hydro-Agricultural Infrastructure for irrigation; departments of Kabbia, Lac Léré, Mayo. The small storage infrastructure will enable both flood and flood recession crops and as well as provide water for aquaculture, drinking water and livestock watering.

Activities

Activities will include: (i) a Preliminary study, mapping of locations of the small hydraulic infrastructure across the 9 countries and detailed designs and ESS studies (ii) Rehabilitating water storage structures to preserve supplies and improve flood protection and (iii) Construction of rainwater harvesting infrastructure across the transhumance corridors.

Output 11: Water points for cattle and development of transhumance pathways

This output includes developing of water infrastructure that is aligned with sustainable forage management in the Agro-pastoral

systems of the Niger basin. It involves construction / development of 134 reservoirs, ponds, wells and boreholes for farming and 134,000ha of transhumance corridors. Warmer temperatures, increased frequency of heatwaves and changes in precipitation resulting in droughts have forced pastoralists to alter traditional migratory corridors in search of new seasonal watering holes and rangelands. These trends are expected to continue, increasing the likelihood of conflicts between the Sahel's farming and pastoralist communities as land degradation and competition over water and land resources increases. Development of water points for cattle along transhumance routes is thus particularly important. Improved water storage will increase resilience to climate change and support water and food security in poor and vulnerable pastoral communities of the basin. The results include, a decrease in severe conflicts over access to grazing, and a decrease in the average distance between functional water points accessible to pastoralists and agro-pastoralists in targeted zones. The project will promote climate resilient pastoral and agriculture activities for women farmers. Measures will be undertaken in all the Niger basin Countries except Nigeria. Measures financed using GCF resources include (i) Benin: Development of water points for cattle and development of transhumance pathways (Alibori, Atacora, and Borgou) in Benin (ii) Development of 16 boudis with pastoral vocation in the provinces of Kouritenga, Namentenga, Soum, Oudalan and Séno (Sahel and Upper-Basins) in Burkina Faso (iii) Development of water points for cattle and development of transhumance pathways (Sikasso, Timbuktu and Gao regions) in Mali and (iv) Rehabilitation of Langui Be dike Pitoa (Far North regions Cameroun) for protection against increased risk of flooding in Cameroun:

Activities

Activities will include: (i) Undertaking a preliminary study/mapping of location for water points across the 9 countries (ii) establishment of corridors for livestock movement to protect farmers' crops and trees, designating grazing and water access areas to ensure resilience of pastoral production systems, (ii) Construction of rainwater harvesting infrastructure across the transhumance corridors and (ii) Formation and support of committees to manage small scale infrastructure.

Output 12: River Sections trained for improved navigation

This output involves river training (through cutting of aquatic plants on 120 km of navigable portions and construction of docks and wharves).and dredging to maintain adequate channel size to accommodate vessels and will be undertaken in close collaboration with agencies responsible for waterways or port management. Climate change is being observed in many of the Niger basin's navigable sections. Increased water temperatures, altered frequencies and intensities of precipitation, result in changes in runoff, sediment distribution and aquatic plant distribution which all affect navigable sections of the Niger river. The increasing episodes of low flows, affect navigation at the landing sites.

Activities

Activities will include building / rehabilitation of navigation sections and landing stages (on Niger, Tinkissu, Sankarani and Milo; (In the prefectures of Mandiana, Siguiri, Kouroussa, Faranah, Dabola and Kankan. The activity will make use of climate information to assist use of current and forecast information on river conditions in relation to transport needs.

Output 13. Fisheries Infrastructure constructed for improved productivity

This output involve restocking of fisheries and protection of spawning grounds in selected wetlands. Freshwater fisheries throughout the Niger basin are based largely on species that migrate to wetland systems within the basin for breeding and spawning. The combination of changes in rainfall and runoff resulting in falling water levels during droughts, is likely to affect habitats and fisheries in the basin, impacting the movement of fish and affecting nursery ground function. Changes may occur in timing and success of migrations, spawning and peak abundance, as well as in sex ratios. In addition there may be potential loss of species or a shift in composition in capture fisheries and impacts on seed availability for aquaculture, thereby threatening livelihoods. Activities for potential adaptation to be financed by the Afdb involve construction of nursery stations and development of spawning & floodplain areas in Benin, Cameroun, Cote D'Ivoire, Mali, •Guinea, Niger and Chad

Activities

Activities include: Construction / rehabilitation of 9 nursery stations and development of 16,000 ha of spawning areas).

Sub-component 2.2. Promotion of income generating activities Building community adaptation capacities

Output 14: Community-based climate adaptation action plans prepared

This output will support the preparation of 65 community catchment based climate change adaptation plans for selected sub catchments. As part of strengthening community resilience, this output will identify and address the impacts of climate change at the local level through the preparation of 65 community-based climate change adaptation plans for selected sub catchments of rural communes. The process will follow a community participatory process, identification of current and projected climate change impacts in combination with other stressors, may impact community infrastructure at sub catchment levels; mapping existing community infrastructure assets; identifying potential risks and identification of options that reduce system vulnerabilities at

community level, prioritizing infrastructure-related adaptation needs; designed to structurally withstand the impacts of climate change and functionally increase community resilience to climate change. This will contribute towards addressing droughts as well as floods. GCF financing will be used to finance the preparation of the community based adaptation action plans. The number of plans to be prepared by country include: Chad (6No); Cameroun (4 No); Benin (10 No); Guinea 5 No, Mali (10 No); Niger (10 No); Burkina (10 No); Cote D'Ivoire (5 No), and Nigeria (5No)

Activities

Activities will include: (i) Selection of target areas and target groups for support; (ii) Community- based surveys of the project area and prioritization of interventions with communities, water and agricultural experts and other stakeholders; (iii) Development of extension training materials that incorporate climate resilient approaches and training of project staff and community volunteers in gender sensitive adaptation planning; (iv) Awareness raising/training of communities in gender sensitive adaptation planning (includes sensitizing an estimated 1,000,000 women on climate change adaptation techniques and climate resilience) (v) Community meetings and planning workshops and (vi) Development of local adaptation plans with zoning of land for agriculture, grazing, perennial crops/grasses etc. (based on sub-catchment plans) with community and agricultural experts;

Output 15. Increase in local economic development arising from the upgraded community based livelihood infrastructure

This output involves diversifying income sources and increasing coping capacity of communities in the basin. It includes the scaling-up of community driven development in the 9 Niger basin countries. Rural communities in the Niger basin are particularly vulnerable to climate change due to their dependence on rain fed agriculture. The nature-based livelihoods in the basin is being challenged and in some cases overwhelmed, by rising water levels, rising air temperatures and shifting rainfall and storm patterns - effects of climate change that are projected to worsen over the next decades. Interventions will cover drought preparedness and natural resource management actions, including new and improved water supplies, range and fodder improvement, ecotourism, bee keeping, livestock product value addition, cage aquaculture, restocking reservoirs with fish and income diversification strategies such as investing in women's' groups. In each region of intervention, a regional approval committee (CRAC) of the activities/micro-projects will be put in place. An activity funding manual will be developed by each country to ensure transparency, equity and the achievement of population and ecosystem resilience goals. An estimated 150,000 jobs will be created for the youth and women (30% women). The indicative criteria for financing micro-projects will include i) vulnerability to the impacts of climate change ii) having been identified as action following a diagnostic analysis with a developed action plan iii) counterpart assistance from the beneficiary provincial to the investment iv) significant economic benefit and v) Have a dimension of adaptation to the climate change of communities.

Activities for climate adaptation to be financed by the GCF include the following:

- Benin: Extension of dyke protecting the irrigated perimeters of Malanville. This includes the installation of flood protection structures through the construction of protective embankments, in particular in certain districts of the city of Niamey and along the river to protect the rice fields in the vicinity of the town of Malanville in Benin against inundation.
- Benin: Building of community adaptation infrastructures (Product Storage facility, Product processing units and market sheds)
- Nigeria: Construction of embankment/dams and levees for protection of cropland and villages from flooding with embankments in upper Benue. The levees will control river floodwater flow and protect communities against costly effects of inundation.
- Niger: Construction of communities adaptation Infrastructures (Building crossing structures of Koris)
- Niger: Upgrading and rehabilitation of dam sections in commune 5 in Niamey for protection against flooding of rice fields
- Guinea: Building CC adaptation community Infrastructure. Activities to be identified from the community adaptation plans prepared under output 14
- Core D'Ivoire: Building of community infrastructures for adaptation to CC. Activities to be identified from the community adaptation plans prepared under output 14

Activities for the entire PIDACC project will include the following:

- (i) Selection and construction of 168 infrastructure upgrading Interventions (Activities) in specific watersheds;
- (ii) Formulation and operationalisation of 292 infrastructure management committees with 30% representation for women;
- (iii) Installation of 110 fish cages installed in river system (50% of beneficiaries will be women); this activity will support cage fish culture technology in selected locations along the Niger River system;;
- (iv) Restocking 270 water reservoirs with fish—incl. integration of aquaculture into existing farming systems
- (v) Provision of access to recovered (previously degraded) land to 9,000 women and 3,000 young people

(vi) Support towards establishment of 100 SMEs for youth employment (30 run by females); (ii) financing of 120 Interventions (Activities) supported by women and (iii) integration of strategies to increase women's participation in the workforce.

Component 3. Programme management and coordination

This component concerns the efficient management, monitoring and evaluation of the programme and dissemination of implementation results. It includes the supervision of activities, the monitoring and evaluation, as well as the annual audits. The PIDACC programme design embraces the “subsidiarity principle,” whereby most interventions will be implemented at the national level. Implementation/institutional arrangements are detailed in Section C.4.

Output 16: Dissemination of good adaptation practices and climate information

This output involves the dissemination of regional/national public awareness materials for the sustainable utilization of Niger Basin resources. It includes outreach activities to seek the support of key policymakers and senior government officials. Generating, disseminating and translating information into a useful form is a key response to uncertainty generated by climate change. The PIDACC project will help to improve climate resilience in the Niger basin, through a number of climate adaptation technologies, practices and innovative solutions which include: Early Warning systems; Agroforestry practices; Forestry practices, soil and water adaptation technologies (reduced tillage, permanent crop cover, mechanical and biological soil defines works); Supplemental irrigation/water harvesting; structural measures like dikes for flood control, Cage aquaculture using natural water bodies; weather-based agro-advisories and use of drought resistant crop varieties to mention but a few. In order to enhance uptake of technologies and enhance future adaptation at scale, the PIDACC project will promote the dissemination and continued use of these improved practices as well as enhance improved access to timely, meaningful and trustworthy climate information and knowledge.

Activities

Activities include: (i) Strengthening of communications systems, based on the existing NBA communication strategy; (ii) Preparation of briefing notes for local and national decision makers, media articles in newspapers, journals, newsletters, radio, participatory videos, etc.; (iii) Preparation of a knowledge management strategy; (iv) Preparation of a gender strategy and action plan²⁷ (v) Preparation of participatory videos and website maintenance (v) conducting ex-post cost-benefit analysis to determine the value added by the project compared with existing practices and (vi) Conducting national workshops on WSM, SFM ,CSA.

Output 17: Strengthened Monitoring and Evaluation (M&E) by the NBA Environment Observatory

This output will provide resources for enhanced M & E of the PIDACC program by the NBA observatory and the NPCUs. Assessment, monitoring and evaluation are crucial for learning and for conducting policy reviews in respect of the PIDACC program implementation. The project log frame in section H1 and H2 of the FP, will provide the primary premise for Impact Evaluation. Climate impact assessments will specify the changes in climate in a historic, current or future context, and establish evidence-based relationships between climate and productivity in the agriculture, forestry and fishery sectors under the PIDACC. The impact assessments will indicate the vulnerability of different stakeholders to a changing climate and the potential impacts of climate change on agriculture. Stakeholders targeted for evaluation will include male and female smallholder farmers and people working in the value chain. Continued evaluation beyond the lifetime of the PIDACC will be undertaken by the NBA observatory to measure the sustainability of the results and the long-term benefits or tradeoffs that are the expected from project. During project feedback and at the end of the project, lessons learned from evaluating the impacts of the project will be used to inform policy makers and future project design.

Activities

Activities will include: (i) Undertaking a baseline data/benefit monitoring surveys (ii) strengthening the regional and national GIS-based M&E and MIS; and (iii) collection, analyses, reporting and dissemination of data and information on the programme's progress with regards to outcomes and impact, based on the indicators provided in the log frame and the theory of change logic (figure 19) and (iii) production of Annual periodic reports

C.4. Background Information on Project / Programme Sponsor (Executing Entity)

C.4.1 Management team. Stakeholders involved in program implementation

See Figure 17: PIDACC implementation arrangements

²⁷ The project gender action plan, will be implemented at a regional level by the NBA and at a national level by the national coordination units in each country.

Regional and inter-regional level

The stakeholders involved in the implementation of the programme are the:

- i) NBA Council of Ministers which approves the strategic actions proposed by the steering committee. It comprises ministers of water and reports to the summit of Heads of State and Government of NBA countries.
- ii) The NBA which has a critical mandate to facilitate the coordinated and cooperative management of the Niger Basin.
- iii) The Programme Regional Steering Committee which will provide the programme's main direction.
- iv) The PIDACC regional coordination Unit (PRCU) —established within the NBA, which will coordinate the regional activities.
- v) NBA environmental observatory, which will coordinate monitoring and evaluation.

National level

Stakeholders in the implementation of the national components in each country include the:

- i) The National Steering Committee (NSC) which will provide country-level direction.
- ii) The National Focal Structure (NFS) comprising all sectoral ministries and the NCU which provides technical coordination and programme monitoring at country level.
- iii) The National program coordinating unit (NPCU) which will carry out the national activities in each country.
- iv) Sectoral ministries that will carry out activities in their respective fields in each country.

Local level

At the level of each intervention site/region/department of the programme, the stakeholders include the following:

- i) Technical Monitoring Committees (TMC), which group the technical services of the local authorities and is responsible for the monitoring of investments made in each region/department of the country.
- ii) Sub-basin management committees, which review and endorse the Interventions (Activities) to be submitted for funding.
- iii) Municipalities, CSOs, grass roots community organizations and communities that initiate and submit Interventions (Activities) to the programme for funding. CSOs will be involved in awareness generation, social mobilization and formation of CBO and interfacing between the government and community activities in source protection.

C.4.2 Overall strategy

The entry points for PIDACC actions will be sub-basin masterplans plans and community climate adaptation plans that will be developed for each intervention site. The approach to watershed management will be based on the use of the Toolkit for Planning, Monitoring and Evaluation of Climate Change Adaptive Capacities which was set up and tested by IUCN,²⁸ in association with OSS, AGRHYMET and UNECA in 2011, with financial support from the International Development and Research Centre (IDRC). Under the approach, the first step is to establish a diagnosis, an assessment of needs and identification of strategies to be implemented. The second step is to plan the implementation of the selected strategies and to put in place tools to monitor the progress. The third step is to ensure that commitments have been well monitored and deliver the expected results.

A capacity building programme will be deployed to improve production techniques and the integration of climate information into the management of agro-pastoral and fish farming activities. The tools for adaptation to the CC will be developed at the regional levels for better cohesion and harmonization. They will then be disseminated in the countries and at the intervention sites. The programme will place emphasis on rural employment creation for women and young people by strengthening the capacities of user associations, as well as village development committees. Monitoring and evaluation will be participatory at regional, national, sub basin, and community levels The inter-regional monitoring will be coordinated by the NBA.

C.5. Market Overview (if applicable)

Describe the market for the product(s) or services including the historical data and forecasts. N/A

C.6. Regulation, Taxation and Insurance (if applicable)

International (e.g. transboundary water ways) treaties or agreements, if any, that need to be complied with for the implementation of the project activities

Approval of Loan and Grant Agreements

Loans to be undertaken by each of the recipient host countries are subject to approvals by national cabinets and /or parliaments. As such Legal agreements "Protocol Agreements" in respect of the PIDACC project will be signed by the African Development Bank and Ministries responsible for finance and economic planning after approval by cabinets/parliaments in each of the countries. Specific details on Protocol Agreements to be signed by Financing Source for the PIDACC project are shown in figure 17 and detailed in section C 7.4.

²⁸ Somda, J., Faye, A. and N'Djafa Ouaga, H. 2011. Handbook and user guide of the toolkit for planning, monitoring and evaluation of climate change adaptive capacities. Niamey, Niger: AGRHYMET Regional Centre. 84 p.

After Board approval, the legal agreements are sent to the borrowing Countries for cabinet authorization. Following the authorization from the cabinet, the loan or grant agreement is signed by AfDB's President or Vice-President and the Representative of the Government. After loan approval a Borrower is allowed a maximum interval of 90 days to sign the loan or grant agreement with the Bank Group. Grants become effective upon signature by the parties. Before disbursement commences the loan or grant must be entered into force, or declared effective²⁹.

Loans take effect once conditions agreed to be precedent to first disbursement are met (loan effectiveness). In addition to "other conditions", the standard conditions precedent to first disbursements include: a) designation of authorized signatories for the loan resources; b) presenting a legal opinion; c) submission of the investment schedule; d) presenting lists of goods and services to be procured; e) opening and maintaining a special project account; f) establishing a project implementation unit; g) any other condition agreed during negotiation as necessary to precede first disbursement. Conditions are limited to the legal requirements as mentioned above, as well as cross-effectiveness of co-financing, and execution of subsidiary loan agreements. The requirements and deadlines for loan effectiveness are stipulated in the loan agreements. The African Development Bank's Legal Counsel and Project Coordinating Officer review if the conditions are met, after which the loan is formally declared effective. Normally, loan documents allow 90 days for the loan agreements to become effective³⁰.

Tax Exemption for the African Development Bank financed operations

Article 57 of the Agreement establishing the African Development Bank (updated in 2016), provides tax exemption and all custom duties for the Banks operations³¹.

Agreements Governing the Niger transboundary waterways

The NBA is governed by the Niger Basin Water Charter (2008) which is designed to ensure that Member States share the river's resources equitably. ³² Additionally, the Heads of State and Government of the NBA States signed the Paris Declaration (2004), which set out principles of management and good governance for the sustainable and shared development of the Niger Basin

Environment and Social Safeguards Clearances

Environmental and social safeguards compliance will be required at national level, prior to undertaking project actions. No other regulatory approval or authorizations are required in respect of the PIDACC program.

C.7. Institutional / Implementation Arrangements

Please describe in detail the governance structure of the project/programme, including but not limited to the organization structure, roles and responsibilities of the project/programme management unit, steering committee, executing entities, as well as the flow of funds structure

C.7.1 Implementation/institutional arrangements

The Niger Basin Authority (NBA)

Implementation of the project most measures are implemented at country level) will be coordinated by the NBA, making use of existing institutional arrangements to ensure national government endorsement and involvement, engagement with relevant line ministries, and appropriate technical oversight. The NBA has been instrumental in preparation of the PIDACC project.

Established in 1980 as successor to the Niger River Commission, the NBA is the regional river basin organization (RBO) mandated to promote cooperation among its 9 member countries to develop and manage the Basin's resources. The mandate of the NBA is to "promote cooperation among the member countries and ensure an integrated development of the Niger Basin in all the fields of energy, water resources, agriculture, livestock, fishing and fish-fanning, forestry and forestry exploitation, transport and communications and industry. "A "Shared Vision Process", which led to the signing of the 2004 Paris Declaration, set the management and governance principles for the sustainable development of the Niger Basin and reaffirmed the central role of the NBA as a coordination, knowledge, and development organization.

Based on the SDAP, an estimated \$7.2 billion 20-year Investment Program was approved in April 2008. The Water Charter, approved in April 2008, confirmed the mandates of the NBA. The NBA monitors the conditions of the Basin (done, in practice,

²⁹ <https://www.Afdb.org/en/projects-and-operations/project-cycle/loan-signing/>

³⁰ <https://www.Afdb.org/en/projects-and-operations/project-cycle/loan-effectiveness/>

³¹ <https://www.Afdb.org/en/documents/document/agreement-establishing-the-african-development-bank-2016-edition-92045/>

³² <http://www.internationalwatersgovernance.com/niger-basin.html>

through the NBA's Observatory of the Environment), provides services to key stakeholders (including flow forecasts) and a neutral analysis of planned water abstractions, and it mobilizes high-level expertise for relevant studies and analyses. Much of the background documentation and analytical work, to the development of the PIDACC has been undertaken with supervision of the NBA Secretariat.

The NBA has so far focused on two types of activities: (1) administration and implementation of activities on the ground together with member countries, and (2) development of frameworks and tools to support sustainable development and IWRM in the Basin, including a repository of the Basin's hydrological historical database and central collection of new data, flow forecasts, and update and running of the basin model to simulate the impact of infrastructure planned by countries on the river.

The NBA has an increasingly central role in facilitating decisions and building consensus among member countries, water users, civil society, and other key partners and is increasingly called upon to link national-level investments to regional processes.. The institutional arrangements of the NBA. The institutional arrangements of the NBA is shown in the figure below.

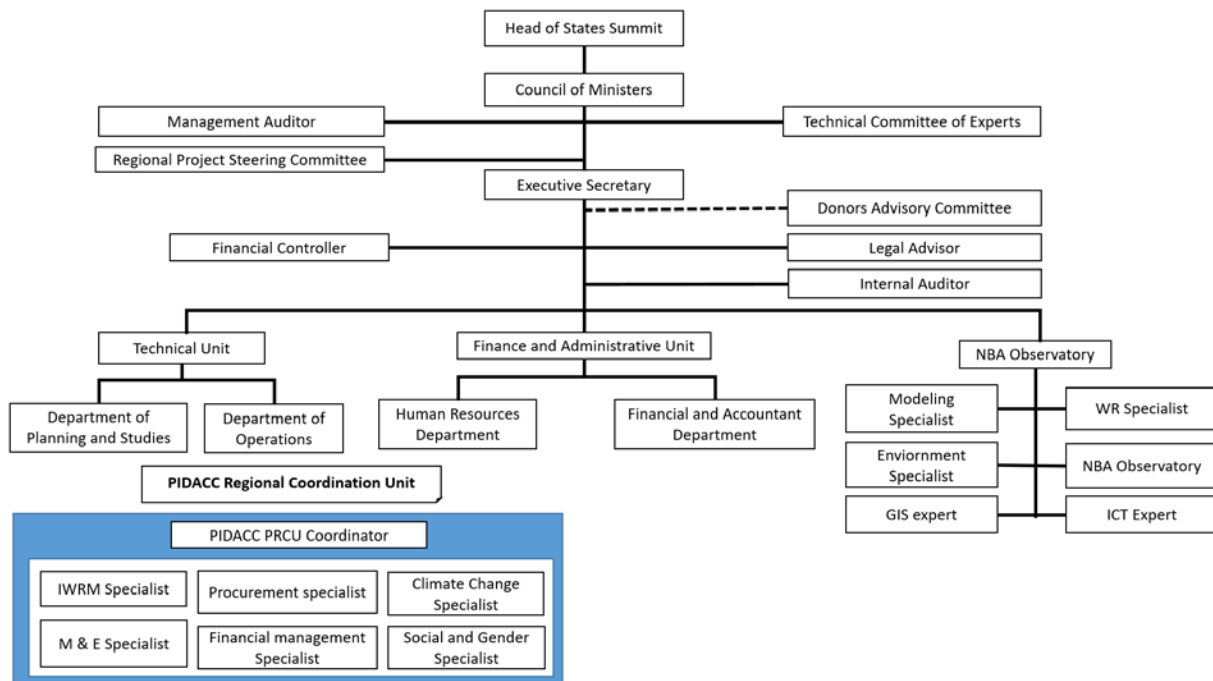


Figure 16: Niger Basin Secretariat Institutional Arrangement

The institutional arrangements of the NBA include the Summit of Heads of State and Government (SHSG), the Council of Ministers (CoM), the Technical Committee of Experts, the Executive Secretariat, and the National Focal Structures (NFS). The SHSG is the supreme body of the NBA. It defines the general orientation of the development policy of the Authority and controls its executive functions to assure the realization of the objectives of the Authority. It meets every two years in an ordinary session hosted by the member state which assumes chairmanship. The mandate of the chairman is two years.

The CoM comprising representatives from the water resources and related ministries is responsible for monitoring of the activities of the Executive Secretariat and reports to the Summit. The CoM provides guidance for issues related to the Niger River Basin and approves strategic actions by supporting the Regional Steering Committee (RSC). The CoM meets once a year in an ordinary session.

Project Steering Arrangements

The Regional Steering Committee (RSC) is the primary decision-making body and serves in a technical and institutional advisory capacity during implementation and review of programme activities. The RSC comprises: the Executive Secretary of the NBA, the coordinators from each NFS; a representative of the Ministry supervising the NBA from the country hosting the RSC meeting; a representative of the technical supervising Ministry of Projects and Programs of the member countries concerned when they are implemented in a ministerial department other than the supervising ministry of NBA; a representative of the Ministry of Finance from each member country; a representative from the existing or planned major facilities in the Basin; and President of the Regional Coordination of Users of natural resources of the Niger Basin. Any development partner can appoint a

representative as observer. The RSC meets at least twice a year to give overall guidance to the NBA and its National Focal Structures, formally review progress, and approve annual project work plans. Their roles will be to: (ii) monitor project progress and take timely actions to resolve implementation constraints; (iii) liaise with different National Focal Points to ensure the national units and the PRCU act in harmony; (iv) review annual project financial reports (v) review and approve Annual Work Plans; and (vi) ensure monitoring and evaluation of programme activities.

The Secretariat is administrated by an Executive Secretary appointed upon the recommendations of the CoM to the SHSG for a period of four years and renewable once. The NBA's institutional reform in 2005 introduced a new change by providing the terms of reference and profile of the Executive Secretariat to guide the CoM's recommendations.

The Technical Committee of Experts comprises representatives of the member states. Its mandate is to prepare the sessions of the COM. The Technical Committee of Experts meets upon invitation of the Executive Secretary according to a work schedule approved by the Council of Ministers.

The NBA monitors the conditions of the Basin (done, in practice, through the NBA's Observatory of the Environment), provides services to key stakeholders (including flow forecasts) and a neutral analysis of planned water abstractions, and it mobilizes high-level expertise for relevant studies and analyses. The observatory will handle the monitoring and evaluation function which includes data collection and all required tasks to track programme indicators, in collaboration with national agencies.

The National Steering Committee (NSC) comprises members of the National Focal Structure (NFS), chaired by the national NBA Focal Point in liaison with the country NDA and participants from the relevant ministries and agencies like the NHMS, civil society. The NSC will provide quality assurance for the national level activities, working directly with the PIDACC Regional Coordination Unit (PRCU) and the National Programme Coordination Unit (NPCU).

Pidacc Implementation Arrangements

Figure 17 shows the project implementation arrangements. PIDACC institutional arrangements build on existing structures within the Niger Basin Authority (regional) and at the national level.

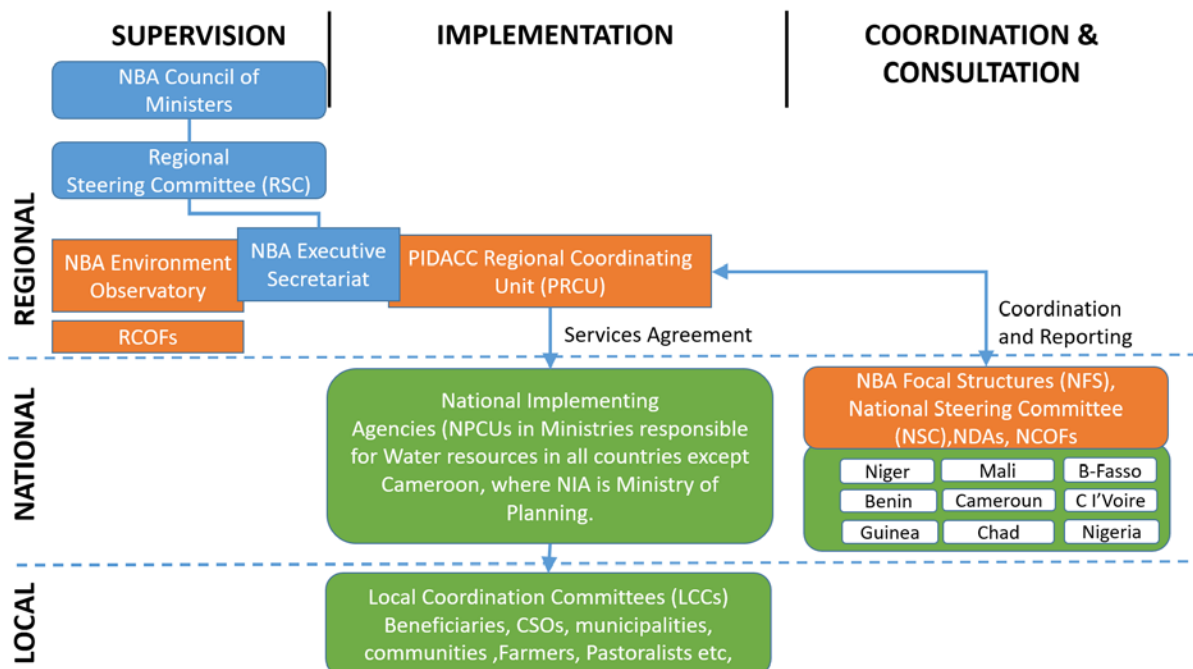


Figure 17: PIDACC implementation arrangements

PIDACC Regional Coordination Unit (PRCU). The PIDACC unit (PRCU) is a unit hosted within the NBA to implement the project. The PRCU will support the NBA in project management; financial, procurement, and administrative responsibilities; and supervision of the technical components of project implementation. Utilizing the PRCU builds on the capacity of the existing implementation structures and allows for implementation to commence in a timely manner. The PRCU will work closely with regional institutions, the RSC, and the NPCU to ensure timely programme implementation and effective capacity building. The

activities of the PRCU will be monitored and directed by the RSC. Experts to support the PIDACC project who will be hired include: (i) Regional Coordinator (Msc natural Resources Management); (ii) IWRM specialist (Msc Water Resources Management) (iii) Climate Change specialist (Msc degree in climate change); (iv) M&E specialist (Msc in Economics or statistics) (v) Social Development and gender expert (MA Social Development) (vi) a Financial management specialist and (vi) Procurement specialist (Master's degree in procurement).

The experts at the PRCU will be regional experts recruited from the NBA countries. This ensures that the regionally built capacity within the basin by the PIDACC is maintained. It is planned to build the capacity of all these national experts to ensure and efficient and effective management of the program at regional and national levels. At the Regional level (PRCU) The 3 experts below will cover the required thematic areas which include Climate change, Water Resources, soil, forests, biodiversity and wetlands. Specifications to suit those attributes will be made in the ToR of the experts

- The Regional Coordinator (Msc in sustainable land and forests management); Natural Resources Expert) with expertise in sustainable land and forests management. The expert will also be responsible for the technical coordination of Sustainable Land Management activities.
- The IWRM expert (Msc WRM) will cover aspects to do with water resources management
- The Climate change specialist (Msc degree in climate change, biology or geography) will cover aspects to do with Climate change, biodiversity and wetlands.

Other experts to support the PRCU will include (i) M&E specialist (Msc in Economics or statistics) (ii) Social Development and gender expert (MA Social Development) (iii) a Financial management specialist and (iv) Procurement specialist (Master's degree in procurement).

Financing for the Regional Project Coordination Unit (RPCU) will be made through proceeds from a number of partners who include: the GCF, the AfDB, the GEF, the FIP and the beneficiary countries.

National Programme Coordinating Unit (NPCU). The Implementing Agency in each country is the NBA's Supervising Ministry which is the Ministry of Water Resources [Environment and Agriculture] in all countries except Cameroon, where the programme will be under the supervision of the Ministry of Economy, Planning and Regional Development. Within each department, a National Programme Coordinating Unit (NPCU) will be established to support the implementation of the country component. The NPCU will be responsible for coordinating and implementing and M & E at the national level. NPCU's will work with decentralized services and locally-based organizations, but also closely with the PRCU. Responsibilities will include: (i) pre-selection of intervention sites; (ii) organization and supervision of all training and workshops; (iii) execution of the ESMF; (ii) development of tools and materials to enable grassroots organizations to implement program actions; and (iii) supervision of consultants and firms for the implementation of activities. Agreements will be signed with the sectoral ministries for the implementation of the sub-components in their respective areas of competence. The GCF, the AfDB, the GEF, the FIP and the partner states, will all make cash contributions towards the costs of the NPCU. Reference for the respective contributions per country are indicated in Budget lines A192; A193; A194; A196; A197 and A198 of "Annex III (sheet "Summary Budget").

The indicative staff composition of the NPCU will comprise the following: (i) National Coordinator (MSc in Natural Resources Management/Land and Water Management) (ii) Water management specialist (Msc Water Resources Management) (iii) Infrastructure Specialist (Msc Civil Engineering) (iv) Social Development and Gender expert (MA in Social Development) ³³. (v) an M&E specialist (vi) a Financial management specialist (MSc Finance) (vii) a Procurement specialist (Master's degree in procurement). Each NPCU will have a National Coordinator with qualifications in Water management and an Expert with qualifications in (Natural Resources Management/Land) who will coordinate water and soil conservation activities as well as sustainable forest management. The profile of the Natural Resources Management expert will focus on Land/ forest management or biodiversity and wetland management dependent the dominant NRM activities of PIDACC in the country. This will cater for the wider environment and natural resources dimensions within the program. All PIDACC coordination units will be in place /operational before the first disbursement [see section C 6].

Local Coordination Committees (LCCs). The collaborative effort of local institutions both public and civil is vital for the PIDACC project success. It will provide opportunities for communities to communicate amongst themselves and with local governments, and be responsible for assisting the implementation and monitoring and evaluation of the interventions. At the local level, working closely with the NPCU for administrative support, the LCCs, as community-based implementation units, will

³³ The specialists will be responsible for incorporating the Gender Action Plan into project planning and implementation including supporting/assisting design (and if required, implementation) of awareness workshops and training, establishment and monitoring of sex disaggregated data and indicators for project performance and monitoring (refer to section H.2)

be key in community awareness and education on climate adaptation programs. They will work with local authorities and communities in developing, regularly monitoring, and evaluating the site-specific on the ground watershed activities.

Beneficiaries. The communities groups (producers, fishermen, breeders, women's groups, agricultural groups, resource management groups and youth groups etc) are the final beneficiaries of the projects. Community groups will work closely with the local authorities, the LCCs and the NPCU. They will: (i) identify site-specific activities (interventions) ; (ii) assess environmental and social impacts; (iii) submit proposals to the LCC for activities to be funded; (iv) sensitize and involve their local communities and capacity building activities; (v) monitor and evaluate pilot and activities using locally adapted M&E systems; (vi) establish or train existing management committees directly responsible for programme monitoring and evaluation; and (vii) identify and articulate priority issues and solutions in their locality as they affect the Basin's sustainable management.

C 7.2 Selection Criteria for the selection of proposals/activities during project implementation

Feasibility studies were carried out for components (1 and 2) of the PIDACC in consultation with all stakeholders. The studies comprised (i) the identification of the project ideas and the sites of intervention and (ii) technical and environmental studies of Sub-component 2.1, Implementation of Climate Resilient Growth Infrastructure (Outputs 10, 11, 12 and 13) are already defined and studied to feasibility level. However detailed design studies will be undertaken during implementation of the PIDACC.

GCF financing will go towards Component 1: Development of ecosystems and natural resources resilience (S/C 1.1 Protection of resources and ecosystems and S/C 1.2: Strengthening shared management of natural resources) as well as S/C 2.2) and Component 3. The activities will be formulated and submitted by the communities to the NPCUs for consideration during the implementation phase. A community support mechanism will be put in place to support the selection of interventions. Measures will include: (i) technical assistance for project development and monitoring; (ii) development of a project finance manual describing the criteria and the review procedure; and (iii) putting in place a technical approval committee of projects in each intervention area.

Candidate Activities for implementation by the Project will follow the basic principles of participatory natural resources management and specifically (i) the request for the activity must emanate from the beneficiaries (ii) the activity is consistent with sub basin masterplans to be developed at country level and located within the Pidacc area of jurisdiction (iii) contribution towards building the resilience of communities, ecosystems and infrastructure to climate risks (droughts and floods); (iv) be in compliance with environmental safeguards (category 3 or 2 projects); (v) be led by groups of communities (producers, breeders, fishermen, women and young people willing to provide the counterpart varying from 5% to 20%); and (vi) justify the capacity of the beneficiary group to ensure a sustainable management and maintenance of the results of the project.

In addition, no activity will be approved that involves any of the following: (i) location in nature reserves; (ii) significant loss of primary forest, mangroves or sensitive wetland; (iii) any permanent negative effect on known rare or endangered species; (iv) any significant impacts on air quality and water quality and (v) involuntary relocation of people. Other criteria may be added during the development of the community interventions implementation manual taking into account the local realities.

C 7.3 Preparation of Community Activities/Interventions

A local consulting firm will be recruited in each country to provide technical assistance for the generation, development and monitoring of projects. This consultant will be responsible for i) sensitizing and informing communities about the activities of PIDACC, ii) developing activities/Interventions based on communities' demand for PIDACC financing and iii) supporting communities for monitoring of projects implementation.

C.7.4. Construction and supervision methodology with key contractual agreements

The contracting and supervision arrangements will be carried out by the NBA, national agencies and promoters of the Interventions (Activities) in line with the established practices acceptable to the African Development Bank. The programme agreements will specify the detailed implementation arrangements budgets, rights and contractual obligations of all parties regarding the project cycle, procurement procedures,³⁴ disbursement procedures³⁵ and reporting requirements, among others.

C.7.5. The African Development Bank's overall role as the Accredited Entity

The African Development Bank's overall role is to provide oversight and quality assurance through its headquarters and country office units. This role includes: (i) project preparation oversight; (ii) implementation supervision, reporting and knowledge-

³⁴ [https://www.Afd.org/fileadmin/uploads/African Development Bank/Documents/Policy-Documents/Procurement_policy_for_bank_group_funded_operations.pdf](https://www.Afd.org/fileadmin/uploads/African%20Development%20Bank/Documents/Policy-Documents/Procurement_policy_for_bank_group_funded_operations.pdf)

³⁵ <https://www.Afd.org/en/documents/document/disbursement-handbook-54419/>

management, and financial management; and (iii) project evaluation oversight. Safeguard specialists (Environment and Social and gender specialists) at the African Development Bank will in addition to other technical experts support program supervision.

C.7.6. Agreements and Funds flow for the grants and loans

Figure 18: shows Agreements, Funds flow and reporting requirements for the PIDACC program at regional and national levels. Separate Protocol Agreements will be signed between the African Development Bank and each of the Co-financiers (GCF, GEF, CIF-FIP, and the EU) for their contributions to the PIDACC project.

Since PIDACC is a public programme, all agreements (loans and grant) will be signed between the AfDB and the governments of the 9 countries and the NBA Executive Secretariat.

At national level, the borrowers will be the Niger Basin Countries through their Ministries in charge of Finance and Economic Planning. The ministries of finance will be the recipients of the loans and grants and will designate the Ministries in charge of Water Resources and Environment as the agencies for the implementation of the relevant project activities.

At regional level the recipient of the funds will be the Niger Basin Authority for regionally implemented activities. Separate protocol agreements will be signed between the African Development Bank and NBA for the regionally implemented component financed by the GCF, the GEF and the African Development Bank. The NBA will make resources available to the regional coordination unit (PRCU) to finance activities.

Financing from the African Development Bank together with co-financing resources to the host countries will be channelled under separate legal agreements which will be signed with the Ministries in charge of Finance and Economic Planning on behalf of the corresponding host countries. The Grant Agreements will require the host countries to confirm the provision of their respective financial contributions for the Project. The Ministries in charge of Finance and Economic Planning will designate the Ministries in charge of Water Resources and Environment as the agencies for the implementation of the relevant project activities.

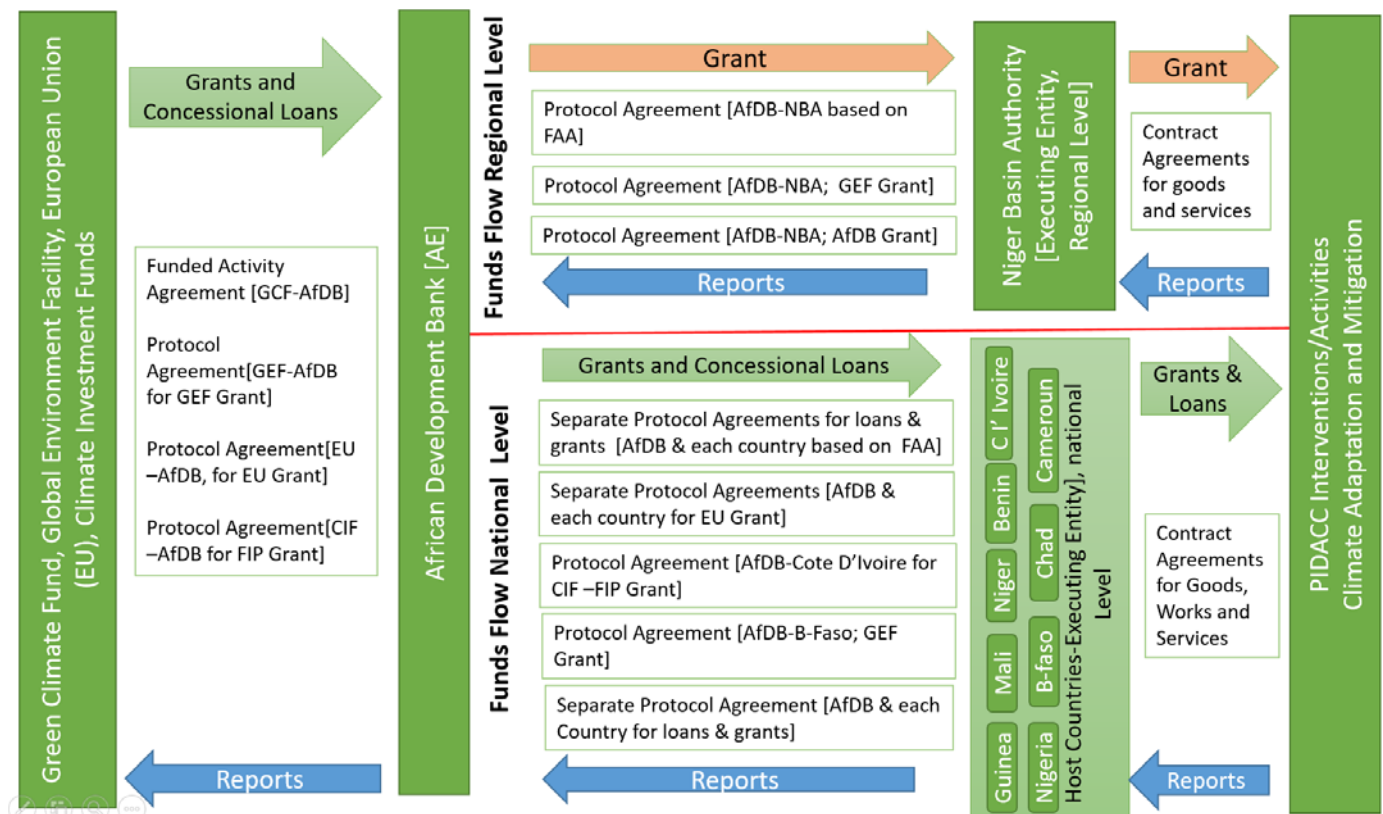


Figure 18: Agreements and Funds flow for the PIDACC Fund

Details on the Specific Agreements by Fund are outlined in the section below.

GCF-Funded Activities

The African Development Bank will enter into a Funded Activity Agreement with the GCF in line with the Accreditation Master Agreement. Separate Grant and Loan Agreements will be entered into by the African Development Bank with the NBA and the Governments of each host country based on the signed FAA.

CIF Funded Activities

The African Development Bank will sign a Protocol agreement with the Trustee of the Climate Investment Funds [CIF] for activities funded with proceeds from the Forestry Investment Program (FIP). The African Development Bank will enter into a separate Protocol Agreement with the Government of Cote D'Ivoire for implementation of the FIP financed activities.

GEF Funded Activities

The African Development Bank will sign a Protocol agreement with the Trustee of the Global Environment Facility (GEF) for activities funded with proceeds from the GEF. Separate Protocol Agreements will be entered by the African Development Bank with the NBA for regionally implemented activities, and with the Government of Burkina Faso for nationally implemented activities.

EU Funded Activities

The African Development Bank will sign a Protocol agreement with the European Commission (EC) for activities funded with proceeds from the EU. Separate Grant Agreements will be entered by the African Development Bank with the Governments of each Host Country for implementation of the FIP financed activities.

African Development Bank Funded Activities

The African Development Bank will sign separate Protocol Agreements for the loans and grants to each country for nationally implemented activities and the Niger Basin Authority for regionally implemented activities.

C.8. Timetable of Project/Programme Implementation PIDACC Program

Please provide a project/programme implementation timetable in section I (Annexes). The table below is for illustrative purposes. If the table format below is used, please refer to the activities as numbered in Section H. In the case of outputs, please mark when all the required activities will be completed.

Component/Sub component	Year 1				Year 2				Year 3				Year 4				Year 5				Year 6				Person in charge/Partners
	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	
Component 1: Development of the resilience of ecosystems and natural resources																									
Sub-component 1.1, Resource and Ecosystems Protection																									
1. Strengthen SLM for reduction of soil erosion and silting to sustainable levels																									
1.1. Undertake consultation with partner states, agronomists, land officers on zoning of land for agriculture etc																									Ministry/CSO/beneficiaries
1.2. Develop 9 sub basin masterplans /watershed management plans																									Ministry/CSO/PIDACC/beneficiaries
1.3 Stabilize dunes; restore degraded land, and mechanical/biological management of ravines																									Ministry/CSO/PIDACC/beneficiaries
1.4. Conduct awareness campaign on erosion control and improved soil management practices																									Ministry/CSO/beneficiaries
1.5. Expand sustainable land and water management interventions and technologies—including gender-responsive CSA technologies																									Ministry/CSO/beneficiaries
1.6. Support the fight against bushfires																									Ministry/CSO/beneficiaries
1.7 Conduct training for men and women farmers in business development and marketing and selected value chains in agriculture/ watershed management																									Ministry/CSO/beneficiaries
1.8. Building of a one lane 60 m span bridge																									Ministry
2. Increasing forest productivity and SFM - targets area 40,000 ha of forests																									
2.1. Support to development of private nurseries																									Ministry //beneficiaries
2.2. Support the provision of nationally available sources of high quality seed and plant material																									Ministry /PIDACC/beneficiaries
2.3. Conduct demonstration /trial forestry plots /re-establishment of forest tree species;																									Consultants/ Ministry /PIDACC /beneficiaries

Component/Sub component	Year 1				Year 2				Year 3				Year 4				Year 5				Year 6				Person in charge/Partners
	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	
2.4. Build capacity in SFM (incl. training of 200 community agents/associations for REDD+																									Consultants/ Ministry /PIDACC /beneficiaries
3. Integration of agro-forestry into farming systems on 26,000ha of agro-forestry ha of the watershed																									
3.1. Development of private nurseries																									Ministry /beneficiaries
3.2. Establish fodder banks with Napier grasses																									Ministry / PIDACC /beneficiaries
3.3. Plant trees/shrubs on existing radical terraces and on progressive terraces as well as on boundaries and within crops																									Ministry /PIDACC/beneficia ries
3.5. Undertake land certification— provide access to restored land for 9,000 women.																									
3.4. Train men and women farmers on gender-responsive CSA technologies on-farm (sensitizing 500,000 women on climate change adaptation techniques)																									Ministry /PIDACC/beneficia ries
4. Strengthening sustainable management of natural habitats																									
4.1. Identify areas for development of the Nature Reserves																									Consultants/ Ministry /PIDACC
4.2. Establish corridors for wildlife to access water and biomass during droughts																									Consultants/ Ministry /PIDACC
4.3. Rehabilitate bare land and spawning grounds across the basin nature conservation areas																									Consultants/ Ministry /PIDACC
4.4. Improve management within protected areas: to reduce degradation of habitats																									Consultants/ Ministry /PIDACC
Sub-component 1.2, Strengthening the shared management of natural resources																									
5. Strengthening hydro meteorological forecasting and early warning systems																									
5.1. Strengthening the decision support tools (Mike Hydro basin, Mike Hydro flood etc) and capacity development through integration of climate risk management																									Consultants/ Ministry /PIDACC/ACMAD
5.2. Long-term and on-demand capacity development and training program for NMHS staff on O&M, data modeling, and forecasting. Includes TA for support to certification of meteorological agencies in line with WMO requirements.																									Consultants/ Ministry /PIDACC/ACMAD

[illegible]

[illegible]

Component/Sub component	Year 1				Year 2				Year 3				Year 4				Year 5				Year 6				Person in charge/Partners
	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	
14.4. Awareness raising /training of project staff and communities in gender based adaptation planning(includes estimated 1,000,000 women)																									Ministry concerned
14.5. Conduct community meetings and planning workshops																									Ministry concerned
14.6. Develop and implement local adaptation plans with zonation of land for agriculture, pasture,																									Ministry concerned
15. Strengthen Community Based Livelihood infrastructure																									
15.1. Selection and construction of 168 infrastructure upgrading Interventions (Activities) in specific watersheds;																									Consultants/ Ministry / Beneficiaries
15.2. Formulation of 292 infrastructure management committees with 30% representation for women;																									Consultants/ Ministry / Beneficiaries
15.3. Installation of 10 floating fish cages in the river system(50% of beneficiaries will be women);																									Consultants/ / Beneficiaries
15.4. Restocking 270 water reservoirs with fish; incl. integration of aquaculture into existing farming systems																									Consultants/ Ministry / Beneficiaries
15.6. Provision of access to recovered (previously degraded) land to 9,000 women and 3,000 young people																									Ministry / Beneficiaries
15.7. Support for the establishment of 100 SMEs for youth employment, of which 30 will be run by females; (ii) financing of 120 integrated Interventions (Activities) supported by women; and (iii) integration of strategies to increase women's participation in the workforce.																									Ministry / Beneficiaries
Component 3; Project Management and Coordination																									
16. Dissemination of good adaptation practices and climate information																									
16.1. Strengthening regional and national communications systems, based on the existing NBA communication strategy.																									Ministry concerned /PIDACC
16.2. Preparation of briefing notes for local and national decision makers																									Ministry concerned /NBA
16.3. Preparation of project knowledge management strategy																									Ministry concerned /PIDACC

Component/Sub component	Year 1				Year 2				Year 3				Year 4				Year 5				Year 6				Person in charge/Partners
	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	
16.4 Preparation of project gender strategy																									
16.5. Preparation of media articles in journals, newsletters, maintain website																									Ministry concerned /NBA
16.6. Conduct an ex-post cost benefit analysis to determine the value added by the project compared with existing practices																									Ministry concerned /PIDACC
16.7. National workshops (WSM, SFM ,CSA)																									Ministry concerned /PIDACC
17. Strengthen M& E system and practice by the NBA Environment Observatory																									
17.1. Undertake project baseline data/benefit monitoring surveys																									Ministry concerned /PIDACC
17.2. Strengthen regional and national GIS-based M&E and MIS incl. gender disaggregated data																									Ministry concerned /PIDACC
17.3. Collection, analyses, reporting, dissemination of data and information on program progress																									/Ministry concerned /PIDACC
17.4. Preparing annual periodic reports																									PIDACC Program coordinating unit
Interim Evaluation																									
Project Completion																									PIDACC Program coordinating unit
Final Evaluation																									PIDACC Program coordinating unit

D.1. Value Added for GCF Involvement

Please specify why the GCF involvement is critical for the project/programme, in consideration of other alternatives.

GCF involvement in the PIDACC is critical as it will contribute towards strengthened resilience of human and natural systems in the Niger Basin to the impacts of climate variability and change and related risks. Without GCF investments that improve resilience in the Basin, PIDACC will have minimal impact.

GCF will contribute to the achievement of Basin country commitments to the UNFCCC conventions—more specifically, the NDCs as committed under the Paris Agreement and the NAPs, NAPA, the national climate change strategies, and the national disaster risk management strategies. Linking PIDACC interventions to NDCs will help align investments with national climate goals and help mainstream climate infrastructure needs into broader growth plans within the Niger Basin. By reducing economic losses and increasing productivity, the programme will directly support climate resilient growth. The contribution of GCF is also essential for the development of a sustainable funding mechanism (PES) to preserve natural resources in the long run through a Regional Adaptation Fund for the CC (RAFCC or FRACC).

D.2. Exit Strategy

The approach for sustainability is drawn on elements from the success of multi country climate adaptation programs like the Bank financed Silt Control project in the Sahelian Zone of the Niger River Basin (Burkina Faso, Mali and Niger and coordinated by the NBA) which strengthens its sustainability beyond project closure. These include (i) a strong participatory approach- emphasizing demand driven interventions by the governments and the beneficiary communities to address climate variability (ii) contribution of the governments and beneficiary communities estimated at 10% and 5% of the project costs respectively—which strengthens ownership, sustainability and replication and (iii) setting up community based infrastructure management committees, which are trained and equipped for O & M of infrastructure. Other dimensions include (i) strengthened mechanisms for O & M of the hydromet infrastructure (ii) strengthened M & E system to document and disseminate best practices and (iii) setting up mechanisms for payment for Ecosystem services (PES) and a Regional Adaptation Fund for Climate Change (RAFCC) to finance downstream measures after project closure. The Bank will ensure that agreements for loans and grants to the recipient agencies include commitments to an exit strategy which ensures the sustainability of project outcomes. Details of the elements are illustrated below.

Strengthened policy and institutional environment

The project is aligned to the national climate change related priorities (including meeting their NDC commitments which strengthens replicability and contributes towards future adaptation interventions and broader project sustainability. At a regional level, the Niger water charter provides for continued promotion of integrated management of the Niger Basin water resources and maintenance of the integrity of ecosystems through protection of aquatic ecosystems against the deterioration of basins and the threat of climate variability among other obligations.

Natural resources interventions

Project activities related to agro-forestry, afforestation, soil and water conservation and natural resources management are community driven type activities. Interventions for example confirming the choice of locations to be protected, fixing of dunes etc, will be implemented with active participation by the beneficiary population (counterpart labor contribution), which strengthens their sustainability after project completion. The technical and financial capacities of the local communities will be strengthened, thereby minimizing the need for recurrent expenditure after project closure. With strengthened management capacity, the communities will be in a better position to own and sustain the project's results.

Sustainable agriculture techniques and technologies introduced to communities have been tested before under the Niger silt control program, and are technically feasible and therefore have the best chance of being self-sustaining and self-replicating. Through the enrichment of this local knowledge and know-how with relatively simple reproducible techniques, the project aims at developing and maximizing the various results by the beneficiary population. Assessments will also be conducted for sustainable harvesting operational plans for communities (including benefit sharing) to build community resilience to climate change impacts through income and livelihoods diversification and enhancement.

Maintenance costs (weeding, raising the fences, replanting, which might be necessary) as well as the protection of the developed sites will be the responsibility of the beneficiaries in partnership with government technical departments and the decentralized communities. The forestry and agro forestry investments will also be supported by the development of training programmes and management protocols for nursery staff and the increased generation of revenue through marketing of tree seedlings, and the development of revenue models to support the sustained operation of each nursery. After project completion, countries will continue to be responsible for the monitoring, training, awareness, technical support for the sustainability of development activities at the request of the population and the decentralized communities.

Community based hydro-infrastructure and pastoral watering points

Strengthening of the technical, managerial and financial capacities of the beneficiary communities (through committees set up for O & M of the small infrastructure) will ensure ownership and the sustainability of community based hydro infrastructure. The management committees will through a benefit and revenue sharing approach ensure revenue collection for O & M of the community infrastructure after project closure. The establishment of catchment based adaptation plans will promote community-based catchment management and serve as an incentive for improving the lifetime of the hydro agricultural infrastructure.

Modernized Climate and Weather information systems

An integrated agency wide approach, through the Niger Hycos—will be undertaken to engage the governments on O & M, focusing on strengthening capacity and ultimately, improving service delivery from the installed hydromet stations. Strengthening of the NMHS, will build onto efforts of the ongoing African Development Bank-ClimDev financed project “system for monitoring water resources and flows forecasts using satellite imagery in the Niger Basin (SATH-ABN)”. Commitment will be obtained for government support beyond the project close to cover the O & M of the new systems and services and retain qualified staff. The African Development Bank through the ClimDev program, will also support long-term planning and capacity development through facilitating knowledge-sharing across borders in the Niger basin.

Strengthened knowledge base and M & E framework

The PIDACC will create dedicated knowledge and learning across project components including the creation of a publicly-available online information portal to increase the availability of information on landscape approaches, as well as seminars to share critical knowledge with key decision makers and community members. The M&E system will generate project information on the socio-economic and environmental benefits, document lessons learned and incorporate them into decision making. Future phases of the PIDACC, will be informed by the experience and knowledge products generated by the project. For sustainability, these M & E systems at the NBA and country level, will be embedded within permanent public agencies. These activities will encourage replication and ensure the sustainability of project results

Financial Sustainability

Country and beneficiary contributions which are estimated at 8% and 4% respectively of the total project costs, reflect ownership of the project outcomes and will contribute to project financial sustainability.

PIDACC also will explore innovative mechanisms for meeting funding needs for sustainability of the natural resources interventions in the basin. Mechanisms will include: raising funds from new markets (such as carbon offsets, water, or other payments for ecosystem services (PES)). PES generates its own funding without substantial budgetary outlays from the governments. The PES mechanism will establish a system for financing the activities for protecting natural resources in the basin watersheds by users of these resources. The PES will be financed through taxes levied on heavy water users, such as mines, industries, large dams and cities in the basin. A regional Adaptation Fund for climate Change (Environment Trust Fund), which is a long term financing instrument, will be put in place and operationalized by year 3 of the PIDACC project implementation to manage the PES. The study to design the RAFCC design is ongoing financed by the European Union. The Trust Fund modalities will be defined during the project to ensure that the fund is created in line with the country's priorities. It is expected that the RAFCC will be catalytic and support scaled up climate adaptation actions leading to a sustainable and virtuous circle of investment in management of the basin's natural resources.

E.1. Impact Potential

In this section, the accredited entity is expected to provide a brief description of the expected performance of the proposed project/programme against each of the Fund's six investment criteria. Activity-specific sub-criteria and indicative assessment factors, which can be found in the Fund's Investment Framework, should be addressed where relevant and applicable. This section should tie into any request for concessionality made in section B.2.

E.1. Impact Potential

Potential of the project/programme to contribute to the achievement of the Fund's objectives and result areas

E.1.1. Mitigation / adaptation impact potential

Specify the mitigation and/or adaptation impact, taking into account the relevant and applicable sub-criteria and assessment factors in the Fund's investment framework.

Building resilience to CC is a priority for poverty reduction and shared prosperity in the Niger basin. Climate Change projections point to higher temperatures, increased frequency and intensity of rainfall and associated flooding, along with more frequent and prolonged droughts. Climate variability, in particular, poses a significant risk to vulnerable communities. Projected impacts threaten many of the basin development priorities (food and energy security in particular).

Expected total number of direct and indirect beneficiaries, number of beneficiaries relative to total population (PMF-A Core 1), particularly the most vulnerable groups. The GCF-financed activities under PIDACC are expected to contribute directly to adaptation by increasing resilience to climate shocks in poor and extreme poor households. The improved resilience will potentially benefit 4 million people directly and 10 million indirectly (~50% female). The community resilience will be supported through strengthening of existing delivery mechanisms including flood and drought prone households through improved storage and conservation of ecosystems.

Expected reduction in vulnerability by enhancing adaptive capacity and resilience for populations focusing on the most vulnerable population groups and applying a gender-sensitive approach. The programme will improve the resilience of the populations and ecosystems of the Niger River through the implementation of measures such as: (i) erosion and silting reduction, dune fixation and watershed management; (ii) protection of biodiversity and wetlands for sustainable management of Ramsar sites; (iii) sustainable rangeland management; and (v) implementation of 33 community climate adaptation plans.

Expected increase in generation and use of climate information in decision-making (PMF-A 6.0 and related indicator(s)).

The PIDACC project will strengthen the decision support tools through integration of climate risk management, strengthening of hydrological forecasting based on rainfall-runoff models driven by projections of future precipitation and application of freeware such as the Soil and Water Assessment tool (SWAT) to assess impacts of landscape measures. The enhanced tools will enable decision-makers to evaluate and implement climate adaptation measures. *The program will also strengthen the use of Climate risk information for decision making, both at the country and basin level.* Examples may include services that improve decisions for resiliency in the short- and long-term, such as early warning systems for climate related disaster risk management, area planning, and provision of services critical for sustaining economic growth.

Expected strengthening of adaptive capacity and reduced exposure to climate risks (PMF-A 7.0 and related indicator(s)).

The programme will have a strong focus on reducing community vulnerability through focusing on the transformation of 'last-mile' community EWS. Ecosystem-based adaptation approaches like agroforestry, IWRM, livelihood diversification, and sustainable forest management interventions will also reduce vulnerability to climate change. An estimated 500,000 people (50% women) will be reached by climate-related EWS systems and other risk reduction measures.

Expected improvement in the management of land or forest areas contributing to emission reductions (PMF-M 4.0 and related indicator(s)). The project places emphasis on assessing, monitoring and reporting on mitigation in the Logical Framework (Section H1) and M&E (Section H 2). The programme will contribute towards reducing GHG emissions and increasing carbon sequestration through sustainable forest and soil management through: (i) REDD+ development of 40,000

ha of forests, and (ii) training of 200 members of community associations in REDD+. ³⁶ Conserving terrestrial carbon represents a “least-cost opportunity” in terms of climate change adaptation and mitigation and is essential to increasing the resilience of agricultural ecosystems. ³⁷ Mitigation benefits have been estimated and analysed ex-ante, and will be assessed and reported in the Project’s Completion Report.

E.1.2. Key impact potential indicator

Provide specific numerical values for the indicators below.

GCF core indicators	Expected tonne of carbon dioxide equivalent (t CO ₂ eq) to be reduced or avoided (mitigation only)	Annual	1.4 million tons CO ₂ [Estimates of reduced/avoided emissions by mid-term/year 3 is 4.2 million tons CO ₂]
		Lifetime	7 million ton CO ₂
	Expected total number of direct and indirect beneficiaries, disaggregated by gender (reduced vulnerability or increased resilience)	Total	Direct beneficiary population i~500,000 households (estimated at 4 million people); (average of 8 persons per household. Indirect beneficiaries estimated at 10 million people (50% women)
	Number of beneficiaries relative to total population, disaggregated by gender (adaptation only)	Percentage (%)	10% (50% are women)
	Expected increase in generation and use of climate information in decision-making	Percentage (%)	50%
	Expected strengthening of adaptive capacity and reduced exposure to climate risks	Percentage (%)	50% of beneficiary population
Other relevant indicators	Enhanced climate resilient landscapes developed as a hedge towards climate extremes	Total	40,000ha — forestry 26,000 ha —agroforestry 10,000 ha of dunes stabilized; 110,000 ha of degraded land restored; - Mechanical and biological management of 45,000 (m3) of ravines undertaken
	Number of households benefiting from last mile early warning systems against climate extremes of floods and droughts		62,500 households (estimated population 500,000~50% being female)

Methodology used for calculating the above indicators

The detailed methodology and data are provided in the preliminary studies for the Programme for the Integrated Development and Adaptation to Climate Change in the Niger Basin (PIDACC/NB). In assessing the potential for emissions reduction, the following hypothesis and basis for calculation was used: (i) Carbon stock: according to FAO estimates (2011), the total carbon of Sahel forests in the living biomass of forests area (above ground + below ground biomass) is about 53 tCo₂/ha (ii) Rate of deforestation: According PRONAGREF (TAED, 2009) is estimated at 107,626 ha/year.

Estimation of potential carbon sequestration, is based on an AGRYMET study. ³⁸ The study estimates the potential carbon sequestration maxima in Senegal, Mali, Burkina Faso and Chad. A sequestration rate of 53 tonnes CO₂ eq. /ha for assisted natural regeneration of forests (80% from carbon above ground biomass, i.e. 42.4 tonnes CO₂ eq. /ha, and 20% from carbon in soil biomass). The estimate for hedgerows is based on 200 stems/km versus 625 stems/ha (e.g. 3 km of hedgerows equals 1 ha in terms of CO₂ sequestration).

³⁶ Preliminary studies of multi-purpose infrastructure of the Programme For The Integrated Development And Adaptation To Climate Change In The Niger Basin (PIDACC/NB); Regional Synthesis Report

³⁷ World Bank 2012. Carbon Sequestration In Agricultural Soils

³⁸ “Preliminary evaluation and mapping of the carbon sequestration potential on the basis of forest species and soil and climate units in the Sahel and particularly in Senegal, Mali, Burkina Faso and Chad” (WMO, Center, AGRHYMET Regional Center), labelled as AGRYMET modelling method in the CEO draft Letter

E.2. Paradigm Shift Potential

Degree to which the proposed activity can catalyse impact beyond a one-off project/programme investment
E.2.1. Potential for scaling up and replication (Provide a numerical multiple and supporting rationale) <i>Describe how the proposed project/programme expected contributions to global low-carbon and/or climate-resilient development pathways could be scaled-up and replicated including a description of the steps necessary to accomplish it.</i> <u>Scaling up low carbon/climate resilient growth pathways</u> A strengthened knowledge base will ensure documentation and dissemination of best practices and this will inform future programming within the wider CRIP programme. Scaling up will also be supported by key partnerships. The project will also pilot low-emission agriculture through increasing tree cover in crop and livestock systems (e.g. agro-forestry~26,000ha) and reducing soil disturbance (e.g. through reduced tillage), peatlands management (low lands), grazing land management, forest management (~40,000 ha) as means of sequestering carbon in agricultural systems in the Niger Basin. Such actions will expand so that by the mid-2020s, land-use change and forestry will become a net carbon sink. The proposed RAFCC mechanism will assist the basin countries in reduction of deforestation and degradation. The Fund will aim to compensate users within the watersheds for GHG reductions through improved protection of forests as carbon stores. The contribution of carbon sequestration would be key, allowing for compensation of GHG emissions by 2050. <u>Potential role women can play in the project as agents of change</u> Under its Gender Strategy, the Niger Basin Authority has committed to mainstream gender considerations in all its operations and support gender-specific activities, especially in areas where gender disparities are most severe. Rural women, mainly active in the primary sector and dependent on weather conditions and natural hazards, would thus be most affected by the effects and impacts of climate change. The improved management of the Niger Basin's natural resources – to be brought about by the Project - will be particularly beneficial to poor rural women.

E.2.2. Potential for knowledge and learning

E.2.2. Potential for knowledge and learning <i>Describe how the project/programme contributes to the creation or strengthening of knowledge, learning processes/ institutions.</i> The PIDACC programme will strengthen the knowledge base and analytical tools related to climate risk management in the Niger Basin. Activities will include the following: • Climate risk assessment tools through enhanced numerical weather prediction models building on existing tools, extended hydrologic forecasting (SWAT); climate risk assessments; integration of spatial information in ArcGIS, climate adaptation factsheets, seasonal forecasts, Web Portal with real-time maps and periodic state of the basin reports; • Creating knowledge products (e.g. reports, briefing papers on high level PIDACC results; interactive knowledge products); documentation of indigenous climate knowledge especially last mile warning; • Knowledge dissemination/experiential learning through outreach (incl. public domain information, technical forums). The diffusion of tailored products will be monitored through gender-sensitive field surveys; and • Strengthening interactions and outreach on climate risk management within the Niger Basin in close collaboration with the NMHS, ACMAD and AGRHYMET (incl. workshops, distance learning, vendor fairs).
--

E.2.3. Contribution to the creation of an enabling environment

E.2.3. Contribution to the creation of an enabling environment <i>Describe how proposed measures will create conditions that are conducive to effective and sustained participation of private and public sector actors in low-carbon and/or resilient development that go beyond the program.</i> The programme will contribute to the creation of an enabling environment for more informed policy making for climate resilient growth in the Niger Basin. Practical examples will include strengthening: (i) awareness raising and information exchange; (ii) national implementation strategies for EWS and community-based adaptation; (iv) climate resilient agriculture, biodiversity, forestry and ecosystems management; and (v) establishment of information systems for GHG inventories.

E.2.4. Contribution to regulatory framework and policies

E.2.4. Contribution to regulatory framework and policies

Describe how the programme strengthens the national / local regulatory or legal frameworks to systematically drive investment in low-emission technologies or activities, promote development of additional low-emission policies, and/or improve climate-responsive planning and development.

Contribution to NBA regional regulatory frameworks

The PIDACC/NB will promote the African Development Bank's 2013-2022 Strategy and other regional policies which include:

- African Development Bank 10 Year Strategy. The strategy emphasizes transition to green growth, responding to the challenges of climate change and reducing the ecological footprint on its natural capital, as a springboard for growth.
- African Development Bank Human Capital Strategy 2014–2018, which places emphasis on improving skills for competitiveness and jobs.
- African Development Bank Feed Africa Strategy. The strategy aims to address food insecurity and climate change by improving resilience against climate impacts, reducing GHG emissions, and increasing productivity in farming.
- African Development Bank HIGH 5. The High5's aim to improve the quality of life for the African people. The strategy recognizes the need for reliable and accurate weather and climate data and information to help African partner states on how to mainstream climate risk management into the water resources sector.
- ECOWAS Regional Agricultural Investment Programme. The programmes focuses on: (i) promotion of early warning systems; (ii) development of crisis management systems; (ii) support for the rehabilitation of zones after crises; and (iv) development of compensation mechanisms/insurance against catastrophes.

E.2.5. PIDACC Theory of Change Diagram

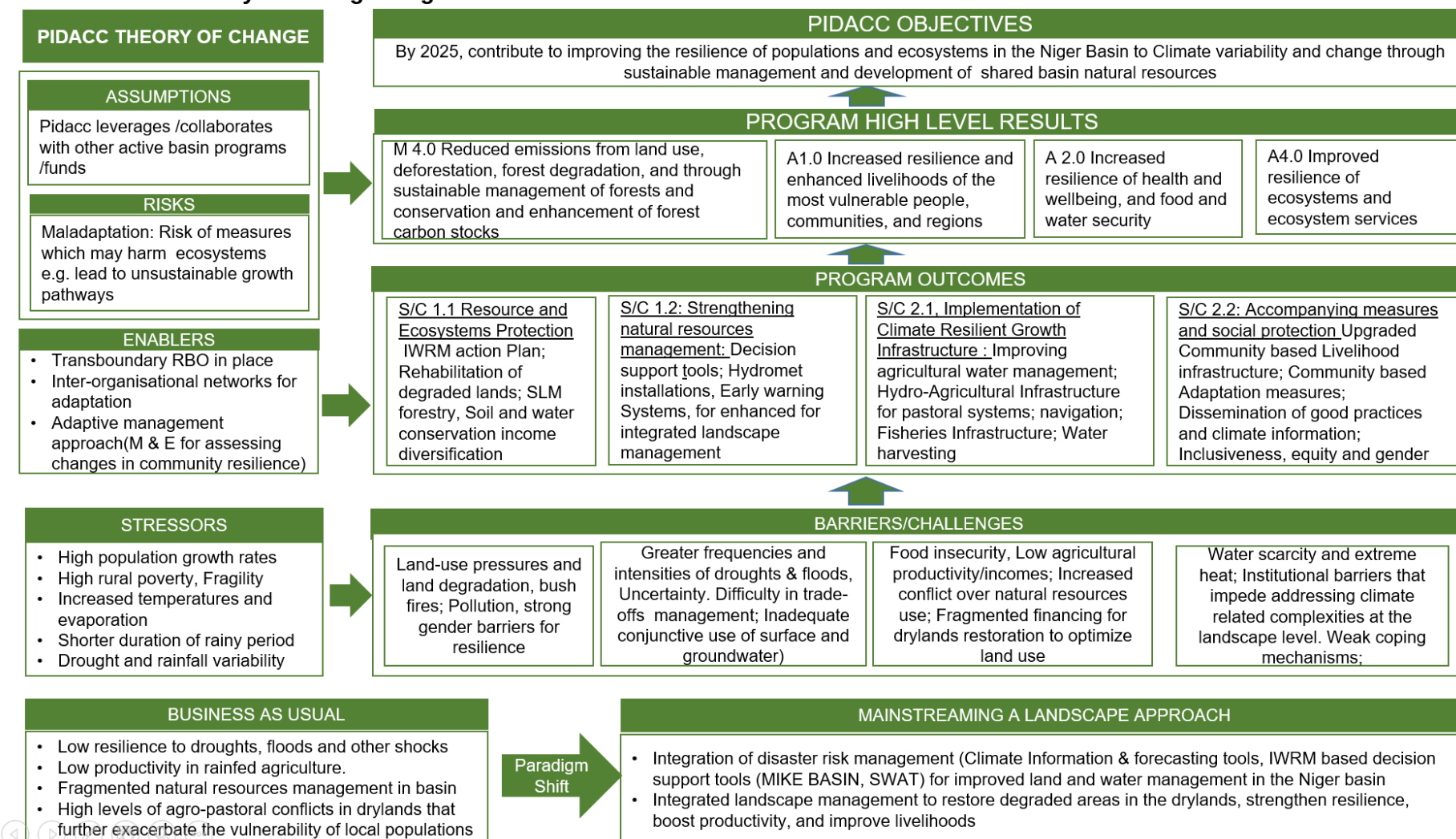


Figure 19: Theory of Change diagram for the PIDACC program showing causal linkage/pathway between the problem statement, barriers and strategic result area inclusive of the assumptions and risks.

E.3. Sustainable Development Potential

E.3. Sustainable Development Potential

Wider benefits and priorities

E.3.1. Environmental, social and economic co-benefits, including gender-sensitive development impact

E 3.1.1. Economic co-benefits

The project economic analysis is based on a 30 year impact given the 7 to 10 year average lifespan of planned climate and weather information systems and a 30 to 40 year lifespan for natural resource investments. Operations and maintenance (O&M) costs are assumed to be between 1% and 5% of programme investments. A discount rate of 10% was used to calculate the Net Present Value (NPV). A 10% discount rate represents an understanding that future costs and benefits are relatively important in comparison to the current situation – concurrent with concerns regarding climate change. The NPV is calculated to be US\$ 250million, at a discount rate of 10% for the base case which includes avoided costs due to climate change expressed as benefits. This represents a benefit-cost ratio of 1.6. The Internal Rate of Return (IRR), which is the interest rate that makes the NPV of all cash flows equal to 0, was calculated to be 18%. A sensitivity analysis was conducted to illustrate the benefits of the programme at varying rates of return. At a discount rate of 12%, the NPV equals US\$ 150million. This implies that the programme would be viable even with higher discount rates.

E 3.1.2. Social co-benefits

Social benefits are derived from increased income reliability. An estimated 150,000 jobs will be created at the end of the programme. Through creation of rural employment for the most vulnerable groups (young people and women), PIDACC/NB will be a model of action promoting gender, social equality and inclusive development. This is in line with the African Development Bank Human Capital Development Strategy,³⁹ which places emphasis on improving demographic dividends through improving skills for competitiveness and jobs. The strategy also emphasizes building financial and social systems to ensure inclusion and social cohesion. Through strengthened climate and weather information systems, PIDACC will also strengthen the productivity of farmers and reduce the number of people who are food insecure in the Niger Basin.

The project will help prevent conflicts between farmers and pastoralists in transhumance transit and arrival areas and build peace among communities. Integrated landscape management will promote the multi-functionality of landscapes and provide a mechanism that enables local stakeholders to reduce conflicts among different types of specialized resource users who differ in their dependencies on a range of ecosystem services (e.g., herders, farmers, or fishers). Examples will include agreements negotiated between farmers and herders to demarcate corridors for livestock movement, which will help to protect farmers' crops and trees from livestock browsing while at the same time safeguarding grazing and water access areas for herders. Investment in transboundary water management (focusing on water and rangeland resources) through hydro infrastructure (Outputs10 and 11) will also help to de-escalate tensions, promote stability, and provide resilience to hydrological shocks that might otherwise act as a trigger for conflict. Targeted areas will include transhumance corridors. This will provide a measure of resilience against water-related disasters and water crises and mitigate against conflict.

E.3.3 Gender-sensitive development impact

Improvements in forecasting and early warning will help minimize the negative impacts on women. People-centred community-based EWS will directly address women's vulnerabilities and exposure to disaster risk. During community sensitization, as well as design and implementation, women beneficiaries will be targeted for their engagement and ownership of the community-based EWS. Specifically, gender will be mainstreamed into activities such as capacity building, communication, project formulation, studies, and assessments. Sharing information to strengthen disaster risk management and operation of the storage infrastructure along the transhumance routes will strengthen resilience.

Section E 3.1.4. Environmental and economic co-benefits

Improved access and utilization of hydromet information and EWS will reduce the climate-related disaster risks through an increase community preparedness for response and recovery, consistent with Sustainable Development Goal (SDG) target 13.1 and SGD target 13.3 on strengthening institutional capacity on climate change mitigation and adaptation.

³⁹ African Development Bank Human Capital Strategy 2014–2018 May 2014

Reforestation, regeneration of tree and forest areas (~40,000 ha), agro-forestry (~26,000ha), erosion and silt control and sustainable agro-sylvo-pastoral management will generate benefits for food security, adaptation to climate change (microclimate), and water management. It will further provide safety nets for rural people in times of economic distress, helping them offset losses in income caused by weather shocks. This is consistent with SDG 13 on climate change, SDG 15 on sustainable forests, and INDC priorities on agriculture and forestry.

Interventions on wetland and flood plain restoration will support many human activities, including fisheries, cropping, and gardening. Protection and wise use of wetland resources will also offer water supply and quality benefits that are often essential to maintaining a basic standard of living in both urban and rural areas. Wetlands will also continue to provide a wide range of other benefits to local communities, including handcraft and building materials, food resources such as wild vegetables and medicine for various ailments. Indirect benefits will include provision of ecological services, that maintain and protect natural and human systems, such as maintenance of water quality and flow, flood control, nutrient retention and climate regulation (micro-climate stabilization), and the production and consumption activities they support; carbon sequestration, etc. This is consistent with SDG Goal 15 on protection, restoration and promotion of sustainable use of terrestrial ecosystems and halting and reverse land degradation and halt biodiversity loss

Integrating protection of natural habitats and improved management of agricultural resources into adaptation plans will contribute to cost-effective strategies for reducing vulnerability to climate change.⁴⁰ Interventions on flood plain/spawning grounds restoration will support fisheries, consistent with SDG 13 on climate change, and SDG 15 (target 15.3) on reducing the degradation of natural habitats, halting the loss of biodiversity and preventing the extinction of threatened species.

Establishment of a Regional Adaptation Fund for Climate Change will provide sustainable funding for conservation actions by prioritizing proposals for funding that involve local people in negotiated resource use and co-management of resources. This is consistent with SDG target 15.a: mobilizing financial resources to conserve and sustainably use biodiversity and ecosystems SDG target 13.a: implement the commitment undertaken by developed-country parties to the UNFCCC.

E.4. Needs of the Recipient

E.4. Needs of the Recipient

Vulnerability and financing needs of the beneficiary country and population

E.4.1. Vulnerability of country and beneficiary groups (Adaptation only)

Describe the scale and intensity of vulnerability of the country/beneficiary groups, and elaborate how the programme addresses issues (e.g. the level of exposure to climate risks for beneficiary country and groups, overall income level, etc).

E.4.1.1 Non climate drivers for vulnerability

The demographic growth in the Niger Basin puts pressure on the natural resources and the environment. The Niger basin's population is growing at ~ 3% per year. By 2030 the number of people living in the drylands of West Africa is expected to increase by 80% (based on current population growth estimates). Over the same period, climate change could result in an expansion of the area classified as drylands by as much as 20% under some scenarios, for the region as a whole, with much larger increases in some countries. Rapid population growth is putting pressure on a deteriorating resource base and creating conditions under which extreme weather events, unexpected spikes in global food and fuel prices, or other exogenous shocks can easily precipitate full-blown humanitarian crises and fuel violent social conflicts⁴¹. Forced to address urgent short-term needs, many households have resorted to unsustainable practices, resulting in severe land degradation, water scarcity, and biodiversity loss. Economic growth will reduce the share of people living in drylands who are sensitive to droughts and other shocks, but probably not fast enough to overcome the effects of demographic growth.⁴²

⁴⁰ 2008 The International Bank for Reconstruction and Development / THE WORLD BANK; Biodiversity, Climate Change and Adaptation Nature-Based Solutions from the World Bank Portfolio

⁴¹ Cervigni, Raffaello, et al. 2016. Confronting Drought in Africa's Drylands: Opportunities for Enhancing Resilience. Africa Development Forum series. Washington, DC: World Bank. doi:10.1596/978-1-4648-0817-3. License: Creative Commons Attribution CC BY 3.0 IGO

⁴² Opportunities for Enhancing Resilience Confronting Drought in Africa's Drylands; Raffaello Cervigni and Michael Morris, Editors A publication of Agence Française de Développement and the World Bank

Poverty and climate change vulnerability in the Niger Basin are strongly correlated. The chronic vulnerability of people living in drylands of the Niger basin, stands at the forefront of the basin member states development challenge. Table 11, shows the prevalence of poverty within the basin. In 2012, the annual per capita income ranged from US\$ 370 in Niger to US\$ 1,430 in Nigeria; the weighted average was US\$ 1251 (PIDACC, African Development Bank, 2014). Of the 275 million inhabitants in the nine NBA countries, about 27.7 million are malnourished, which is 10% of the total population (FAO 2010 Statistics). Widespread poverty limits the resources with which communities can adapt to and recover from climate events. Economic reliance on rain fed farming and pastoralism mean that livelihood and food security is intimately linked with weather trends and environmental conditions. The average poverty rate in the Niger Basin is estimated at 55 % (2011). While there has been a substantial drop in poverty, there has been an increase in the number of poor, due to demographic expansion in the basin. The Basin's rural poor represent a significant population in the riparian countries.

Table 11: Prevalence of Poverty in the Niger basin

- In Mali, poverty declined from 55.4% (2001) to 48.4% (2011). By 2013, the poverty rate was estimated at 50.9%.
- In Chad, 47% people live below the national poverty line and 68% are considered vulnerable.⁴³
- In Cote d'Ivoire, the poverty rate declined to 28% (%1.90 a day poverty line, 2011 PPP) in 2015, down from an estimated 34% in 2011. The poverty rate using the national poverty line was down to 46%.^{Ibid}
- In Nigeria, extreme poverty was estimated at 53.4% (2009). Poverty is estimated to have reduced to 46.8% (2015).
- In Benin, estimates of the 2011 PPP US\$ 1.90 per day indicate poverty rates dropping from 61.4% to 49.6% (2006-2015).
- In Guinea, poverty remains widespread, with a poverty rate (national poverty line) stagnating at around 55% (2002-2012).
- In Niger, from 2005 and 2014, the incidence of poverty (based on poverty line of US\$ 1.90 per day) fell from 50.3% to 45.7%.
- In Burkina Faso, the percentage of people living below the poverty line declined from 52.7% in 2003 to 40.1% in 2014.
- In Cameroon, poverty is concentrated in the northern parts of the country. The number of poor in that region has more than doubled between 2001 and 2014 from 2.1 million to 4.5 million (56% of all poor in the country) in 2014.⁴⁴

Insecurity and fragility. The vulnerability of people in the Basin is exacerbated by political instability, notably recent conflicts in Mali and other parts of the Sahel, and the resurgence of Boko Haram and the resulting violence and lack of security. Security problems and displaced populations (Mali, Niger/Chad/Cameroon/Nigeria, etc.). The impacts of climate change and recurrent natural disasters also lead to mass migration of people in the region. In 2003 more than 25,000 refugees settled in the Fouta Djallon and Mount Nimba regions of Guinea. This led to increased degradation of the highlands, with rapid deforestation and associated land degradation.

Conflicts over shared natural resources use: Some of the biggest issues facing agro-pastoral systems that could undermine resilience building efforts in the Niger basin include farmer-herder conflicts and land tenure and rights⁴⁵. The pastoral systems are trans-boundary by nature, and pastoralists migrate across borders as a coping strategy against droughts and conflicts. Competition over natural resources, especially land and water, along with the erosion of traditional range management and tenure systems, have inflamed conflicts among pastoralists (like the Fulani) and between pastoralists and settled farming populations over the past three decades. Threats related to climate change and other forms of instability, make traditional transhumance routes less accessible, marketing corridors more dangerous, and livestock production more risky. Strengthening resilience means working on a continuum of actions that combine preventive interventions and social safety nets with development efforts (World Bank 2014).

Gender Inequality. The vulnerability to climate change and variability, food insecurity and food crises have a significant gender dimension, because women are among the most vulnerable groups, owing to their limited social and economic power in the Niger basin. Differentiated power relations between men and women and unequal access to and control over assets mean that men and women in the Niger basin do not have the same adaptive capacity. Instead, women have distinct vulnerability, exposure to risk, inadequate coping capacity, and inability to recover from climate change impacts (Masika 2002⁴⁶). Gender-based inequalities in access to livelihood assets, division of labour, and participation in decision-making processes result in women's and girls' increased vulnerability to the risks of natural disasters like droughts. Resilience is inconceivable without rural women. In the Niger basin countries, women represent only 8% of landowners and access only

⁴³ Reference: <http://povertydata.worldbank.org/poverty/country>

⁴⁴ "World Bank Group. 2016. Republic of Cameroon: Priorities for Ending Poverty and Boosting Shared Prosperity. Country Diagnostic

⁴⁵ Gray, Erin, Norbert Henninger, Chris Reij, Robert Winterbottom et al. 2016. Integrated Landscape Approaches for Africa's Drylands. World Bank Studies. Washington, DC: doi: 10.1596/978-1-4648-0826-5. License: Creative Commons Attribution; CC BY 3.0 IGO

⁴⁶ Masika, Rachel. 2002. "Gender and Climate Change." Gender and Development Journal 10 (2): 2–9.

10% of the credit available in West Africa, although they provide at least 50% of agricultural labour. Reducing inequalities based on gender and empowering women to participate more fully in social-economic growth are thus recognized as essential to reducing poverty and achieving development goals within the Niger Basin Authority (NBA) countries

E.4.2. Financial, economic, social and institutional needs

Describe how the project/programme addresses the following needs: • Economic and social development level of the country and the affected population; • Absence of alternative sources of financing; • Need for strengthening institutions and implementation capacity.

Strengthening the institutions and implementation capacity through the PIDACC program

The NBA and the NBA National Focal Points still lack the institutional depth to perform some important activities. Core staffing is thin relative to the size of the NBA's work program and insufficient to respond to the increasing and emerging demands placed on the institution (e.g. strategic planning, social development and gender or responding to basin management issues, such as climate risk management). Thus capacity strengthening will be undertaken at the regional, national and user level, following previous capacity assessments for African Development Bank financed projects. Activities for strengthening are indicated in the sections which follow.

(i) Strengthening the NBA Capacity

Strengthening commitment and core capacity at the regional level will have many other benefits, such as: increasing knowledge sharing; and strengthening institutional capacity, especially for the management of transboundary challenges like climate risk management. Activities will include the following:

- i) Strengthening the NBA's program coordination skills for enhanced M & E systems under the PIDACC;
- ii) Strengthening the NBA's communication and practices for sharing information, knowledge and experience, implementing its communication's strategy and reinforcing the NBA's Information Technology;
- iii) Strengthening the NBA capacity in regional hydro-meteorological and Early Warning Systems; and
- iv) Creation of innovative public-domain analytical toolkits, portals, and development of a strong working relationship with regional academia through support for internships and targeted research studies.

(ii) Strengthening national level capacity

The capacity development interventions at national level will comprise the following:

- i) Strengthening the NBA's National Focal Structures including additional skills, resources and staff trainings for coordination with National Implementing Agencies (NIAs) and supervision of programme activities;
- ii) Strengthening capacity through technical help desk services for the model user community (national level); as well as publication of a series of technical briefs to provide information and analysis on strategic water resource issues; and
- iii) National level-capacity building and on-the-job training of those involved in downscaling of RCOF forecasts to inform various sectors and communities as part of last mile early warning.

(iii) Strengthening capacity of local communities

Capacity development interventions will include the following:

- i) Strengthening communities in gender-responsive CSA technologies on-farm or at functional farmer training centres sensitizing 500,000 women on climate change adaptation techniques;
- ii) Programs addressing climate change impacts on specific activities (livestock management practices, etc.); and
- iii) Field visits and farmer –farmer exchanges as experiential learning opportunities.

E.5. Country Ownership

Beneficiary country (ies) ownership of, and capacity to implement, a funded project or programme

E.5.1. Existence of a national climate strategy and coherence with existing plans and policies, incl. NAMAs, NAPAs ,NAPs

Please describe how the project/ contributes to country's identified priorities for low-emission and climate-resilient development, and the degree to which the activity is supported by a country's enabling policy and institutional framework.

All Niger basin countries are parties to the United Nations Framework Convention on Climate Change (UNFCCC) and have signed and ratified the Kyoto Protocol. By ratifying the UNFCCC, the Niger Basin states have committed to the implementation of measures to adapt to climate change.

The PIDACC programme will contribute to the implementation of objectives of the three Rio conventions ratified by all NBA countries, including the Convention on Climate Change (UNFCCC), the Convention to Combat Desertification (UNCCD) and the Convention on Biological Diversity (CBD).

PIDACC interventions are aligned with operationalization of national commitments on climate mitigation and adaptation through NAPAs, the National Climate Change Policies and Strategies, NDCs, National Communications (NCs), SDGs, and the National Strategies for Disaster Risk Reduction. Niger Basin partner states NCs are also motivated through the lens of poverty alleviation and drought control, considering that many countries suffer from recurring drought. The proposed program is consistent with the NAPAs for 6 of the Niger Basin countries (Burkina Faso, Guinea, Niger, Mali, Chad and Benin). Cote D'Ivoire and Cameroon have not developed NAPAs. Nigeria is not classified as a least developed country, but produced a similar document in 2010 which it has called its National Adaptation Strategy and Plan of Action (NASPA).

Measures in the country national climate change commitments which are consistent with the PIDACC include: sustainable management of forests; forecasting and EWS; restoration of degraded land; fixation of dunes; promoting improved farming techniques; bush fire defence; preserving the quality and quantity of water resources, managing ecosystems, development of agro-sylvo-pastoral techniques; exploitation of fisheries; implementation of soil and water conservation techniques and rainwater harvesting and storage. A summary of the NBA's country commitments to the INDCs is shown in Table 12 below.

Table 12: Niger River Basin INDCs targets and Climate Adaptation Measures

Country	Reference Million tons CO2 eq.	Target (Business as Usual = BAU).	Agriculture, Forestry and Other Land Uses
Cameroon	40 M tons CO2 eq. (2010) Agric.: 62.5%	71 M Tons CO2 eq. (2035); Reduction of 32% (BAU, 2035)	46.5 M tonnes CO2 eq. (2035)
Côte d'Ivoire	15. 96 M tonnes CO2 eq. (2012) (Agriculture: 6.14 Mt CO2 eq.)	24.58 M Tons CO2 eq. Reduction of 28% (BAU, 2030)	4 .72 M tonnes CO2 eq. (2030)
Niger	0,031; (SCN), 2000)	Unconditional reduction 2.5% (BAU, 2020) and 3.5% (2030) - Conditional reduction: 25% (BAU, 2020) and 34.6% (2030, or reduction of 33,400 GgCO2 eq.).	89% of total GHG emissions
Benin	6.3 (2000); Agriculture : 68%	Conditional reduction 21.4 % (BAU 2021-2030)	9.5 (2030)
Guinea	15.12 (1994) Agriculture: 62%	-13% greenhouse gas (GHG) emissions in 2030, excluding land use and forestry	
Nigeria	2 Tons CO2 eq. per capita 214.21 (2000); 263 (2010)	2030 BAU: around 3.4 Tons CO2 eq. 2030 Conditional: around 2 Tons CO2 eq.	
Chad	8,380 Gg CO2 eq. (2010), incl. lands, forests, Agriculture: 18 448,00 Gg CO2 eq. (2010),	Unconditional reduction of 18.2% (BAU 2030, 23,449 GgCO2 eq.). Conditional reduction of 71%	30,399 Gg CO2 eq. (2030).
Burkina Faso	75,633 Gg CO2 eq. (2015)	BAU: 118,323 Gg CO2 eq. (2030); Conditional reduction: 13,766 Gg CO2 eq. (2030)	Reduction 7,236 Gg CO2 eq. (2030)
Mali	-192 M Tons CO2 eq. (2010) Agriculture: 48.5 (2010)	-85 M Tons CO2 eq.	Agriculture 59.5 (2030) Forestry ,Land use,-153 M Tons CO2 eq. (2030)

.....

E.5.2. Capacity of accredited entities and executing entities to deliver

Please describe experience and track record of the accredited entity and executing entities with respect to the activities that they are expected to undertake in the proposed project/programme.

The African Development Bank, as the GCF-AE, will oversee appropriate implementation of the programme in line with African Development Bank procedures and standards, and specific requirements in the AMA agreed with the GCF. The African Development Bank brings unique qualities to the shared effort to support the Niger Basin. The African Development Bank has supported implementation of climate resilient growth programs in the lake Chad Basin, the Nile Basin, and the Songwe River Basin. Others include the implementation of regional programmes to build resilience in Africa (the 2015-2020 P2RS Sahel Resilience Programme and the Horn of Africa Resilience Programme) Africa (Phase II and III). The African Development Bank plays a role in preparing projects, identifying partners with the right capacity and convening them to

provide solutions to development challenges as a complement to its own investments. Finally, the Bank's continued support is warranted as a strategy to manage and reduce risks to other elements of its programmes in Niger Basin RMCs.

The Project will be implemented by the NBA and the 9 partner states (Executing Entities - EEs). The EEs have significant experience in implementation of programs in the Niger Basin. However the EEs still lack the institutional depth to perform certain important activities. Thus, the programme will equip the EEs with robust skills needed to deliver the project outcomes effectively.

E.5.3. Engagement with NDAs, civil society organizations and other relevant stakeholders

Please provide a full description of the steps taken to ensure country ownership, including the engagement with NDAs on the funding proposal and the no-objection letter.

E.5.3.1 Endorsement by NDAs

Following consultations with all nine GCF NDAs in July 2017, PIDACC GCF was endorsed by all Niger River Basin partner states. Programme design gives a sharper focus on strengthening the climate resilience of the Basin's natural resources and ecosystems on the one hand, and strengthening the resilience of its population on the other hand.

Stages for preparation of the PIDACC

- i) Identification of actions on the basis of the Sustainable Development Action Plan (SDAP) (2012)
- ii) Validation of the proposed actions and areas of intervention of the programme (2014)
- iii) Niger River Basin Climate Risk Assessment. A Joint Initiative of the NBA and the World Bank (2014)
- iv) Investment Plan for the Strengthening of Resilience to Climate Change in the Niger Basin Nov— 2015
- v) Endorsement of NBA Ministers' Programme (2015)
- vi) Endorsement of the programme by the Summit of Heads of State 2016
- vii) Consultation with civil society stakeholders at the level of the programme and the intervention strategy 2016
- viii) Consultation meetings with stakeholders in each country within the framework of the feasibility study (2016)
- ix) Validation of the feasibility reports of the programme by the beneficiaries and NBA 2016 consultation meeting with GCF NDAs of Niger Basin countries in July 2017

On 1 June 2016, a donor roundtable was held in Abuja, Nigeria between representatives of NBA, Niger basin member countries and development partners, to mobilize financing for the PIDACC. During this meeting, 80% of the required funding was pledged, including US\$ 136 million from the African Development Bank, US\$ 12 million from GEF and US\$ 55 million from member countries.

E.5.3.2 Specific climate change strategies that informed project preparation

The PIDACC project is consistent with national commitments of the various member countries to the UNFCCC, including NAPAs, climate communications, INDCs, as well as the African Development Bank Group Second Climate Action Plan 2016-2020⁴⁷. Specific Climate resilient investment plans that informed interventions include (i) Niger River Basin Climate Risk Assessment— Niger River Basin Sustainable Development Action Plan; A Joint Initiative of the Niger Basin Authority (NBA) and the World Bank , 2014 and (ii) Investment Plan for the Strengthening of Resilience to Climate Change in the Niger Basin Final version 7.3 November 2015 (CRIP)

E.5.3.3 Engagement and consultations with civil society

Programme activities were identified in a participatory manner through consultations with communities, local groups, civil society and national services. Consultation for CSOs and NGOs from the Niger Basin on the CRIP and PIDACC programme was organized from 12 to 13 September 2015, which enabled CSO concerns to be taken into account before the presentation of the CRIP at COP21 in December 2015 in Paris. A follow up consultation was undertaken during appraisal with various stakeholders including the CSOs, NDAs and private sector, and a stakeholder engagement plan elaborated.

⁴⁷ file:///C:/Users/eolet/Desktop/AfDB-CCAP/AfricanDevelopmentBankClimateChangeActionPlan2016-2020.pdf

E.5.3.4 Multi stakeholder engagement and consultations

The proposal has been developed using a coordinated approach with engagement of multiple partners like the EU, the World Bank and KfW, to enhance synergy. National workshops for the validation of feasibility studies for the national components of PIDACC were organized in October 2016. Subsequently, a regional workshop was held from 7 to 9 November 2016 in Conakry, Guinea, which validated the technical and ESS studies of PIDACC at the regional level. The views and suggestions of all stakeholders were duly taken into account in the final design of the project, as well as some particularities specific to the context of each country. The implementation of the programme will be based on a participatory approach that will make communities the owners of financed Interventions (Activities). Stakeholder engagement plans will be prepared as part of the national level ESMFs, and will guide subsequent consultations and public participation. The ESMFs, will also include grievance and redress mechanisms, in line with the African Development Bank's integrated safeguards.

E.6. Efficiency and Effectiveness

Economic and, if appropriate, financial soundness of the project/programme

E.6.1. Cost-effectiveness and efficiency

Describe how the financial structure is adequate and reasonable in order to achieve the proposal's objectives, including addressing existing barriers; providing the least concessionality; without crowding out private and other public investment.

E 6.1.1. Financial structure

The proposed financing structure is adequate and comprises concessional loans and grants that will allow tackling both short and medium-term objectives related to climate resilient growth in the River Basin. GCF Grant proceeds will not be blended with the African Development Bank financing. GCF Grant proceeds will be directed to TA and to non-revenue generating interventions, while GCF loan proceeds will be directed towards implementation of community livelihood based infrastructure. Without GCF resources, the holistic approach proposed by the present project will not be possible.

E 6.1.2. Efficiency and effectiveness

Mainstreaming climate risk management into the PIDACC will enhance the effectiveness, and efficiency of planned interventions by reducing climate-related risks while contributing to sustainable growth. Planned interventions are efficient given that the value of economic benefits (i.e. economic benefits) are greater than the value of what society has to give up for their implementation (i.e. economic costs). Refer to section E 6.3 for details on the economic analysis.

E.6.2. Co-financing, leveraging and mobilized long-term investments (mitigation only)

Please provide the co-financing ratio (total amount of co-financing divided by the Fund's investment in the project/programme) and/or the potential to catalyse indirect/long-term low emission investment.

Please make a reference to E.6.5 (core indicator for the expected volume of finance to be leveraged).

Table 13: Showing financing by source

Source of funding	Amount Millions USD	% Contribution	Funding Instruments
African Development Bank (ADB)	35.924	17%	Loan
African Development Bank (ADF)	42.171	20%	Grant
Global Environment Fund (GEF)	12.980	6%	Grant
Green Climate Fund (GCF)	10.000	5%	Loan
Green Climate Fund (GCF)	57.774	28%	Grant
European Union (EU)	18.081	9%	Grant
Forest Investment Fund (FIP)	9.000	4%	Grant
Countries	16.186	8%	In-Kind
Beneficiaries	7.786	4%	In-kind
Total	209.903	100%	

With a total co-financing amount estimated at USD 142 million, the Co-financing ratio is estimated at 2: 1

E.6.3. Financial viability

Please specify the expected EIRR and FIRR with and without the Fund's support, based on the analysis conducted in F.1. The EIRR without the Considering Climate Change is estimated at 15.4%, while that with the funds support (consideration of CC is estimated at 18%. See table 14 below (details in section F1 and Annex IV).

Table 14: EIRR without and with consideration of Climate Change

Indicators for Financial Returns	IRR without consideration of CC	IRR with consideration of CC
Base Case	15.4%	18.0%

Financial viability in the long run beyond the Fund intervention.

Use of grants and loans will ensure long-term financial viability of PIDACC. Its scalability and contribution to long-term transformation at basin and country levels are important determinants for viability. Viability after financing ends is assumed to go beyond that of economic returns and encompasses socioeconomic metrics, including, improvements in the resilience of communities, and alignment with national priorities in the climate and economic growth policies of the Basin partner states [Please refer to section D.2. Exit Strategy for details on financial sustainability]

E.6.4. Application of best practices

Please explain how best available technologies and practices are considered and applied.

Investment actions will be proposed in terms of: (i) sub-basin conservation; (ii) development infrastructure; and (iii) capacity building. This will include industry best practices in community planning frameworks for addressing environmental issues.

New technologies for climate-smart agriculture have shown great potential and impact within Bank-financed programs and have potential for replication within this programme. These include farm ponds, micro irrigation, drought-resistant seeds, solar pumping, shade netting, and even a few cutting-edge concepts like use of remote sensed products to address monitor effects of improved watershed management. Use of remote sensing and geospatial data and information products is an important part of the information-sharing and capacity-building efforts promoted by the Bank under various programs. Many of the river basin investment projects undertaken have incorporated geospatial data gathering and built institutional capacity to manipulate geospatial data. The programme will enhance the NBA spatial knowledge base with the best available local, regional, and global datasets. This will in turn inform the spatial analysis on erosion modelling to help prioritize watersheds.

E.6.5. Key efficiency and effectiveness indicators

GCF core indicators	Estimated cost per t CO ₂ eq, defined as total investment cost / expected lifetime emission reductions (mitigation only)	
	(a) Total project financing (mitigation component)	30.97 million US\$ ⁴⁸
	(b) Requested GCF amount	10.39 million US\$ _____
	(c) Expected lifetime emission reductions overtime	7 million tCO ₂ eq
	(d) Estimated cost per tCO₂eq (d = a / c)	US\$ 4.4 / tCO₂eq
	(e) Estimated GCF cost per tCO₂eq removed (e = b / c)	US\$1.48 / tCO₂eq
	<i>Describe the detailed methodology used for calculating the indicators (d) and (e) above.</i>	
	Sequestration rate of 53 tonnes CO ₂ eq./ha; for assisted natural regeneration and for trees planted for the purpose of dune fixation It is 50 tonne CO ₂ eq./ha (FAO, AGRHYMET).	
Expected volume of finance to be leveraged by the proposed project/programme and as a result of the Fund's financing, disaggregated by public and private sources (mitigation only) N/A		
<i>Describe the detailed methodology used for calculating the indicators above. N/A: Please describe how the indicator values compare to the appropriate benchmarks established in a comparable context. N/A</i>		

⁴⁸ Output 2: Strengthened Sustainable forest management ; Output 3: Strengthened Sustainable agroforestry forest management ; Output 4: Strengthened sustainable protection of natural habitats

* F.1. ECONOMIC AND FINANCIAL ANALYSIS

The information can be drawn from the project/programme appraisal document.

F.1. Economic and Financial Analysis

Please provide the narrative and rationale for the detailed economic and financial analysis (including the financial model, taking into consideration the information provided in [section E.6.3](#)).

F.1.1. Programme cost-benefit analysis

The detailed economic analysis is presented in Annex IV.

The 9 NBA countries have variable inter-country and intra-country agro-climatic characteristics. Thus, economic conditions vary between these different agro-climatic zones. To this end, the economic analysis made is specific to each of these countries. Since PIDACC actions will be spread over the entire national portion of the Niger Basin in each country, the economic analysis is of interest to the country's entire basin. While cost-benefit analysis provides a useful process and resultant metrics to help steer investment decision-making, it is not the only factor considered. The results of the economic and financial analysis of the programme and country components are presented in Annex VIII. The EIRRs for the country components range from 15 to 26% (Table 15). The overall IRR is estimated at 18% for the base case with Climate Change considerations.

Table 15: IRR By Country

Region / Country	Global	Burkina Faso	Mali	Niger	Chad	Benin	Cameroon	Côte d'Ivoire	Guinea	Nigeria
ERR	18%	16%	18%	15%	15%	15%	18%	16%	15%	26%

Table 16: Sensitivity Analysis

Indicators for Financial Returns			IRR without consideration of CC	IRR with consideration of CC
Base Case			15.4%	18.0%
Sensitivity test	Costs	-20%	19.6%	22.5%
		-10%	17.4%	20.0%
		10%	13.7%	16.2%
		20%	12.3%	14.6%
	Benefits	-20%	11.6%	13.9%
		-10%	13.6%	16.0%
		10%	17.2%	19.8%
		20%	18.8%	21.6%

The programme implementation period is estimated at five years, while the project impact duration is estimated at 30 years. This is based on the premise that planned climate and weather information systems would have an average lifespan of 7-10 years, while natural resources investments would have a much longer life-span, in the range of 30-40 years. O&M costs are assumed at between 1-5% of project investments. For the case with climate change; the approach was to add avoided costs to the base case benefits, without climate change. The benefits of taking action (the avoided costs of damages) are further in the future. The estimated benefits are then compared to the costs of taking action.

A discount rate of 10% is used to calculate the NPV. A 10% discount rate represents an understanding that future costs and benefits are relatively important in comparison to the current situation – concurrent with concerns regarding climate change. The Internal Rate of Return (ERR) was calculated to be 18% for the base case with avoided costs due to Climate Change considered. The NPV was calculated to be US\$ 250 million, with a discount rate of 10%. This represents a benefit-cost ratio of 1.6. Since it is impossible to accurately predict future per capita consumption growth, a sensitivity analysis was conducted to illustrate the benefits of the project at varying rates of return. At a discount rate of 12%, NPV equals US\$ 150 million. This demonstrates project viability even with higher discount rates.

F.2. Technical Evaluation

Please provide an assessment from the technical perspective. If a particular technological solution has been chosen, describe why it is the most appropriate for this project/programme. N/A

F.3. Environmental, Social Assessment, including Gender Considerations

Describe the main outcome of the environment and social impact assessment. Also describe how the gender aspects are considered in accordance with the Fund's Gender Policy and Action Plan.

F.3.1. Environment and Social Assessment

PIDACC is classified in category 1 according to the African Development Bank's Integrated Safeguard System (ISS) because of its potential negative impacts on the human environment and on the ecosystems of the Niger River. A Strategic Environmental and Social Assessment (SESA) was prepared and comprised a systematic examination of environmental and social risks and impacts, and issues, associated with the planned PIDACC at Basin level. An environment and social management framework [ESMF] was also prepared. The SESA and ESMF can be accessed using the link <https://www.afdb.org/en/documents/document/niger-strategic-environmental-and-social-studies-of-the-integrated-program-for-development-and-adaptation-to-climate-change-in-the-niger-basin-PIDACC-bn-100875/>. A summary SESA and ESMF can be accessed through the following link: <https://www.afdb.org/en/documents/document/abn-multinational-integrated-development-and-climate-change-adaptation-in-the-niger-basin-program-PIDACC-nb-97280/>

The PIDACC ESMF seeks to: (i) enhance positive and sustainable environmental and social outcomes associated with project preparation and implementation; integrate environmental and social aspects associated with the numerous projects into the decision making process; (ii) minimize environmental degradation as a result of either proposed individual projects or their cumulative effects; and (iii) minimize impacts on ecosystems. The ESMF provides guidance and measures with clear roles and responsibilities, along with capacity strengthening measures for effective implementation and monitoring. Table 17 shows Bank Safeguard Policies Triggered by the Project.

Table 17: Bank Safeguard Policies Triggered by the Project

Bank Safeguard Policies Triggered by the Project	Yes	No
<i>Operational safeguard 1 – Environmental and social assessment</i> —Applicable because the program will support investments in the watersheds. Since the CDD-type activities and exact locations have not been identified ex-ante, the safeguard instrument used in each country is the ESMF.	X	
<i>Operational safeguard 2 – Involuntary resettlement: land acquisition, population displacement and compensation</i> — Not Applicable because the project will not limit access to land and forest resources by the local communities. Refer to the African Development Bank ISS, Guidance Materials, and Volume 2: Guidance on Safeguard Issues ⁴⁹ , in particular Involuntary Resettlement and Operational Safeguard (OS) 2.		X
<i>Operational safeguard 3 – Biodiversity, renewable resources and ecosystem services</i> —Not Applicable because PIDACC will not implement activities in protected or transboundary areas. So the project doesn't trigger OS.3. The ESMFs will address issues pertaining to this policy.		X
<i>Operational safeguard 4 – Pollution prevention and control, hazardous materials and resource efficiency</i> —Applicable. The PIDACC will involve, preparing a pollution control plan.	X	
<i>Operational safeguard 5 – Labor conditions, health and safety</i> —Not Applicable S5 is not triggered by PIDACC because it will finance only small infrastructure to be built by communities and SMEs. Security measures will be included the SMEs contracts.		X

Formulation of country level ESMFs

All participating Niger Basin countries shall formulate their own ESMFs that are consistent with the African Development Bank's Integrated Safeguards System (ISS). Section 9 of the SESA includes the Project ESMF. The ESMF outlines (i) Measures for the Mitigation of Negative Impacts (ii) Environment and Social Procedures for Selection of Interventions (Activities) (iii) Land Use Restrictions and Involuntary Resettlement (iv) Stakeholder Engagement Plan (SEP) (v) Grievance and Redress Mechanism(vi) Environmental Monitoring and Surveillance Program(vii) Capacity Building Program and (viii) the Implementation And Monitoring Schedule. Country ESMFs shall be reviewed and approved by the African Development Bank to ensure that they are consistent with CGF's environmental and social performance standards. Specific attributes of Operational safeguard 1 – Environmental and social assessment are shown in table 18.

⁴⁹ [https://www.afdb.org/fileadmin/uploads/African Development Bank/Documents/Publications/SSS_-_IESIA_Volume_2_-En.pdf](https://www.afdb.org/fileadmin/uploads/African%20Development%20Bank/Documents/Publications/SSS_-_IESIA_Volume_2_-En.pdf)

Table 18: Specific ESMF procedural measures applicable to the PIDACC

Measure	Description
Activity Screening Protocol	A screening protocol exists in the form of the African Development Bank ISS—Guidance Materials Volume 3: Sector Key sheets (2014), checklists, which guide the borrowers /grantees in, among others, identifying eligible projects and assessing social and environmental risks and the associated required assessments studies and requirements shall be developed (SESA Section 9.2, under Stage 1: Identification, environmental and social selection and Intervention (Activity) classification).
Stakeholder Engagement Plan to guide grantees in the development of their respective country level ESMFs	In consultation with the Bank, the NBA (regional) and National Governments in the Niger basin, will develop and implement a Stakeholder Engagement Plans (SEP) commensurate to the nature and scale of the interventions (activities) and potential risks and impacts (SESA section 9.2). Reference will be made to the African Development Bank ISS, Guidance Materials, and Volume 2: Guidance on Safeguard Issues in the drafting of the SEPs as part of the ESMFs. An indicative generic Stakeholder engagement Matrix is included as well, that will guide the countries during the drafting process.
Project-level and country level grievance redress mechanism	Grievance And Redress Mechanism, provides relevant guidance and references (SESA Section 9.5). A grievance and redress mechanism will be included as part of the National project ESMFs, compliant with the national safeguards and the African Development Bank ISS (Operational Safeguard 1: Environmental and Social Assessment). Such grievance mechanisms will utilize existing formal or informal grievance mechanisms suitable for project purposes, supplemented with project-specific arrangements designed to resolve disputes in an impartial manner.

Supervision of the ESMF Implementation

The project teams at the NBA EE and the National Executing Entities [EE] will ensure the provision of technical support to develop and improve understanding of the GCF's safeguard policies and the effective implementation of the African Development Bank Group's Integrated Safeguards. African Development Bank supervision teams will include environmental and social safeguard experts. Regular monitoring reports (twice per year) on the implementation of environmental and social safeguards provisions will be provided to the African Development Bank for approval. These reports will be verified during project supervision missions, which will include environmental and social safeguard experts. At the national level, the national EEs will be in charge of the monitoring and evaluation of safeguards.

F.3.2. Gender considerations

The status of women in the Niger Basin has remained poor, with little progress evident on gender indicators over the years. Gender dynamics vary across the area based on ethnicity, livelihood systems and socio-economic status.⁵⁰ In the Niger basin, large gender disparities persist. Women and girls are disadvantaged in many areas and don't enjoy the same opportunities as their male counterparts. This is true for almost all public sectors, ranging from unequal access to basic social services, unequal property rights and persistent gender gaps in the labour market and in the public sphere.

Some of the highest rates of maternal mortality in the world are observed in the Sahel region, for example in countries including Niger (590 per 100,000), Nigeria (630), and Chad (1,100). As a consequence, women are exposed to climate risk. The social structure of Sahelian communities has changed considerably over the years. This change is chiefly manifested in the redefinition of roles and responsibilities within households. The 2012 food crisis has revealed that women have taken on increasing responsibilities from a number of standpoints. Their contribution to maintaining their households in times of crisis is tacitly expected, even though this increased responsibility within the household is not always reflected in increased power within the community. Indeed, because they have only a subsidiary role in relation to household responsibilities and are relegated to the side-lines when it comes to community-based management in general, women continue to have limited access to factors of production.

Climate extremes particularly impact women in the Basin. As a result of droughts women and girls face a triple burden in some cases: to survive, care for their families and evade sexual violence in the process. Droughts in the Niger basin also frequently result in the degradation of water sources. Decreased water resources also causes women's health to suffer as a result of the increased work burden and reduced nutritional status. This decreases the time available for food production and preparation, with consequences for household food security and nutritional well-being. Investments are required in interventions that reduce the risks (such as losing access to land) that present formidable barriers to women wanting to adopt new technologies and practices and actively participate in markets.

Since the mid-2000s, however, almost every basin country has adopted a gender policy or strategy and is increasingly mainstreaming gender issues in different policy sectors⁵¹. The weak implementation of gender-focused policies is, however, not just a funding problem. There are other obstacles and risk factors: strong individual and institutional resistance to gender initiatives, deep-rooted cultural issues and traditions, general under-representation of women in the public sphere, illiteracy. In practice, gender is still considered mostly as an afterthought and gender policies are often not implemented effectively. Budgets allocated to gender-specific issues within sectoral policies remain tiny; disaggregated gender data is missing etc.⁵².

Interventions are included as part of the gender action plan GAP in Annex VIII. As part of the Gender Action Plan (GAP), the project will promote women's representation on community level action committees and involve women in planning and undertaking adaptation activities. The project will also Increase women's access to agricultural extension and land resources within the context of climate smart agriculture. The project will also promote gender-sensitive budgeting in local administrations.

Implementation and supervision of the Gender Action Plan (GAP)

The program GAP, will be implemented at a regional level by the NBA and at a national level by the national coordination units in each country. A project baseline will be established in Year 1, (i) that allows for monitoring changes in activities, perceptions and attitudes that relate to women's participation (ii) that allows for monitoring distributional issues within the household (iii) that allows for monitoring changes in women's time budgets and labour burden and (iv) that allows capture of women's access to land for farming / gardening.

The project will monitor gender actions through a GAP, including the number of beneficiaries (gender disaggregated) as well as gender sensitive project output, outcome and impact indicators. Supervision arrangements will comprise: (i) Safeguard specialists (Social and gender at the African Development Bank for implementation support supervision)—AE; (ii) Social Development and gender specialists at the regional level by the NBA—EE and (iii) Social Development and gender specialists at the NPCU who will cover community development and gender aspects as well.

The specialists will be responsible for incorporating the Gender Action Plan into project planning and implementation including supporting/assisting design (and if required, implementation) of awareness workshops and training, establishment and monitoring of sex disaggregated data and indicators for project performance and monitoring. Engaging with different gender groups and grassroots organizations from the beginning will help the identification and implementation of gender-equitable activities.

A number of measures will be put in place at the start to help guide implementation of gender equitable activities, particularly when collecting gender-disaggregated data. These include: (i) Ensuring there are sufficient funds to support gender-related activities and facilitate the participation of traditionally marginalised communities members (ii) Establishing mixed-gender field teams (iii) Scheduling activities at times and places that are convenient for different gender groups (iv) Making sure that field coordinators, facilitators and translators communicate with women and men participants in a language in which they feel comfortable (v) Managing group dynamics and promoting the equal participation of all participants and (vi) After discussions with women's and men's groups separately, bringing groups together to exchange and learn from people's different perspectives.

The PIDACC RPCU will report on progress of Gender Action Plan activities in quarterly progress reports to the African Development Bank and the government. Other training providers, non-government organizations may be hired to implement different Gender Action Plan activities under the guidance of the specialists.

⁵⁰ Towards Climate Resilient Agricultural and Pastoral Production Systems; A synopsis of programme design considerations under constraints of select natural resources, capacity and climate in Burkina Faso, Mali and Niger Monica L. Wrobel, Tony Hill, Ellen A. Bean, Mark Mulligan, Sophia Burke, Josh Allen, Felicity Roos and Angie Dazé (© CARE International UK, RBM, SNV, TREE AID, August 2014)

⁵¹ <http://www.oecd.org/swac/topics/gender/>

⁵² <http://oecdinsights.org/2017/03/08/gender-equality-in-west-africa/>

F.4. Financial Management and Procurement

Describe the project/program's financial management and procurement, including financial accounting, disbursement methods and auditing.

Details on grant and loan agreements as well as Funds Flow arrangements are outlined in section C 7.4 and figure 17.

Due Diligence

The African Development Bank, as the GCF-accredited entity, will oversee implementation of the programme in line with the African Development Bank procedures, standards and requirements in the AMA/FAA to be agreed with the GCF. The AfDB team responsible for project origination will carry out due diligence and assessments of financing proposals for each sub-project. The origination team's findings and recommendations undergo a rigorous internal review process before they are cleared by Senior Management to be presented to the Bank's Board of Directors for approval. This includes various interdepartmental committee reviews.

The AfDB will, through its Anti-Corruption and Integrity department, provide Integrity Due Diligence (IDD) for the project operations through a structured, systematic analysis to identify, assess, mitigate, manage and monitor potential loss from integrity risks and riskier exposure. This is to ensure that funds are used for their intended purposes and with due attention to considerations of economy, efficiency and competitive trade. The Bank will use the following assessment criteria to safeguard its investment. (See also the Bank's IDD Policy for further information.)

- Identification of beneficial ownership
- Assessment of civil and regulatory backgrounds
- Identification of sanctioned persons and entities
- Identification of Politically Exposed Persons (PEPs) and other high risk relationship

Corporate financial transactions are closely monitored through the MDB harmonized treatments of corporate groups and also through the AfDB's established guidelines on anti-fraud, anti-corruption and anti-money laundering policies (AMLCFT). The AfDB continues to ensure that its financing operations and investments are not used for illegal or tax-evasion purposes. In addition, measures are currently in place to address the issue of tax havens and the accompanying risks. The AfDB will lead standard KYC due diligence process including anti-money laundering and other evaluations of sponsors.

F.4.1 Financial management

The NBA PRCU and the national coordinators will oversee the project financial management aspects, including the preparation of the financial statements and quarterly Interim Unaudited Financial Reports, monitoring financial transactions and making the necessary arrangements for the annual project financial audits. The PRCU is coordinating the implementation of the GEF-financed Niger Basin Integrated Development for Increased Rural Climate Resilience Programme (Guinea, Ivory Coast, Mali, Burkina Faso, Niger, Benin, Chad, Cameroun, Nigeria). Please refer to section C.7.4. Agreements and Funds flow for the grants and loans, for the Funds flow arrangements.

Financial and accounting management will be based on the principles and rules of the OHADA accounting system and will operate on a financial accounting management software comprising at least the budgetary, analytical and general accounting modules. The principles and rules of the OHADA accounting system are compliant with the Bank's policies and international accounting standards on financial reporting. A Project Implementation Manual will be developed to guide the budgeting, accounting and financial procedures.

F.4.2. Disbursements

In line with African Development Bank disbursement handbook⁵³, the NBA and the partner states will be required to open special Accounts denominated in US\$ in a commercial Bank acceptable to the African Development Bank. Evidence to this effect will be submitted by the Executing Entities [EEs] ahead of project disbursement together with the signatories to the accounts in accordance with the Financing Agreements.

⁵³ <https://www.afdb.org/en/documents/document/disbursement-handbook-54419/>

Disbursement of GCF funds will be transaction-based in line with the African Development Bank disbursement handbook. Disbursements will be made through: reimbursement, direct payment and working capital. The NBA and National EEs will open accounts in national banks acceptable to the Bank. All supporting documents will be provided for Statement of Expenditures for all transactions submitted for African Development Bank prior review. The others will be retained at the PRCU and National EEs and be made available for periodic reviews by the African Development Bank missions and external auditors.

The PRCU will produce Interim Unaudited Financial Reports on a quarterly basis together with the quarterly progress reports to be submitted to the African Development Bank and the annual financial statements of the Programme for financial and accounting audit. The Interim Financial Reports will include sources and uses of funds by project expenditures classification. They will also include a comparison of budgeted and actual project expenditures (commitment and disbursement) to date and for the quarter. The PRCU and National EEs will submit the Interim Financial Reports to the Bank within 45 days following the end the calendar quarter.

The annual financial statements will comprise a balance sheet, a statement of employment and resources showing annual and cumulative data, notes to the financial statements, physical inventory of acquired assets and reconciliation of the Bank's disbursements with the amounts received from the Programme, and the reconciliation statements of the special account.

F.4.3. Audit

Audits will be undertaken in line with the African Development Bank Guidelines for Financial Reporting and Auditing of Projects⁵⁴. The Financial Agreement with the NBA and the National EEs, will require the submission of Audited Financial Statements to the African Development Bank within six months after each year-end. An independent external auditor will be recruited based on Terms of Reference acceptable to the Bank (not later than four months after effectiveness) for the entire duration of the project. The Financial Statements will be audited in accordance with international auditing standards. The external auditor will prepare a Management Letter to provide observations, comments, and recommendations for improvements in accounting records, systems, controls and compliance with financial covenants in the Financial Agreements. The cost of the audit will be met from project resources and paid by the Bank after validation of the annual reports. In line with Clause 16.01 of the AMA, the African Development Bank will make available to the GCF, copies of audited financial statements and/or, annual audited reports of the Funded Activities by the Bank from the Executing Entity pursuant to the relevant Subsidiary Agreement, which shall not be later than six months after the end of the Executing Entity's fiscal year.

F.4.4. Procurement

Procurement of goods, works and consulting services, financed under the programme, will be in accordance with the "Procurement Policy for Bank Group Financed Operations" (2016).⁵⁵ The procurement plan will be prepared at project appraisal. Procurement by International Competitive Bidding (ICB) and procurement of consultant services under the Programme shall be in accordance with the Bank's Rules and Procedures for the Procurement of Goods and Works or, the Bank's Rules and Procedures for the use of the Consultants, using the Bank's appropriate standard bidding documents. Procurement by National Competitive Bidding (NCB) of works and goods will be carried out in accordance with the national procurement procedures of each of the nine countries and the provisions of the letters of agreement relating to the application of national procurement procedures. To strengthen the NBA EE and National EEs capacity on the African Development Bank procedures, a procurement training program will be undertaken at programme launch.

⁵⁴ https://www.afdb.org/fileadmin/uploads/African_Development_Bank/Documents/Procurement/Project-related-Procurement/GUIDELINES%20FOR%20FINANCIAL%20REPORTING%20AND%20AUDITING%20OF%20PROJECTS.pdf

⁵⁵ https://www.afdb.org/fileadmin/uploads/African_Development_Bank/Documents/Policy-Documents/Procurement_policy_for_bank_group_funded_operations.pdf

G.1. Risk Assessment Summary

Please provide a summary of main risk factors. Detailed description of risk factors and mitigation measures can be elaborated in G.2.

The main risks that may affect the implementation of the programme are:

- Some countries in the Basin have been confronted with socio-political and security crises in recent years. This may hinder or delay the implementation of programme activities.
- The success of the programme is also conditioned on the effective mobilization and support of the beneficiary populations who are the main stakeholders of the programme.
- Land conflicts could hamper the implementation of certain activities, such as watershed treatment.
- The programme targets the development of agro-pastoral activities. It can be a source of conflict between farmers and livestock breeders regarding the delimitation and occupation of land.
- The coordination of large geographical areas and the diverse nature of the target audience require a concentration site approach to avoid under-powdering.
- The project may encounter the challenges of international waters and competing national agendas.

G.2. Risk Factors and Mitigation Measures

G.2. Risk Factors and Mitigation Measures

Please describe financial, technical and operational, social and environmental and other risks that might prevent the project/programme objectives from being achieved. Also describe the proposed risk mitigation measures.

Selected Risk Factor 1

Description	Risk category	Level of impact	Probability of risk occurring
Insecurity	Social and environmental	High (>20% of project value)	High

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

Countries facing conflicts are Mali, Niger, Nigeria, Chad and Cameroon. The sites selected for the programme are secured sites in the countries concerned. However, each country will propose a plan of action for the safety of the intervention sites that will be taken into account during the evaluation mission.

Selected Risk Factor 2

Description	Risk category	Level of impact	Probability of risk occurring
Effective mobilization and adherence of the beneficiary populations	Social and environmental	High (>20% of project value)	Medium

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

The implementation of a participatory and inclusive approach will facilitate this social mobilization. Recovery of degraded land and intensive development of fertile land is an important motivation.

Selected Risk Factor 3

Description	Risk category	Level of impact	Probability of risk occurring
Land conflicts	Social and environmental	High (>20% of project value)	High

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

In order to minimize the risks of land conflicts, the site for each of the intervention (activity) will require un-encumbered access for project activities. This will require that a certificate of land security be attached to each intervention (activity) document. The PIDACC program will provide support to local land commissions in each of the countries to facilitate access to land certificates. It should be noted that the project does not provide for the acquisition of land or change of land status. Awareness-raising campaigns will help to smooth out any potential land conflict before the start of programme activities. Support will be provided to local land commissions. Awareness-raising actions will be implemented in relation to existing coordination mechanisms.

Selected Risk Factor 4

Description	Risk category	Level of impact	Probability of risk occurring
Coordination of wide geographical areas and the diverse nature of the target public	Technical and operational	Medium (5.1-20% of project value)	Medium

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

Accountability of national coordination for the implementation of the investments of each country-component.

Selected Risk Factor 5

Description	Risk category	Level of impact	Probability of risk occurring
Insufficient synergy of actions between programme partners	Technical and operational	Medium (5.1-20% of project value)	Medium

Mitigation Measure(s)

Establishment of a unique coordination for the programme and organization of joint partners' missions. A mechanism exists at regional level for donor coordination and country coordination through the NBA PIDACC Regional Steering Committee. The NBA strategy and individual country strategies for coordination will be applied to the project.

Selected Risk Factor 6

Description	Risk category	Level of impact	Probability of risk occurring
General project complexity including the challenges of international waters and competing national agendas	<i>Technical and operational</i>	<i>Medium (5.1-20% of project value)</i>	<i>Medium</i>

Mitigation Measure(s)

To ensure successful implementation, the following measures have been taken: (i) implementing NBA's communication strategy to foster common understanding, collaboration and coordination among riparian countries; (ii) identifying activities and beneficiaries at national and local levels to ensure key stakeholders and vulnerable groups are positively impacted by Basin Program activities; and (iii) clarifying roles and responsibilities of each agency in terms of financial management and procurement to avoid delays and increase accountability.

* Please expand this sub-section when needed to address all potential material and relevant risks.

H.1. Logical Framework.

Please specify the logic framework in accordance with the GCF's Performance Measurement Framework under the Results Management Framework.

H.1.1. Paradigm Shift Objectives and Impacts at the Fund level ⁵⁶						
Paradigm shift objectives						
The PDO is to contribute to improving the resilience of populations and ecosystems in the Niger River Basin to Climate variability and change through sustainable management and development of shared basin natural resources	This proposal addresses existing climate risks in the Niger Basin but also implements early actions that begin preparing for future climate change in key fast expanding economic sectors such as agriculture, energy and construction. GCF Finance will also strengthen institutional and regulatory systems for climate-responsive planning and development in the Niger Basin through supporting climate mitigation and adaptation measures in the NDCs of the Niger River Basin partner states. The project has a high potential to cost effectively scale-up, replicate and sustain the results of project interventions beyond project completion. The project includes a strong focus on best practices, while the technologies promoted by the programme are financially viable, utilize local natural resources, are transferable, and will be delivered through community-based approaches. There is strong potential for these activities to continue beyond the end of the project. The Payment for Ecosystem Services (PES) and adaptation mechanism to be created with financing from the GCF will further enhance continuity of project benefits beyond project closure.					
Expected Result	Indicator * = Core	Means of Verification (MoV)	Baseline ⁵⁷	Target		Assumptions
				Mid-term	Final	
Fund-level Impacts						
	☐ *Tonnes of carbon dioxide equivalent (t CO2eq) reduced as a result of Fund funded projects/ programmes	NBA reports, national GHG emissions surveys, Lima REDD+ Information Hub	Baseline to be established in year 1	4.2 MMtCO2e ⁵⁸	7 MM TCO2e	
*	☐ *Total Number of direct and indirect beneficiaries; Number of beneficiaries relative to total population	NBA observatory Reports and Surveys	0	2 million people directly and 5 million indirectly (~50% female).	4 million people directly and 10 million indirectly (~50% female).	
M 4.0 Reduced emissions from land use, deforestation, forest degradation, and through sustainable management of forests and	☐ 4.1 Tonnes of carbon dioxide equivalent (t CO2eq) reduced or avoided (including increased removals) as a result of Fund-funded projects/programmes	NBA reports, national GHG emissions surveys, Lima, REDD+ Information Hub	Baseline to be established in year 1	4.2 MMtCO2e	7 MM TCO2e	Niger Basin governments and private landowners make land available for project interventions. High uptake of improved practices.

⁵⁶ Information on the Fund's expected results and indicators can be found in its Performance Measurement Frameworks available at the following link (Please note that some indicators are under refinement): http://www.gcfund.org/fileadmin/00_customer/documents/Operations/5.3_Initial_PMF.pdf

⁵⁷ Baseline is to be confirmed by socio-economic baseline survey at appraisal and before project effectiveness

⁵⁸ Estimates based on annual reduction/avoided emissions of 1.4 MMtCO₂e, which comes to 4.2 MMtCO₂e by mid-term/year

conservation and enhancement of forest carbon stocks	☐ 4.3 Number of technologies and innovative solutions transferred or licensed to support low-emission development as a result of Fund support.	NBA Observatory Reports	Baseline to be established in year 1	4	9 ⁵⁹	
A1.0 Increased resilience and enhanced livelihoods of the most vulnerable people, communities, and regions	☐ 1.2 Number of males and females benefiting from the adoption of diversified, climate resilient livelihood options (including fisheries, agriculture, tourism, etc.)	NBA Observatory Reports	400,000 (10% women)	1,600,000 (50% women)	4,000,000 (50% women)	Vulnerable groups and women are effectively targeted and are enabled to participate in the project.
A 2.0 Increased resilience of health and wellbeing, and food and water security	☐ 2.2 Number of food-secure households (in areas /periods at risk of climate change impacts)	NBA Observatory Reports National reports FAO Reports	Food Insecurity Experience Scale (26% of population) ^{60,61}	Food Insecurity Experience Scale (23% of population)	Food Insecurity Experience Scale (20% of population)	
A4.0 Improved resilience of ecosystems and ecosystem services	☐ 4.1 Coverage /scale of ecosystems protected and strengthened in response to climate variability and change ⁶²	Project reports	5,400 ha	50,000ha	100,000 ha	

H.1.2. Outcomes, Outputs, Activities and Inputs at Project/Programme level

H.1.2. Outcomes, Outputs, Activities and Inputs at Project/Programme level						
Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
Project/programme outcomes	Outcomes that contribute to Fund-level impacts					
7.0 Strengthened adaptive capacity and reduced exposure to climate risks	☐ 7.2: Number of males and females reached by [or total geographic coverage of] climate related early warning	Project reports	80,000	200,000 (50% women)	500,000 (63% women)	

⁵⁹ CSA practices and technologies to be introduced within the PIDACC include: Early Warning systems; Agroforestry practices; soil and water adaptation technologies (reduced tillage, permanent crop cover); Supplemental irrigation/water harvesting; Irrigation techniques to maximize water use; Modification of cropping calendar; Cage aquaculture using natural water bodies; weather-based agro-advisories and Use of drought resistant crop varieties

⁶⁰ FAO. 2017. Regional Overview of Food Security and Nutrition in Africa 2016. The challenges of building resilience to shocks and stresses. Accra.

⁶¹ The prevalence of moderate or severe food insecurity [FAO, 2017] is measured based on the Food Insecurity Experience Scale (FIES). The FIES generates estimates of food insecurity at the individual level, based on analysis of self-reported occurrence of experiences during a period of 12 months

⁶² Indicator captures changes in agricultural land (ha) as well as forest land (ha) and protected areas (ha) in targeted areas.

⁶³ 62500, Households (average household size estimated at 8 people per household)

H.1.2. Outcomes, Outputs, Activities and Inputs at Project/Programme level						
Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
	systems and other risk reduction measures established/ strengthened					
8.0 Strengthened awareness of climate threats and risk-reduction processes	□□8.1: Number of males and females made aware of climate threats and related appropriate responses	Project reports climate service providers surveys	80,000	800,000 (50% women)	2,000,000 (50% women)	
9.0 Improved management of land or forest areas contributing to emissions reductions	□□9.1 Hectares of land or forests under improved and effective management that contributes to CO2 emission reductions	Project reports	9,000 ha	12,000ha — forestry 12,000 ha — agroforestry 4,000 ha of dunes stabilized; 40,000 ha of degraded land restored Mechanical and biological management of 6,000 m3 of ravines undertaken	40,000ha — forestry 26,000 ha — agroforestry 10,000 ha of dunes stabilized; 110,000 ha of degraded land restored; Mechanical and biological management of 45,000 (m3) of ravines undertaken	Communities supportive and adopt improved practices
Project/programme outputs	Outputs that contribute to Fund-level impacts					
Component 1-1						
Output 1: Strengthened sustainable land management (SLM)	1.1.No of Hectares (ha) of degraded area restored under participatory Sustainable Land Management	Project reports		40,000	100,000	
	1.2.No of new green jobs created for vulnerable groups	NBA observatory Reports and Surveys	None	80,000 (30% women)	150,000 (30% women)	
	1.3.No of farmers in project areas using climate resilient technologies (disaggregated by gender)	IP reports, project reports	Baseline to be established in year 1	40,000 (30% women)	100,000 (30% women)	
Output 2: Strengthened sustainable forest management	2.1 No of Hectares (ha) of forest area under participatory management	Project reports	3,000ha	10,000	40,000	
	2.2 % increase in yield productivity ⁶⁴	Project reports	Baseline to be established in year 1	10%	20%	

⁶⁴ Forestry yield productivity measured in terms number of trees per hectare

H.1.2. Outcomes, Outputs, Activities and Inputs at Project/Programme level						
Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
	2.3 No of community agents that are using REDD+ (No)	Project reports	12	100 (50% women)	400 (50% women)	
Output 3: Strengthened sustainable agroforestry management	3.1 No of Hectares (ha) of land area covered by agro-forestry	Project reports	1,000ha	10,000	26,000	
	3.2 No of farmers with skills in sustainable natural resources management (agroforestry, forestry, etc)	Project reports	Baseline to be established in year 1	800,000 (50% women)	2000,000 (50% women)	
	3.3 Number of farmers with additional incomes	Project reports	Baseline to be established in year 1	800,000 (50% women)	2000,000 (50% women)	
Output 4 Strengthened sustainable protection of natural habitats	4.1 No of Hectares (ha) of natural habitat improved/conserved	Project reports	2,000	2000	4,200	
	4.2 No of Hectares (ha) of spawning grounds rehabilitated (ha)	Project reports	2,000	2,000	4,700	
Component 1-2						
Output 5 Strengthened climate and weather information services including Last Mile EWS	5.1 Hydromet stations with accessible data shared in real time	Project reports		53	53	
	5.2 No of people with skills on climate and weather information (disaggregated by gender by sex and age group)	Project reports	0	1000 (30% women)	2000 (30% women)	
Output 6: Regional pollution prevention plan developed	6.1 Operational Pollution control plan at basin level	Project reports	Existing pollution study	1	1	Surveys are successfully conducted.
Output 7: Payment mechanism for environmental services (PES) and FRACC Fund set up	7.1 Operational RAFCC Fund and PES mechanisms	Project reports Final study reports	Decision of heads of state to establish RAFCC	1	1	Required data is available & surveys are successfully conducted.
Output 8: Strengthened Capacity for IWRM implementation at national/ regional (NBA) levels	8.1 No of basin/sub basin IWRM plans developed and operationalised	Project reports Final study reports	0	9	9	
	8.2 Number of NBA, national focal structures and associations' executives trained	Project reports	0	250	500	
Output 9: Master Plan for the river navigability developed	9.1 Operational Masterplan for river navigability in place on the Benoué river	Final study reports		1	1	Required data is made available

H.1.2. Outcomes, Outputs, Activities and Inputs at Project/Programme level						
Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
	(Nigeria and Cameroun)					
Component 2-1						
Output 10 Strengthened Climate Resilient Hydro-Agricultural Infrastructure	10.1 No of Hydro-Agricultural Infrastructure constructed (small hydraulic infrastructure)	Project reports	15	60	134	
	10.2 Number of Hectares (ha) of irrigation schemes developed (ha)	Project reports	1,000	10,000	19,000	
	10.3 % increase of household revenue	Project reports	To be established at year 1	10%	25%	
Output 11 Water points for cattle and development of transhumance pathways	11.1 Number of Reservoirs, ponds, wells and boreholes for farming constructed along 1425km of Transhumance corridors	Project reports	31	50	130	
	11.2 Number of Hectares (ha) of pastoral land developed	Project reports	20	60000	134,000	
Output 12 River Sections trained for improved navigation	12.1 Distance of Navigation sections rehabilitated (km)	Project reports	none	50	120	
Output 13 Fisheries Infrastructure constructed for improved productivity	13.1 Number of Rehabilitated docks/landing sites	Project reports	2	5	30	
	13.2 Number of Developed spawning and floodplain areas	Project reports	To be established at year 1	20	40	
	13.3 Percentage (%) increase in fishery production	Project reports	To be established at year 1	10%	20%	
Component 2-2						
Output 14 Community-based Climate Adaptation Plans prepared and implemented	14.1 Community-based Climate Adaptation Action Plans prepared and implemented	Project reports	None	30	65	
Output 15 Increase in local economic development arising from the upgraded	15.1 Number of youths with gainful employment (self-employed, # of start-ups, etc.)	Project reports	To be established at year 1	75,000(30% women)	150,000 (30% women)	

H.1.2. Outcomes, Outputs, Activities and Inputs at Project/Programme level						
Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
Community based Livelihood infrastructure	15.2 Operational Infrastructure management committees, with at least 30% women representation	Project reports	10	100 (30% women)	292 (30% women)	
	15.3 Number of SMEs for youth employment established	Project reports	10	50 (30% women)	100(30% women)	
	15.4 Dykes for flood control constructed (Distance-km)	Project reports	None	100	122	
Component 3						
Output 16 Dissemination of good adaptation practices and climate information	16.1 Number of knowledge products produced and disseminated'	Knowledge products national workshops	0	10 Knowledge Products	20 Knowledge Products	
Output 17 Strengthened monitoring and evaluation by the NBA Environment Observatory	17.1 Functional M&E system 17.2 Functional MIS system	Annual M&E reports	0 0	Mid-term report, Annual reports, Annual audit reports	Project Completion Report, Annual reports, End of project audit	

Activities and Inputs

Activities	Description	Inputs	Description
Component 1: Development of the resilience of ecosystems and natural resources			
Component 1.1 Resource and Ecosystems Protection			
1. Strengthening sustainable land management for reduction of soil erosion and silting to sustainable levels	Reducing risks related to water shortages and land degradation. Includes promotion of soil conservation and stabilization measures	1.1. Consultation with partner states, agronomists, land officers on zoning of land for agriculture etc 1.2. Development of 9 sub basin masterplans/watershed management plans 1.3. Stabilization dunes; restore degraded land, and mechanical/biological management of ravines 1.4. Awareness campaign on erosion control and improved soil management practices 1.5. Expansion of sustainable land and water management interventions and technologies — includes gender-responsive CSA technologies 1.6. Supporting the fight against bushfires 1.7. Training for men and women farmers in business development and marketing and selected value chains in agriculture/ watershed management 1.8. Building of a one lane 60 m span bridge (Cameroun)	The interventions in each area will take into account climatic information soil type, plant species and competitive ability, aspect and topography.
2. Increasing forest productivity and sustainable forest management - targets area	Support to regeneration of dry forests and woodlands through assisted natural regeneration of trees and increasing density of trees on farms through farmer-	2.1. Support to development of private nurseries 2.2. Provision of nationally available sources of quality seed and plant material 2.3. Establishment of demonstration/trial forestry plots and re-establishment of forest tree species	Includes a suite of capacity strengthening approaches and activities to ensure skills and

Activities	Description	Inputs	Description
40,000 ha of forests	managed natural regeneration	2.4. Building capacity in SFM approaches, includes training of 200 community agents and associations for REDD+	management competency
3. Integration of agro-forestry into farming systems on 26,000ha of agro-forestry ha of the watershed	Support to increase the use of agro-forestry systems to reduce soil erosion and increase soil fertility and soil water holding capacity, thereby enhancing climate resilience and productivity	3.1. Support to development of private nurseries 3.2. Establishment of fodder banks with Napier grass 3.3. Planting trees/shrubs on as well as on boundaries/crops and in home gardens 3.4. Training men and women farmers on gender-responsive CSA technologies on-farm (sensitizing 500,000 women on climate change adaptation techniques) 3.5. Undertaking land certification— includes provision of access to restored land for 9,000 women.	
4. Strengthening sustainable management of natural habitats	Habitat expansion under conservation management beyond park boundaries in buffer zones, biological corridors, and ecological stepping-stones, to build connectivity & resilience to climate change	4.1. Identifying areas for development of the nature reserves 4.2. Establishment of corridors for wildlife to access water and biomass during droughts 4.3. Rehabilitation of bare land and spawning grounds across the basin nature conservation areas 4.4. Improvement of management within protected areas to reduce degradation of habitats	Interventions carried out with the aim of protecting biodiversity and the sustainable management of wetlands
Component 1.2 Strengthening the shared management of natural resources			
5. Strengthening hydro meteorological forecasting and early warning systems	This output is aimed at strengthening climate risk management in the Niger Basin to support poverty alleviation and sustainable development within the PIDACC. It builds on ongoing programs by the NBA Environment observatory, ACMAD and AGRHYMET.	5.1. Strengthening the decision support tools (Mike Hydro basin, Mike Hydro flood etc) through integration of climate risk management and capacity development 5.2. Long-term and on-demand capacity development and training program for NMHS staff on O&M, data modeling, and forecasting. Includes TA for support to certification of meteorological agencies in line with WMO requirements. 5.3. Expansion of coverage of Hydrological and meteorological monitoring networks 5.4. Establishing Community based Early Warning Systems 5.5. Development of new information and value-added products ⁶⁵ 5.6. Undertaking capacity-building, in the generation, & dissemination of climate products ^{66,67}	Effective consultant selection process, Effective consultation with stakeholders
6. Technical assistance for preparation of the Niger regional pollution	Interventions will comprise updating the existing study, supporting the operation of existing pollution control stations and installing new pollution stations	6.1. Diagnostic study leading to preparation of plan for strengthening the conventional network for water quality monitoring in the Niger River. Study includes (i) updating sources of diffused pollution in each country; (ii) characterization of pollutants (iii) preparation of plan for strengthening WQ monitoring in the Niger River	Effective consultant selection process, Effective consultation with stakeholders

⁶⁵ The project will ensure that women are adequately involved in the implementation of the EWS and that they benefit from climate information which is relevant to them, presented and transmitted in a way that is accessible and easy to understand

⁶⁶ The project will ensure that women are adequately involved in the implementation of the EWS and that they benefit from climate information which is relevant to them, presented and transmitted in a way that is accessible and easy to understand

⁶⁷ The project will improve the understanding of gender and climate change. Gender impact analyses will be conducted to identify gender-specific needs and protection measures related to floods, droughts, and other climate change related disasters in the basin. A toolkit for the dissemination of early warnings will be developed and it will include a gender-based analysis of the national and local media that are used to disseminate weather and climate alerts. Based on gender-sensitive consultations with end-users, the project will aim to supply climate information that is tailored to the specific needs of decision makers and local communities.

Activities	Description	Inputs	Description
1. Payment for Ecosystem Services (PES) and a Regional Adaptation Fund	The Environment Trust Funds will be designed to establish a system for the funding of watershed restoration	1.1. Diagnosis study and FRACC and PES feasibility studies 1.2. Establishing payments for biodiversity conservation, wildlife corridors and, in steep areas, watershed protection	Effective consultation with stakeholders
2. Capacity building of NBA executives and partner institutions functional countries	Effective water resource management through the capacity building of NBA executives, partner institutions and user associations and IWRM devices of functional countries	2.1. Preparation of Regional IWRM action Plan 2.2. Operationalization of IWRM plans in the nine countries and improved regional coordination in the implementation of the OP (2016-2025); 2.3. Detail a capacity development program that includes on-site and off-site training 2.4. Undertake capacity building of NBA executives, partner institutions to plan and implement climate resilient land and water management regimes	Leveraging development partner resources through the capacity development program
3. Technical assistance for preparation of the feasibility study leading to the Navigation masterplan (Benoué river, Nigeria and Cameroun)	A feasibility study for an investment program for installation of maritime infrastructure and identification of policy and institutional frameworks to ensure sustainability of the priority investments	3.1. Diagnostic analysis including (i) Sediment yield analysis (ii) Hydrologic and hydraulic analysis of the Niger and Benoué course and (iii) Definition of a regional soil conservation policy	Effective consultant selection process Effective consultation with stakeholders
Component 2: Development of population resilience			
Sub-component 2.1, Implementation of Climate Resilient Growth Infrastructure			
4. Implementation of Hydro-Agricultural Infrastructure (rehabilitation / construction of small hydro-agricultural infrastructure)	External water harvesting systems to mitigate dry spells, recharge groundwater, and permit multiple uses of water. Storage and irrigation will provide an important buffer against droughts, particularly in the Sahelian part of the Niger basin	4.1. Preliminary study and mapping of locations of the small hydraulic infrastructure across the 9 countries 4.2. Rehabilitation of water storage structures to preserve supplies and improve flood protection 4.3. Construction of rainwater harvesting infrastructure across the transhumance corridors	To ensure the sustainability of the small-scale infrastructure, the project will establish committees at the sector and cell level to select the sites, manage the structures.
5. Development of water points for cattle and develop transhumance pathways	Developing water infrastructure that is aligned with sustainable forage management in the Agro-pastoral systems of the Niger basin. It supports activities that reduce sensitivity to climate related Shocks in poor and vulnerable countries of the Niger basin.	5.1. Preliminary study and mapping of location for water pints across the 9 countries 5.2. Establishment of corridors for livestock movement to protect farmers' crops and trees, designating grazing and water access areas to ensure resilience of pastoral production systems 5.3. Construction of rainwater harvesting infrastructure across the transhumance corridors 5.4. Formulation and support of committees to manage small scale infrastructure	To ensure the sustainability of the small-scale infrastructure, the project will establish committees at the sector and cell level to select the sites, manage and maintain the structures.
6. Training of River Sections for improved navigation	River training and dredging (river training on 120 km of navigable portions, construction of docks and wharves).to maintain adequate channel size to accommodate vessels ; undertaken with agencies responsible for waterways or port management	6.1. Build / rehabilitate navigation sections and landing stages (on Niger, Tinkissu, Sankarani and Milo; (In the prefectures of Mandiana, Sigui, Kouroussa, Faranah, Dabola and Kankan).	

Activities	Description	Inputs	Description
7. Constructing Fisheries Infrastructure for improved productivity	Ecosystem approach to fisheries and aquaculture management that integrates ecological, socio-economic, governance dimensions.	7.1. Construction of 9 nursery stations and develop of 16,000 ha of spawning and floodplain areas).	
Component 2.2 Accompanying measures and social protection			
8. Development of Community Action Plans for adaptation to climate change	Support to increase the understanding by sub basin communities of the impacts of climate change and the value of climate resilient adaptation for proposed interventions.	8.1. Selection of target areas for support 8.2. Community based survey of project area and prioritization of interventions 8.3. Development of extension training materials that incorporate climate resilient approaches 8.4. Awareness raising/training of communities in gender sensitive adaptation planning (includes sensitizing an estimated 1,000,000 women on climate change adaptation techniques and climate resilience) 8.5. Community meetings and planning workshops 8.6. Development of local adaptation plans with zonation of land for agriculture, pasture, etc.	Participation of local communities will foster ownership of planned interventions and provide an opportunity for consensus building.
9. Strengthening community-based livelihood infrastructure	Diversifying income sources and increasing coping capacity. Livelihood activities will cover drought preparedness and natural resource management activities, including new/ improved water supplies, natural resource management activities and income diversification strategies such as investing in women groups	9.1. Selection and construction of 168 infrastructure upgrading Interventions (Activities) in specific watersheds; 9.2. Formulation and operationalisation of 292 infrastructure management committees with 30% representation for women; 9.3. Installation of 110 fish cages installed in river system(50% of beneficiaries will be women); this activity will support cage fish culture technology in selected locations along the Niger River system;; 9.4. Restocking 270 water reservoirs with fish—including integration of aquaculture into existing farming systems 9.5. Provision of access to recovered (previously degraded) land to 9,000 women and 3,000 youth 9.6. Support towards establishment of 100 SMEs for youth employment, of which 30 will be run by females; (ii) financing of 120 integrated Interventions (Activities) supported by women; and (iii) integration of strategies to increase women's participation in the workforce.	
Component 3: Project management and coordination			
10. Dissemination of good adaptation practices and climate information (includes awareness raising and advocacy to support replication)	Analysis of project results and production of a range of knowledge products including policy briefs and Case studies to demonstrate the benefits of project interventions.	10.1. Strengthening regional and national communications systems, based on the existing NBA communication strategy. 10.2. Preparation of briefing notes for local and national decision makers 10.3. Preparation of program knowledge management strategy 10.4. Preparation of program gender strategy 10.5. Preparation of media articles in newspapers, journals, newsletters, participatory videos, maintain website 10.6. Conduct an ex-post cost benefit analysis to determine the value added by the project	Dialogue and learning at workshops will ensure horizontal Scaling up and the project will use a website, site visits, and the media

Activities	Description	Inputs	Description
		compared with existing approaches and practices 10.7. Hold National workshops on WSM, SFM ,CSA	
11. Strengthen M&E system and practice by the NBA Environment Observatory	Improved management of the project by implementing agencies and local communities	11.1. Undertake project baseline data/benefit monitoring surveys 11.2. Strengthening regional and national GIS-based M&E and MIS incl. gender disaggregated data 11.3. Collection, analyses, reporting, dissemination of data and information on the project's progress 11.4. Preparing annual periodic reports	

H.2. Arrangements for Monitoring, Reporting and Evaluation

H.2. Arrangements for Monitoring, Reporting and Evaluation

Besides the arrangements (e.g. semi-annual performance reports) laid out in AMA, please provide project/programme specific institutional setting and implementation arrangements for monitoring and reporting and evaluation. Please indicate how the interim/mid-term and final evaluations will be organized, including the timing.

Project monitoring, reporting and evaluation

Monitoring and reporting will follow (i) the Theory for change (section E2-5) and (ii) the log frame matrix in Section H 1. The log frame will be used to assess progress towards the PDO through key indicators. The M&E approach will involve a combination of field-based data collection and remote sensing/GIS. In addition, the project monitoring, reporting and evaluation arrangements will be fully compliant with the GCF monitoring and accountability policy requirement and AMA.

Project progress will be monitored by the African Development Bank based on official data sources monitored directly by the NBA and national executing agencies. In addition, a number of individual evaluations will gauge progress towards the PDO, assess the impact of the programme on the beneficiaries, assess the quality of the work carried out by its different components, and evaluate overall project efficiency. The Grant Agreements between the African Development Bank and the NBA and executing agencies will detail the reporting requirements in line with the provisions of the African Development Bank procedures and on the basis of the indicators acceptable to the World Bank.

Milestone	Expected Dates
Start of Project Implementation	02/01/2019
Interim Evaluation	31/12/2022
Project Completion	31/12/2024
Final Evaluation	01/07/2025

Detailed reporting will be as follows:

Inception phase

A baseline study will be undertaken at programme inception to generate an updated assessment of the context, particularly with respect to the various indicators and targets. The results of the baseline study will be used as the reference point by which progress towards achievement of the targets will be assessed.

Annual performance report

Annual performance reports will be elaborated by the NBA and the National EEs during the lifespan of the programme. The reports will highlight risks that require the attention of the African Development Bank and GCF. The reports will include a narrative on implementation progress based on the log frame submitted in the FP, against the relevant criteria of the investment framework as set out in the FP. In line with the AMA, the APR shall be submitted to the GCF Secretariat on an annual basis for the period ending on 31 December within sixty days after the end of the relevant annual period.

The first APR will be submitted following the end of the calendar year and the last APR required to be submitted within six months of the end of the relevant reporting period.

Mid-term evaluation

The mid-term evaluation (MTE) will provide an independent assessment of programme performance at mid-term against the Fund's investment framework criteria, to analyse whether it is on track, what problems and challenges it is encountering, and which corrective actions are required so that the programme can achieve its intended outcomes by programme completion in the most efficient and sustainable way.

Final project evaluation

A final evaluation will be implemented six months after project closure and will examine the overall performance of the project against the investment criteria and other evaluation criteria that will be defined in the evaluation guidelines including financial/economic performances as part of the "Funded Activity" efficiency and effectiveness criterion, as well as the sustainability and scalability of results and impacts and lessons learned, during the relevant period. The objective of the final evaluations will be to learn lessons and apply those lessons in order to upscale and replicate successful programmes.

Independent evaluators

Interim and final evaluation reports will be prepared by an independent evaluator selected by the African Development Bank and will be submitted on the dates or according to the schedule set forth therein. Copies of these reports shall be forwarded by the African Development Bank to the Fund. The African Development Bank will retain all M&E records for this project for up to five years after project financial closure in order to support ex-post evaluations. A detailed M&E budget, monitoring plan and evaluation plan will be included in the programme evaluation prior to African Development Bank Board approval.

Methodologies for monitoring and reporting on key outcomes of the programme

The NBA observatory under the NBA Secretariat will be responsible for the overall project based M & E and will coordinate the establishment of a management information system (MIS) and M&E plan with the support of partner government agencies, contracted service providers, NGOs, and communities. A major effort will be dedicated to improving, updating and modernizing M&E systems at different levels and for different purposes. The M & E system will (i) enhance the periodic assessment of project performance; and (ii) enhance evaluation of their results in terms of relevance, efficiency, effectiveness, impact, and sustainability.

A highly qualified third party institution will be contracted for the duration of the project to provide M & E support to the NBA observatory and national agencies. The support would encompass tracking and analysis of inputs and outputs, key implementation processes, periodic impact assessments, and targeted studies/ analyses where implementation problems are identified. The third party institution will develop a comprehensive MIS to capture field information and allow for effective data analysis and reporting to guide implementation. The project-wide MIS would mainly address input-output monitoring related to the various activities proposed under the three PIDACC project components as a means of tracking implementation progress.

Sources of data for feeding the M&E system will include: (a) administrative data collected through the project MIS, such as progress, technical, and financial reports; (b) specially designed qualitative and quantitative household survey instruments; (c) existing and newly collected geo-referenced data (including global datasets); and (d) scientifically collected environmental and ecosystem health data. A baseline survey will be prepared. Data collected will be as much as possible disaggregated by gender.

GIS techniques (vegetative cover and land-use monitoring systems, erosion modelling using SWAT and related tools) combined with field based data collection and surveys will be used to provide baseline data, track program inputs and outputs, key processes, institutional performance, impacts and outcomes for spatially oriented landscape interventions (example Output 1: Strengthened sustainable land management).

For knowledge oriented services, client satisfaction surveys will be used to assess the performance of NBA and national agency services based on client experience (e.g. Output 5: Strengthened climate and weather information services including Last Mile EWS). The surveys will shed light on the constraints clients face in accessing information services, their views about the quality/adequacy of services, and the responsiveness of government. The surveys will be conducted by the NBA observatory supported by national agencies.

For capacity strengthening outputs (for instance, Output 8: Strengthened capacity for IWRM implementation at national and regional (NBA) levels, formal surveys will be used to monitor outputs. Findings from the sample of people interviewed will be applied to the wider target group or the sub basin population as a whole. Quantitative estimates will also be made for the size and distribution of impacts.

A Core Welfare Indicators Questionnaire will be used to measures changes in social indicators for different population groups at household level—specifically relating to indicators of access, utilization, and satisfaction with social and economic services, like access to rehabilitated agricultural land for women and youth.

Rapid appraisal methods will be used to gather views and feedback of beneficiaries and other stakeholders, in order to respond to national decision-makers' needs for information (e.g. Output 14 —Community based Climate Adaptation Plans preparation and implementation). Methods will include: key informant interviews; community group interviews and mini surveys. The M & E system will regularly track the performance of the community-based interventions (activities) and contribute to reporting on the national PIDACC indicators.

Beneficiary surveys and assessments will be undertaken (example Output 15 Community-based livelihood infrastructure implemented) through systematic consultation with project beneficiaries and other stakeholders to identify and design development initiatives, signal constraints to participation, and provide feedback to improve services. For activities implemented at national level (including the village/community levels), a monitoring system will be applied that will strengthen participatory methods and processes for data collection.

Beyond tracking and monitoring progress, an impact evaluation would be undertaken to better understand the extent to which GCF funded activities reach the poor and the magnitude of their effects on people's welfare. An ex-post comparison of the programme and non-equivalent control group will be undertaken. The methods will range from large-scale sample surveys in which project populations and control groups are compared before and after, and possibly at several points during program intervention to small-scale rapid assessment and participatory appraisals. Baseline and end surveys will help to gather baseline data, as well as evaluate progress across variables such as incomes, health, access to land, access to natural capital, gender dimensions and adaptive capacities.

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

I. Supporting Documents for Funding Proposal

- ☒ NDA No-objection Letter
- ☒ Feasibility Study
- ☒ Integrated Financial Model that provides sensitivity analysis of critical elements (xls format, if applicable)
- ☒ Confirmation letter or letter of commitment for co-financing commitment (If applicable)
- ☐ Project/Programme Confirmation/Term Sheet (including cost/budget breakdown, disbursement schedule, etc.) – see *the Accreditation Master Agreement, Annex I*
- ☒ Environmental and Social Impact Assessment (ESIA) or Environmental and Social Management Plan (If applicable)
- ☒ Appraisal Report or Due Diligence Report with recommendations (If applicable)
- ☒ Evaluation Report of the baseline project (If applicable)
- ☒ Map indicating the location of the project/programme
- ☒ Timetable of project/programme implementation

Appendix 1: Acronyms and Abbreviations

Acronyms	Definition
AfDB's	African Development Bank Group
ADF	African Development Fund
AMA	Accreditation Master Agreement.
AMLCFT	AfDB's established guidelines on anti-fraud, anti-corruption and anti-money laundering policies
AGRHYMET	Agriculture, Hydrology, Meteorology (research center; Niamey, Niger)
ACMAD	African Centre of Meteorological Applications for Development
ACPC	African Climate Policy Centre
AE	Accredited Entities
CSA	Climate Smart Agriculture
COM	Council of Ministers
CILSS	Permanent Inter-State Committee against Drought in the Sahel
CIF	Climate Investment Funds
CSOs	Civil Society Organizations
CBO	Community Based Organizations
COP	Conference of the Parties— supreme decision-making body of the UNFCCC Convention
ECOWAS	Economic Community of West African States
ECCAS	Economic Community of Central African States
EIRR	Economic Internal Rate of Return (Economic decision Criteria)
ESMF	Environmental and Social Management Framework
EC	European Commission
EE	Executing Entities
FEWS	Flood Early Warning System
FRACC	Regional CC adaptation fund
FAA	Funded Activity Agreement
FIP	Forestry Investment Program
GAP	Gender Action Plan
GCM	General Circulation Models
GHG	Green House Gases
GEF	Global Environmental Facility
GCF	Green Climate Fund
HIGH 5	African Development Bank Priorities for accelerating Africa's economic transformation: Light up and power Africa, feed Africa, industrialize Africa, integrate Africa and improve the quality of life for the people of Africa
IDD	Integrity Due Diligence
IDRC	International Development and Research Centre
IPCC	Intergovernmental Panel on Climate Change
IRR	Internal rate of Return
IUCN	International Union for Conservation of Nature
INDC	Nationally Determined Contribution
IWRM	Integrated Water Resources Management
LCCs	Local Coordination Committees
M&E	Monitoring and Evaluation
MoFP	Ministry of Finance and Economic Planning
MIS	Management Information System
NAPs	National Adaptation Plan— established under the Cancun Adaptation Framework
NDAs	National Designated Authorities

Acronyms	Definition
NAPA	National Adaptation Programme of Action
NASPA	National Adaptation Strategy and Plan of Action
NCs	National Communications submitted as part of the UNFCCC process
NCOFs	National Climate Outlook Forums
NFS	National Focal Structure
NMHS	National Meteorological and Hydrological Services
NSC	National Steering Committee
NPCU	National Programme Coordinating Unit
NPV	Net Present Value (economic decision support criteria)
O&M	Operation and Maintenance
OS	Operational Safeguard
OSS,	Sahara and Sahel Observatory (OSS)
PEPs	Politically Exposed Persons
PES	Payment mechanism for environmental services
PIDACC	Programme for Integrated Development and Adaptation to Climate Change in the Niger Basin
PME	Participatory monitoring and evaluation
PMF	Performance Measurement Frameworks (PMF)
PRCU	PIDACC Regional Coordination Unit
REDD+	Countries' efforts to Reduce Emissions from Deforestation and Forest Degradation,
RSC	Regional Steering Committee
RCOFs	Regional Climate Outlook Forums
SCP/NB	Silting Control Programme in the Niger Basin
SLM	Sustainable Land Management
SDAP	Sustainable Development Action Plan — Niger Basin
SDG	Sustainable Development Goals
SESA	Strategic Environmental and Social Assessment
SWAT	Soil and Water Assessment tool
SMEs	Small and medium-sized enterprises
SFM	Sustainable Forestry Management
TMC	Technical Monitoring Committees
UNECA	United Nations Economic Commission for Africa
USAID	U.S. Agency for International Development
UNCCD	United Nations Convention to Combat Desertification
UNCBD	United Nations Convention on Biological Diversity
UEMOA	West African Economic and Monetary Union,
WMO	World Meteorological Organization
WSM	Watershed Management
WRM	Water Resources Management



N° 538 /DGEC/MCVDD/SA

To: The Green Climate Fund ("GCF")

Cotonou, Benin, May 31, 2017

Re: Funding proposal for the GCF by the African Development Bank Group regarding GCF Project Proposal by the African Development Bank Group regarding the "Integrated Development and Adaptation to Climate Change in the Niger River Basin (PIDACC/NB)"

Dear Madam, Sir,

We refer to the programme, **"Integrated Development and Adaptation to Climate Change in the Niger Basin (PIDACC/NB)"** in **Benin** as included in the funding proposal submitted by the African Development Bank Group to us on May 31, 2017.

The undersigned is the duly authorized representative of Aina Martin Pepin, the National Designated Authority/focal point of Benin.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of **Benin** has no-objection to the as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with **Benin's** national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Name: Mrs. Aina Martin Pepin

Title: Ministry of Living Environment and Sustainable Development

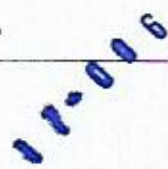
PRIME MINISTRY

CABINET

DESIGNATED NATIONAL
AUTHORITY OF THE GREEN
CLIMATE FUND

BURKINA FASO
Unité-Progrès-Justice

Ouagadougou, le 05/06/2017

N°  /PM/AND-FVC

The Focal Point

To

Green Climate Fund (GCF) Executive Director
Songdo International Business District
G-Tower, 175 Art Center daero
Yeonsu gu, Incheon 406 840
Republic of Korea

Subject: Funding proposal for the
GCF by the African Development
Bank Group regarding GCF Project
Proposal by the African Development
Bank Group regarding the "Integrated
Development and Adaptation to
Climate Change in the Niger River
Basin (PIDACC/NB)"

Dear Executive Director,

We refer to the programme, "**Integrated Development and Adaptation to Climate Change in the Niger Basin (PIDACC/NB)**" in **Burkina Faso** as included in the funding proposal submitted by the African Development Bank Group to us on May 31, 2017.

The undersigned is the duly authorized representative of Mamadou HONADIA, the National Designated Authority of Burkina Faso.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of **Burkina Faso** has no-objection to the as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with **Burkina Faso's** national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Ampliations:

- BAD
- DGRE
- MEEVCC
- Chrono



Mamadou HONADIA

Autorité Nationale Désignée FVC (AND)
Burkina Faso, Premier Ministère
(Ouagadougou - BF)

Tél : +226 70 24 02 40 / +226 78 24 02 40

E-mail: m60honadia@gmail.com

To: The Green Climate Fund ("GCF")

Abidjan, Côte d'Ivoire, 16 June 2017

Re: Funding proposal for the GCF by the African Development Bank Group regarding the "Integrated Development and Adaptation to Climate Change in the Niger River Basin (PIDACC/NB)".

Dear Madam, Sir,

We refer to the programme **"INTEGRATED DEVELOPMENT AND ADAPTATION TO CLIMATE CHANGE IN THE NIGER RIVER BASIN (PIDACC/NB)"** in Côte d'Ivoire as included in the funding proposal submitted by the African Development Bank to us on 31 May 2017.

The undersigned is the duly authorized representative of the Ministry of Salubrity, Environment and Sustainable Development, the focal point of Côte d'Ivoire.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Côte d'Ivoire has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with Côte d'Ivoire's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Name: Richemond Agré ASSIE

Title: Côte d'Ivoire GCF Focal Point

MINISTERE DE L'ENVIRONNEMENT,
DES EAUX ET FORETS



REPUBLIQUE DE GUINEE

Travail-Justice-Solidarité

Point Focal du Fonds Vert pour le Climat

Conakry, le 05.06.2017

No. 008/FVC/MEEF/PFFVC/20..

Le Point Focal

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by the African Development Bank Group regarding GCF Project Proposal by the African Development Bank Group regarding the "Integrated Development and Adaptation to Climate Change in the Niger River Basin (PIDACC/NB)"

Dear Madam, Sir,

We refer to the programme, "Integrated Development and Adaptation to Climate Change in the Niger Basin (PIDACC/NB)" in Guinea as included in the funding proposal submitted by the African Development Bank Group to us on May 31, 2017.

The undersigned is the duly authorized representative of Mohamed Lamine Doumbouya, the National Designated Authority/focal point of Rep. of Guinea.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Guinea has no-objection to the as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with Guinea's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Name: Mr. Mohamed Lamine Doumbouya

Title: Focal Point



Agence de l'Environnement et du
Développement Durable (AEDD)

Bamako, le 02 JUIN 2017



N° 1016 /MEADD/AEDD-FVC

*Le Directeur Général de l'Agence de
l'Environnement et du Développement
Durable*

A

**Monsieur Howard Bamsey Directeur Exécutif du
Fonds Vert pour le Climat
24-4 Songdo- Dong, Yeonsu-gu Incho City, Republic
of Korea
Tel: +82 32 458 60 59, Fax : + 82 32 458 60 94
Email: countrie@gcfund.org / secretariat@gcfund.org**

Re: Funding proposal for the GCF by the African Development Bank Group regarding GCF Project Proposal by the African Development Bank Group regarding the "Integrated Development and Adaptation to Climate Change in the Niger River Basin (PIDACC/NB)"

Dear Monsieur,

We refer to the programme, "Integrated Development and Adaptation to Climate Change in the Niger Basin (PIDACC/NB)" in Mali as included in the funding proposal submitted by the African Development Bank Group to us on May 31, 2017.

The undersigned is the duly authorized representative of Boureïma CAMARA, the National Designated Authority/focal point of Mali.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Mali has no-objection to the as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with Mali's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Le Directeur général,

Boureïma CAMARA
Chevalier de l'ordre national

REPUBLIQUE DU NIGER



Fraternité – Travail – Progrès

CABINET DU PREMIER MINISTRE

CONSEIL NATIONAL DE L'ENVIRONNEMENT POUR UN DEVELOPPEMENT DURABLE
SECRETARIAT EXECUTIF

To: The Green Climate Fund ("GCF")

Niamey, Niger, June 05, 2017

Re: Funding proposal for the GCF by the African Development Bank Group regarding GCF Project Proposal by the African Development Bank Group regarding the "Integrated Development and Adaptation to Climate Change in the Niger River Basin (PIDACC/NB)"

Dear Madam, Sir,

We refer to the programme, **"Integrated Development and Adaptation to Climate Change in the Niger Basin (PIDACC/NB)"** in **Niger** as included in the funding proposal submitted by the African Development Bank Group to us on May 31, 2017.

The undersigned is the duly authorized representative of Kamayé Maâzou, the National Designated Authority/focal point of Niger.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of **Niger** has no-objection to the as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with **Niger's** national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Name: Dr. Kamaye Maâzou

Title: Executive Secretary





FEDERAL MINISTRY OF ENVIRONMENT

HEADQUARTERS, MABUSHI, ABUJA.

Ref No. **FMENV/DCC/GCF/003**

Date. **5 June 2017**

To: The Green Climate Fund ("GCF")
Songdo International Business District
175, Art Center-daero
Yeonsu-gu, Incheon 406-840
Republic of Korea

Dear Sir,

Re: Funding proposal for the GCF by the African Development Bank Group regarding GCF Project Proposal by the African Development Bank Group regarding the "Integrated Development and Adaptation to Climate Change in the Niger River Basin (PIDACC/NB)"

We refer to the programme, "**Integrated Development and Adaptation to Climate Change in the Niger Basin (PIDACC/NB)**" in **Nigeria** as included in the funding proposal submitted by the African Development Bank Group to us on May 31, 2017.

The undersigned is the duly authorized National Designated Authority/focal point of Nigeria.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate Nigeria's no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of **Nigeria** has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with **Nigeria's** national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Name: Dr. Yerima Peter Tarfa
Title: GCF Focal Person Nigeria/Director, Department of Climate Change

REPUBLIQUE DU CAMEROUN
Paix – Travail – Patrie

MINISTERE DE L'ENVIRONNEMENT,
DE LA PROTECTION DE LA NATURE ET DU
DEVELOPPEMENT DURABLE

REPUBLIC OF CAMEROON
Peace – Work – Fatherland

MINISTRY OF ENVIRONMENT, PROTECTION
OF NATURE AND SUSTAINABLE
DEVELOPMENT

Yaoundé, Cameroon, May 31, 2017

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by the African Development Bank Group regarding GCF Project Proposal by the African Development Bank Group regarding the "Integrated Development and Adaptation to Climate Change in the Niger River Basin (PIDACC/NB)"

Dear Madam, Sir,

We refer to the programme, "Integrated Development and Adaptation to Climate Change in the Niger Basin (PIDACC/NB)" in Cameroon as included in the funding proposal submitted by the African Development Bank Group to us on May 31, 2017.

The undersigned is the duly authorized representative of Valentin WagnounTchonkap, the National Designated Authority/focal point of Cameroon.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of **Cameroon** has no-objection to the as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with **Cameroon's** national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

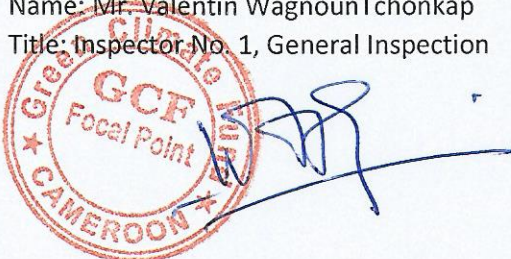
We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Name: Mr. Valentin WagnounTchonkap
Title: Inspector No. 1, General Inspection



MINISTERE DE L'ENVIRONNEMENT
ET DE LA PECHE

=====

FONDS VERT POUR LE CLIMAT

=====

CABINET DU MINISTRE

N° 025 /CAB/PNFVC/2017



REPUBLIQUE DU TCHAD
Unité*Travail*Progrès

=====

N'Djamena, le 09 juin 2017

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by the African Development Bank Group regarding GCF Project Proposal by the African Development Bank Group regarding the "Integrated Development and Adaptation to Climate Change in the Niger River Basin (PIDACC/NB)"

Dear Madam, Sir,

We refer to the programme, "Integrated Development and Adaptation to Climate Change in the Niger Basin (PIDACC/NB)" in Chad as included in the funding proposal submitted by the African Development Bank Group to us on May 31, 2017.

The undersigned is the duly authorized representative of Koularambaye Koundja Julien, the National Designated Authority/focal point of Chad.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Chad has no-objection to the as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with Chad's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Name: Mr. Koularambaye Koundja Julien

Title: Director



Environmental and social report(s) disclosure

Basic project/programme information	
Project/programme title	Programme for Integrated Development and Adaptation to Climate Change in the Niger Basin (PIDACC/NB)
Accredited entity	African Development Bank Group (AfDB)
Environmental and social safeguards (ESS) category	Category A

Environmental and Social Impact Assessment (ESIA) (if applicable)	
Description of report/disclosure	Strategic Environmental and Social Studies of The Integrated Program for Development and Adaptation to Climate Change in the Niger Basin (PIDACC/BN)*
Date of disclosure on accredited entity's website	2018-03-23 (English) and 2018-06-27 (French)
Language(s) of disclosure	English and French
Link to disclosure	English: https://www.afdb.org/en/documents/document/niger-strategic-environmental-and-social-studies-of-the-integrated-program-for-development-and-adaptation-to-climate-change-in-the-niger-basin-pidacc-bn-100875/ French: https://www.afdb.org/fr/documents/document/niger-strategic-environmental-and-social-studies-of-the-integrated-program-for-development-and-adaptation-to-climate-change-in-the-niger-basin-pidacc-bn-100875/
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	The summary of the ESMF below contains an ESMP consistent with the requirements for a category A project.
Resettlement Action Plan (RAP) (if applicable)	
Date of disclosure on accredited entity's website	Not Applicable
Any other relevant ESS reports and/or disclosures (if applicable)	
Description of report/disclosure	Summary of the Environmental and Social Management Framework (ESMF)
Date of disclosure on accredited entity's website	2018-02-16
Language(s) of disclosure	English and French

**This disclosure form is updated from the version circulated to the Board and active observers on 2 March 2018 (attached).*

Link to disclosure	English: https://www.afdb.org/en/documents/document/abn-multinational-integrated-development-and-climate-change-adaptation-in-the-niger-basin-program-pidacc-nb-97280/ French: https://www.afdb.org/fr/documents/document/abn-multinational-integrated-development-and-climate-change-adaptation-in-the-niger-basin-program-pidacc-nb-97280/
--------------------	--